Transport-Stokes Equation as a Mean-Field Limit

Manuel Heger

16th June 2026

Abstract

This thesis establishes improved convergence results for the microscopic model of spherical and inertialess particles sedimenting in a Stokes fluid to the transport-Stokes equation in the spririt of a mean-field limit in Wasserstein spaces. Based on existing work the Stokes interaction kernel is utilized to apply a version of Hauray's method in combination with approximation obtained by the method of reflections. In order to overcome some of the non-physical separation assumptions, we employ the non-attractivity of the Stokes interaction as it was illustrated in a recent paper. While the result presented there is just valid for interactions expressed by an explicit kernel, we demonstrate the compatability with the approximation generated by the method of reflections. In addition, we discuss the possiblity of extending this idea and enlight some of the difficulties arising while considering the behaviour of three sedimenting particles.

Contents

We consider a system of N spherical particles $\{X_i\}_{i=1}^N \subset \mathbb{R}^3$ with radius R. We assume the particles to be mutually disjoint, i.e., $B_i \cap B_j = \emptyset$ for $i \neq j$, where $B_i := B_R(X_i)$. Moreover, we assume these particles to be contained in a viscous fluid with a fluid velocity u_N satisfying the following system of equations:

$$\begin{cases} -\Delta u_N + \nabla p = 0 & \text{in } \mathbb{R}^3 \setminus \bigcup_{i=1}^N B_i, \\ \operatorname{div}(u_N) = 0 & \text{in } \mathbb{R}^3 \setminus \bigcup_{i=1}^N B_i, \\ |u_N(x)| \to 0 & \text{as } |x| \to \infty, \\ eu_N = 0 & \text{in } B_i \text{ for } i = 1, \dots, N, \\ \int_{\partial B_i} \sigma(u_N, p) \cdot n \, \mathrm{d}\mathcal{H}^2 = \frac{1}{N} g & \text{for all } i = 1, \dots, N, \\ \int_{\partial B_i} (x - X_i) \times (\sigma(u_N, p) \cdot n) \, \mathrm{d}\mathcal{H}^2 = 0 & \text{for all } i = 1, \dots, N, \end{cases} \tag{0.1}$$

where $eu := \frac{1}{2}(\nabla u + (\nabla u)^T)$ denotes the symmetric gradient, $\sigma(u, p) := 2eu - p\mathrm{Id}$ refers to the Cauchy stress tensor, and n is the inward unit normal of the respective ball. In this scenario, the fifth equation encodes that the gravitational force acting along a vector $g \in \mathbb{R}^3$ on a particle with mass $1/N$ equals the drag force on the particle's boundary. In the same manner, the last equation is imposed to force the system to be torque-free.

direkt ver
System ei

Beschreib

This system can be understood as a model for the instantaneous fluid velocity in a viscous fluid in the regime of small Reynolds numbers. This is extendable to a model for the sedimentation of inertialess particles by adding the dynamics

$$\dot{X}_i = u_N(X_i).$$

Accordingly, defining the empirical distribution of the particle centers at time t by $\rho_N(t) = \frac{1}{N} \sum_{i=1}^N \delta_{X_i(t)}$, a natural question is whether one can deduce the macroscopic dynamics of ρ_N, i.e., the collective behaviour of a sedimenting cloud of particles.

The system introduced above has been part of mathematical research for more than two decades, yet there is still room for further insights. The rigorous mathematical analysis started with the paper [JO04], in which the authors worked out separation assumptions on the interparticle distance sufficient to neglect the interaction effects between the particles on a relevant time scale. In other words, whenever the proposed condition is fulfilled, the cloud of particles moves approximately like a collection of single particles, eaxg independent of the presence of the others. Although the stated condition is unlikely to be fulfilled by a randomly generated particle constellation in nature, the paper introduced important conceptual techniques and a formalism to study the problem of sedimenting particles.

Based on this prework, the non-dilute case was brought into focus in [Hö18]. Here, a first mean-field result for system (0.1) was introduced and proven: if the initial empirical measure $\rho_N(0)$ converges to a given measure ρ_0 with an L^∞ density with respect to the Lebesgue measure, then the dynamics can be approximated by a solution to the transport-Stokes system given by

$$\begin{cases} \partial_t \rho + \left(K * \rho + \frac{1}{6\pi N R} g\right) \cdot \nabla \rho = 0 & \text{in } [0, \infty) \times \mathbb{R}^3, \\ \rho(0) = \rho_0 & \text{on } \mathbb{R}^3, \end{cases} \tag{0.2}$$

using the kernel $K = \Phi g$ with the Oseen tensor

$$\Phi(x) := \frac{1}{8\pi} \left(\frac{\mathrm{Id}}{|x|} + \frac{x x^T}{|x|^3} \right) \quad \text{for } x \neq 0.$$

However, the presented result was still restrictive in the sense that certain separation assumptions were needed, essentially forcing the minimal distance to scale like $N^{-1/3}$, thereby preventing any uncontrollable effects caused by the kernel's singularity. In light of the observation that for randomly placed particles the expected scaling behaviour for the minimal distance is $d_{\min} \sim N^{-2/3}$, the imposed assumptions remain restrictive. Further progress in this direction was achieved in [HS21], where the setup was reformulated in terms of Wasserstein spaces. Consequently, Hauray's method for proving mean-field results became applicable, leading to an improvement in the allowable scaling behaviour up to $d_{\min} \sim N^{-1/2}$.

Taking a step back, one observes that these improvements and modifications all went hand in hand with an advanced understanding of a simplified model that only takes into account binary interactions, given in terms of K as

$$\dot{X}_i(t) = \frac{1}{6\pi RN} g + \frac{1}{N} \sum_{j \neq i} K(X_i - X_j). \tag{0.3}$$

Indeed, the result in [JO04] indicated under which conditions the self-interaction term $\frac{1}{6\pi RN} g$ essentially dominates the influence of the interaction term, whereas the results in [Hö18] and [HS21] are based on proving that (0.2) is a mean-field approximation for $\rho_N(t)$ with particle centers following (0.3). Working with (0.3) is advantageous as its more explicit character allows to analyse the solution of the binary system more concrete.

Motivated by this, it is natural to examine the properties of the simplified system to obtain further insights into the behaviour of the original problem. Following this agenda, it was illustrated in [HS25] that by utilizing the non-attractivity of the Stokes interaction - a property encoded by symmetry properties of the kernel K- one can push the boundaries up to a scaling behaviour of $N^{-2/3+\varepsilon}$ with a small $\varepsilon > 0$ depending on further parameters. However, the presented technique was only carried out for the binary interaction, and so the task to lift the result to the actual system remains open — which is the goal of this paper.

To have all the ingredients at hand for solving this, it would be unforgivable to withhold an important tool that connects these two worlds: the actual non-binary system and the simplified binary system. This tool is an approximation technique called the method of reflections, which is frequently used in microhydrodynamics. The method allows to quantify the distance between a solution of the actual microscopic system and a solution of the simplified model and hence is a fundamental key for proving the aforementioned results. In order to lift the findings presented in [HS25], we need to revisit the existing estimates forming the basis of the method of reflections. Being sufficient for the scenarios considered in [Hö18] and [HS21], the existing results are slightly too weak to handle the situation in [HS25]. Thus, we illustrate how to modify the method, inspired by the insights generated in [HS25], and how to finally apply Hauray's method to lift the result from the simplified binary case to the full problem.

1 Preliminaries

In this section, we introduce basic notational conventions and facts used throughout the whole text. Moreover, general well-posedness results are mentioned and will be assumed later on.

In general, we use the notation $A \lesssim B$ to indicate the existence of a constant $C > 0$ such that $A \leq CB$. This constant is independent of any relevant quantity, e.g. time or the specific particle constellation, and may change from line to line. Similarly, we say $A \sim B$ if $A \lesssim B$ and $B \lesssim A$. In situations, where this convention is not applicable, we will use C as a generic constant that may change from line to line.

Particle Configurations

A particle configuration is given by a set of particle centers

$$\{X_i\}_{i=1}^N \subset \mathbb{R}^3 \quad \text{with } X_i \neq X_j \text{ for } i \neq j.$$

The empirical distribution associated to a particle configuration is defined as $\rho_N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}$ with δ_x being the unit point mass located in x. Given the particle radius R we identify the i-th particle with $B_i := B_R(X_i)$. By $\phi_N = R^3 N$ we denote the particle volume fraction.

Given a particle configuration we abbreviate the particle distances by $d_{ij} = |X_i - X_j|$ for $i, j \in \{1, \dots, N\}$. For a fixed index i we denote by i_{nn} the index of a nearest neighbour, i.e. $i_{nn} \in \operatorname{argmin}_{j \neq i} d_{ij}$. The minimal interparticle distance is defined as $d_{min} := \min\{d_{ij} \mid j \neq i\}$, whereas the minimal next-to-nearest distance $d_{min,1}$ satisfies $d_{min,1} = \min\{d_{ij} \mid j \notin \{i, i_{nn}\}\}$. The same procedure can be applied with $i_{nnn} \in \operatorname{argmin}_{j \neq i, i_{nn}} d_{ij}$ to define $d_{\min,2}$.

Moreover, sums of inverse distances will arise frequently and we will denote their supremum for $k \in \mathbb{N}$ by

$$S_{k,0} = \sup_{i=1,\dots,N} \frac{1}{N} \sum_{j \neq i} d_{ij}^{-k}.$$

By a standard sphere packing argument, one obtains the classical bounds

$$S_{2,0} \leq N^{-2/3} d_{min}^{-2}, \tag{1.1}$$

$$S_{3,0} \leq \log(N) N^{-1} d_{min}^{-3}, \tag{1.2}$$

which are proven in [NS20, Lem. 4.8].

Function Spaces

To describe the fluid velocity and incorporate the assumption that the fluid rests at infinity we are going to work with homogeneous Sobolev spaces. They are defined by

$$\dot{H}^1(\mathbb{R}^3) = \overline{C_c(\mathbb{R}^3, \mathbb{R}^3)}^{\|\nabla \cdot\|_{L^2(\mathbb{R}^3)}}$$

$$\dot{H}^1_\sigma(\mathbb{R}^3) = \{u \in \dot{H}^1(\mathbb{R}^3) \mid \operatorname{div}(u) = 0 \text{ in } \mathbb{R}^3 \text{ in the distributional sense}\}$$

endowed with the gradient norm $\|u\|_{\dot{H}^1} = \|\nabla u\|_{L^2}$. Given a sequence $u_n \in L^q_{\text{loc}}(\mathbb{R}^3)$ and $u \in L^q_{\text{loc}}(\mathbb{R}^3)$ we abuse notation and write $\|u_n - u\|_{L^q_{\text{loc}}} \to 0$, whenever $\|u_n - u\|_{L^q(\Omega)} \to 0$ for all $\Omega \Subset \mathbb{R}^3$.

In the following, for S being a metric space $\mathcal{P}(S)$ refers to the space of probability measures on $(S, \mathcal{B}(S))$ and the notation $L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ is used to denote all the probability measures on $(\mathbb{R}^3, \mathcal{B}(\mathbb{R}^3))$, which are absolutely continuous with respect to the Lebesgue measure and whose densities are in $L^\infty(\mathbb{R}^3)$. For $\rho \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ the symbol ρ refers to both the measure and the density at once.

For $F : \mathbb{R}^n \to \mathbb{R}^m$ measurable and $\mu \in \mathcal{P}(\mathbb{R}^n)$ the pushforward measure $F_\# \mu$ is defined via $F_\#\mu(A) = \mu(F^{-1}(A))$. Given $\mu, \nu \in \mathcal{P}(\mathbb{R}^3)$ the set of couplings of μ and ν is denoted by

$$\Gamma(\mu, \nu) := \{\pi \in \mathcal{P}(\mathbb{R}^3 \times \mathbb{R}^3) \mid T^{(1)}_\# \pi = \mu, \ T^{(2)}_\# \pi = \nu\},$$

where $T^{(i)}((x_1, x_2)) = x_i$. For $p \in [1, \infty]$ we consider the space

$$\mathcal{P}_p(\mathbb{R}^3) = \{\mu \in \mathcal{P}(\mathbb{R}^3) \mid \|x\|_{L^p(\mu,\mathbb{R}^3)} < \infty\}$$

and set the p-Wasserstein distance between $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^3)$ to be

$$\mathcal{W}_p(\mu, \nu) = \inf\{\|x - y\|_{L^p(\pi,\mathbb{R}^3\times\mathbb{R}^3)} \mid \pi \in \Gamma(\mu, \nu)\}.$$

Stokes Equation

First, it is appropriate to clarify what to understand as a weak solution to the Stokes problem

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{in } \Omega_N, \\ \operatorname{div}(u) = 0 & \text{in } \mathbb{R}^3, \\ u(x) \to 0 & \text{as } |x| \to \infty, \\ u = V_i & \text{in } B_i, \forall i = 1, \dots, N, \end{cases}$$

with $V = (V_i)_{i=1}^N \in (\mathbb{R}^3)^N$ and $\Omega_N = \mathbb{R}^3 \setminus \cup_{i=1}^N B_R(X_i)$. A pair $(u, p) \in \dot{H}^1_\sigma(\mathbb{R}^3) \times L^2(\mathbb{R}^3)$ is a weak solution to the previous problem, if $u = V_i$ in $B_R(X_i)$ and

$$\int_{\Omega_N} \nabla u : \nabla\varphi - p \operatorname{div}(\varphi) dx = 0$$

for all $\phi \in C_c(\Omega_N, \mathbb{R}^3)$. Existence and uniqueness of weak solutions follow by standard arguments using the Lax-Milgram Lemma, see for example [Gal11, V.2.1].

The following maximum modulus theorem will be useful in the subsequent analysis.

Lemma 1.1. *Given $R > 0$ and $x \in \mathbb{R}^3$ let $\Omega = \mathbb{R}^3 \setminus B_R(x)$ be an exterior domain, $g \in C(\partial\Omega)$ with $\int_{\partial\Omega} g \cdot n \, d\mathcal{H}^2 = 0$, then the unique solution $u \in \dot{H}^1_\sigma(\Omega, \mathbb{R}^3)$ to the Dirichlet problem*

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{in } \Omega, \\ u = g & \text{on } \partial\Omega, \end{cases}$$

satisfies

$$\|u\|_{L^\infty(\Omega)} \le C\|g\|_{L^\infty(\partial\Omega)}$$

for some constant C independent of R.

Proof. This statement is proven for general exterior domains in [MR94, Thm 2.2]. □

The foundation of most of the subsequent analysis is the explicit solution to the one-particle problem

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{in } \mathbb{R}^3 \setminus B_R(0) \\ \operatorname{div}(u) = 0 & \text{in } \mathbb{R}^3, \\ u(x) \to 0 & \text{as } |x| \to \infty, \\ \int_{\partial B_R(0)} \sigma(u, p) \cdot n \, d\mathcal{H}^2 = \frac{1}{N} g, \end{cases}$$

for some fixed $g \in \mathbb{R}^3$. The unique solution of this problem is given by

$$U(x) = \begin{cases} \frac{1}{6\pi RN} g & \text{for } x \in B_R(0), \\ \frac{1}{N}(\Phi g)(x) + \frac{R^2}{6N}\Delta(\Phi g)(x) & \text{for } x \notin B_R(0), \end{cases} \tag{1.3}$$

where Φ is the fundamental solution for the Stokes equation, known as the Oseen tensor, defined by

$$\Phi(x) := \frac{1}{8\pi}\left(\frac{\mathrm{Id}}{|x|} + \frac{xx^T}{|x|^3}\right) \qquad \text{for } x \neq 0.$$

A proof of this solution formula can be found in [GMP11, 2.1.2]. Setting $K = \Phi g$, we obtain the bounds

$$|K(x)| + |\nabla K(x)||x| + |\nabla^2 K(x)||x|^2 \le |x|^{-1}.$$

Well-Posedness

Well-posedness of the microscopic problem is discussed in [Hö18] and can be summarized in the following theorem.

Theorem 1.1 ([Hö18, Thm. A.1])**.** *For any initial configuration of particles $(X^0) \in (\mathbb{R}^3)^N$ s.t. the closed balls $\bar{B}_R(X_i^0)$ are disjoint, there exists a time $T_* > 0$ such that (**??**) has a unique solution on $[0, T_*)$. Moreover, $T_* = \infty$ or*

$$\liminf_{t \to T_*} d_{min}(t) = 2R.$$

In addition, a detailed analysis of the well-posedness of the macroscopic system can be found in [HS21]. The proof is based on the fact that the solution is the fixed point of some contractive operator.

Theorem 1.2 ([HS21, Thm. 2.7])**.** *Assume $\tau_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$, then for (**??**) there exists a unique solution pair $(\tau, v_N) \in L^\infty_T(L^\infty \cap \mathcal{P}) \times L^\infty_T(W^{2,p})$ for $p \in (3, \infty)$. In particular,*

$$\|\tau\|_{L^\infty_T(L^\infty)} \leq \|\tau_0\|_{L^\infty} \quad \text{and} \quad \|v_N\|_{L^\infty_T(W^{2,p})} \leq C_p \|\tau_0\|_{L^\infty}$$

with a constant C_p just depending on p.

Remark 1.1.

- For $\rho_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ we can assume for $t \geq 0$ that the fluid velocity satisfies $v_N(t, \cdot) \in W^{1,\infty}$ by a Sobolev embedding. In this case, $\rho(t)$ can be represented as the pushforward along the flow associated to $v_N + \frac{1}{6\pi RN} g$.

2 Introducing Hauray's Method

In this section we are going to illustrate a general approach used for proving mean-field results in Wasserstein spaces for dynamics given in terms of interaction kernels. We will emphasize on the current obstacles arising in the case of sedimenting particles. Originally introduced by Hauray in [Hau09] for the case of the Biot-Savart kernel and based on the work of Dobroushin [Dob79], the presented approach forms the foundation of several mean-field results as the ones in [HS21] with the Oseen tensor or in [HS25] for a more general class of kernels.

Since in the case of system (**??**) the velocity field is not given directly in terms of a kernel, three essential steps must be performed before applying Hauray's method.

- Find an approximation $u_N^{(0)}$ of u_N in terms of the Oseen tensor and establish bounds on the approximation error.
- Bound $S_{2,0}$ in terms of d_{min} and $\mathcal{W}_p(\rho_N, \rho)$.
- Establish an Lipschitz-type estimate on the driving dynamics, i.e. a estimate like

$$|u_N(X_i) - u_N(X_j)| \leq C|X_i - X_j|$$

  with explicit dependencies of C on the actual particle configurations.

So far, these points might seem unreasoned, but we will illustrate their importance by performing the proof of the following theorem, basically adapted from [HS21, Thm.1.1].

Theorem 2.1. *Let $\rho_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ and assume $\mathcal{W}_p(\rho_N(0), \rho_0) \to 0$ for some $p \in [1, \infty]$. Furthermore, assume that the sequence of particle configurations fulfills*

$$d_{min}(0) \sim N^{-1/3}, \tag{A1}$$

$$\log(N)\phi_N \to 0. \tag{A2}$$

Then, there exists some $C > 0$ depending on ρ_0 such that for all $T > 0 : \exists N(T) \in \mathbb{N} : \forall N \geq N(T)$:

$$\mathcal{W}_p(\rho_N(t), \rho(t)) \leq (\mathcal{W}_p(\rho_N(0), \rho_0) + E_N)e^{Ct} \quad \text{for all } 0 \leq t \leq T$$

and some sequence $E_N \to 0$ for $N \to \infty$. Moreover, for $N \geq N(T)$

$$d_{min}(t) \geq d_{min}(0)e^{-Ct} \quad \text{for } 0 \leq t \leq T.$$

In addition, for $q < 3/2$ and $\Omega \Subset \mathbb{R}^3$ it holds $\|u_N(t) - v_N(t)\|_{L^q(\Omega)} = C_\Omega e_N(t)$ as $N \to \infty$. Here, the constant C_Ω is just depending on diam(Ω) *and the error term $e_N(t)$ is independent of* Ω.

Before diving into the technicalities of the proof, we will state the results regarding the key properties introduced previously.

Approximation via the Method of Reflections

We are going to use the kernel-based approximation $u_N^{(0)}$ given by the superposition of the solutions to the single-particle problems, i.e. $u_N^{(0)}(x) = \sum_{i=1}^N U(x - X_i)$ for

$$U(x) = \begin{cases} \frac{1}{6\pi RN} g & \text{for } x \in B_R(0), \\ \frac{1}{N} K(x) + \frac{R^2}{6N} \Delta K(x) & \text{for } x \notin B_R(0). \end{cases}$$

This approximation replaces the non-binary, highly implicit hydrodynamic interaction by an explicit binary interaction, which is more accessible for direct calculations. Moreover, using an expression in terms of the kernel can be motivated by observing that the leading order term of $u_N^{(0)}$ in R is given by

$$u_N^{(0)}(t, X_i) \approx \frac{1}{6\pi RN} g + K * \rho_N(t)(X_i) + \dots .$$

Consequently, there is an obvious similarity to the macroscopic driving vector field given by $\frac{1}{6\pi RN}g + K * \rho(t)$.

The quality of approximation is ensured by the method of reflections - an approximation technique, for which a more detailed discussion will follow in subsequent sections. The relevant result is the following estimate.

Lemma 2.1 ([Hö18, Prop. 3.12]). *There exists some $\theta > 0$, such that for any particle configuration with $d_{min} > 4R$ and $\phi_N S_{3,0} < \theta$ it holds*

$$\|u_N - u_N^{(0)}\|_{L^\infty(\mathbb{R}^3)} \lesssim S_{2,0}(S_{2,0}\phi_N + R). \tag{2.1}$$

Bounds on the Sums of Inverse Distances

As it becomes clear from (2.1) the terms $S_{k,0}$ for $k = 2, 3$ play an important role in quantifying the approximation errors. Besides the estimates (1.1) and (1.2), there are more advanced estimates utilizing the fact that the empirical distribution $\rho_N(0)$ of the particle positions tends to ρ_0. Remarking that the estimates in (1.1) and (1.2) are based on a sphere packing argument relying on a worst case assumption, in which the particles are highly concentrated around one point, one can make use of the the regularity of ρ_0 to weaken the possible concentration of mass. By this approach the Wasserstein distance is incorporated into the estimate and one obtains an inequality of the following form, which is the statement of [HS25, Lem. 2.1].

Lemma 2.2. *Given $Y_1, \dots, Y_N \in \mathbb{R}^3$, set $\tau_N = \frac{1}{N}\sum_{i=1}^N \delta_{Y_i}$ and assume $\tau \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$, then for $p \in [1, \infty]$*

$$\frac{1}{N}\sum_{i=1}^N |Y_i - Y_j|^{-2} \lesssim \|\tau\|_\infty^{2/3} + \left(1 + N^{-2/3} d_{min}^{-2}\left(\|\tau\|_\infty^{\frac{p}{3(3+p)}} + \|\tau\|_\infty^{1/3}\right)\right)\mathcal{W}_p(\tau_N, \tau)^{\frac{p}{3+p}}, \tag{2.2}$$

reducing to

$$\frac{1}{N}\sum_{i=1}^N |Y_i - Y_j|^{-2} \lesssim \|\tau\|_\infty^{2/3} + \left(1 + N^{-2/3} d_{min}^{-2}\|\tau\|_\infty^{1/3}\right)\mathcal{W}_\infty(\tau_N, \tau)$$

in the case $p = \infty$.

Lipschitz-type Estimate for u_N

Up until now, it is clear that neither (1.1), (1.2) nor (2.2) are useful, if we are not able to bound the temporal development of d_{min} and thus control the behaviour of $S_{k,0}(t)$. One natural approach is to analyze

$$\frac{d}{dt}|X_i - X_j|^2 = 2(X_i - X_j)^T\left(u_N(X_i) - u_N(X_j)\right) \geq -2|u_N(X_i) - u_N(X_j)||X_i - X_j|$$

and hence a Lipschitz-estimate for u_N would be desirable. In fact, according to [Hö18, Lem. 3.16] it holds

Lemma 2.3. *Consider a particle configuration with $d_{min} > 4R$. There exists $\theta > 0$, such that in the case $NR^3 S_{3,0} < \theta$ the particle velocities satisfy*

$$|u_N(X_i) - u_N(X_j)| \lesssim S_{2,0}|X_i - X_j| \qquad \textit{for } i \neq j. \tag{2.3}$$

However, the Lipschitz constant in (2.3) again depends highly on the inverse of the distances. This fact seems to undermine the goal of bounding the minimal distance using this estimate. To overcome this difficulty, a bootstrap argument will be performed in the following proof of Theorem 2.1.

Proof of Theorem 2.1:

Step 0: *Defining the bootstrap setup:*

To reduce case distinctions, we are going to give the proof for $p \in [1, \infty)$, the case $p = \infty$ is performed in [HS21]. We start the proof by setting up the bootstrap environment. Given $N \in \mathbb{N}$, we define $T^*(N)$ to be the maximal time such that

$$d_{ij}(t) \geq \tfrac{1}{2} d_{ij}(0) e^{-C_1 t}, \tag{B1}$$

$$S_{2,0}(t) \leq L, \tag{B2}$$

are satisfied for all $0 \leq t \leq T^*(N)$ with fixed constants $L, C_1 > 0$, which will be determined later. Here $S_{k,0}(t)$ refers to the value of $S_{k,0}$ associated to the particle configuration $\{X_i(t)\}_i$ at time t. The overall goal is to prove that for any $T > 0$ there is some $N(T) \in \mathbb{N}$ with $T^*(N) \geq T$ for $N \geq N(T)$. This will be done by some continuity argument based on proving that the strict versions[1] of the inequalities (B1) and (B2) hold true for all $s \leq T \wedge T^*(N)$. The bound on the Wasserstein distance will be a necessary side product of the proof. According to this strategy, we consider from now on $T > 0$ arbitrary, but fixed.

Step 1: *Bounding $S_{3,0}(t)\phi_N$:*

In order to apply both Lemma 2.1 and 2.3, we need to ensure that for N large enough $S_{3,0}(t)\phi_N < \theta^*$ for some $\theta^* > 0$ and all $0 \leq t \leq T \wedge T^*(N)$. By (B2) and (1.2), we obtain for $0 \leq t \leq T \wedge T^*(N)$ the bound

$$S_{3,0}(t)\phi_N \leq e^{3C_1 t} S_{3,0}(0)\phi_N \leq e^{3C_1 T} \log(N) N^{-1} d_{min}^{-3}(0)\phi_N \to 0 \quad \text{as } N \to \infty$$

according to assumptions (A1) and (A2). In consequence, we will assume the validity of the estimates given in Lemma 2.1 and 2.3 for the ongoing discussion.

Step 2: *Differential inequality for the Wasserstein distance:*

Let $\gamma \in \mathcal{P}(\mathbb{R}^3 \times \mathbb{R}^3)$ be an optimal coupling between $\rho_N(0)$ and ρ_0, i.e.

$$\mathcal{W}_p(\rho_N(0), \rho_0) = \left(\int_{\mathbb{R}^3 \times \mathbb{R}^3} |y - z|^p d\gamma(y, z) \right)^{1/p}.$$

According to Remark 1.1, we know that v_N is Lipschitz and thus $\rho(t)$ is the pushforward of ρ_0 along the flow of $v_N(t) + \frac{1}{6\pi N R} g$, which we want to denote by $Z(t, 0, \cdot)$. In addition, $\rho_N(t)$ is the pushforward of $\rho_N(0)$ along the flow map $Y(t, 0, \cdot)$ of the vector field $u_N(t)$. Hence, we find a coupling γ_t between $\rho_N(t)$ and $\rho(t)$ by considering

$$\gamma_t = (Y(t, 0), Z(t, 0))_{\#}\gamma$$

and by definition of the Wasserstein distance we end up with the inequality

$$\mathcal{W}_p(t)^p := \mathcal{W}_p(\rho_N(t), \rho(t))^p \leq \int_{\mathbb{R}^3 \times \mathbb{R}^3} |y - z|^p d\gamma_t(y, z) =: \eta(t).$$

Taking the derivative of $\eta(t)$ in t leads to

$$\begin{aligned} \tfrac{d}{dt}\eta(t) &\leq p \int_{\mathbb{R}^3 \times \mathbb{R}^3} |Y(t, 0, y) - Z(t, 0, z)|^{p-1} \, |\tfrac{d}{dt} Y(t, 0, y) - \tfrac{d}{dt} Z(t, 0, z)| d\gamma(y, z) \\ &\leq p \int_{\mathbb{R}^3 \times \mathbb{R}^3} |y - z|^{p-1} |u_N(t, y) - v_N(t, z) - \tfrac{1}{6\pi R N} g| d\gamma_t(y, z). \end{aligned}$$

[1] Given an inequality $A \leq B$, we call $A < B$ its strict version.

By introducing the approximation $u_N^{(0)}$ we obtain

$$\begin{aligned}
\tfrac{d}{dt}\eta(t) &\lesssim \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|u_N(t,y)-u_N^0(t,y)|d\gamma_t(y,z) \\
&\quad + \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|[K*\rho_N(t)](y)-[K*\rho_N(t)](z)|d\gamma_t(y,z) \\
&\quad + \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}\tfrac{R^2}{6}|[\Delta K*\rho_N(t)](y)|d\gamma_t(y,z) \\
&=: I_1 + I_2 + I_3.
\end{aligned}$$

By Lemma 2.1, Young's inequality, and estimate (1.1), I_1 satisfies

$$I_1 \lesssim \eta(t) + \left(S_{2,0}(t)(S_{2,0}(t)\phi_N + R)\right)^p \lesssim \eta(t) + \left(Le^{2C_1 t}\left(\phi_N + R\right)\right)^p.$$

Based on

$$\frac{R^2}{6}[\Delta K*\rho_N(T)](y) \lesssim R^2 S_{3,0}(t) \lesssim e^{3C_1 t}\, R^2 S_{3,0}(0) \lesssim e^{3C_1 t} R S_{2,0}(0) \quad \gamma_t\text{- a.e.}$$

one deduces the upper bound

$$I_3 \lesssim \eta(t) + \left(e^{3C_1 t} R S_{2,0}(0)\right)^p.$$

Establishing a bound on I_2 is more extensive. In a first step, we remark

$$\begin{aligned}
|[K*\rho_N(t)](y)-[K*\rho(t)](z)| &\le \int_{\mathbb{R}^3\times\mathbb{R}^3} |K(y'-y)-K(z'-z)|d\gamma_t(y',z') \\
&\lesssim \int_{\mathbb{R}^3\times\mathbb{R}^3} \left(|y'-y|^{-2}\mathbb{1}_{y\neq y'} + |z'-z|^{-2}\right)\left(|y'-z'|+|y-z|\right)d\gamma_t(y',z').
\end{aligned}$$

Using Young's inequality to estimate

$$|y-z|^{p-1}|y'-z'| \lesssim |y-z|^p + |y'-z'|^p,$$

we realize that after applying Fubini I_2 decomposes in terms of the form

$$I_{2,1} = \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^p \int_{\mathbb{R}^3\times\mathbb{R}^3} |y'-y|^{-2}\mathbb{1}_{y'\neq y} d\gamma_t(y',z')d\gamma_t(y,z)$$

and

$$I_{2,2} = \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^p \int_{\mathbb{R}^3\times\mathbb{R}^3} |z'-z|^{-2} d\gamma_t(y',z')d\gamma_t(y,z).$$

Since $\operatorname{supp}(\gamma_t) \subset \{X_i(t)\}_i \times \mathbb{R}^3$, we can bound $I_{2,1}$ by

$$I_{2,1} \le \eta(t)S_{2,0}(t). \tag{2.4}$$

To handle $I_{2,2}$ we first remark that $\|\rho(t)\|_{L^\infty} \le \|\rho_0\|_{L^\infty}$ by the conservative properties of the transport equation with a Lipschitz vector field. Based on this, we can establish for arbitrary $y \in \mathbb{R}^3$

$$\begin{aligned}
\int_{\mathbb{R}^3\times\mathbb{R}^3} |y'-y|^{-2}d\gamma_t(x',y') &= \int_{\mathbb{R}^3} |y'-y|^{-2}\rho(t,y')dy' \\
&= \int_{B_1(y)} |y'-y|^{-2}\rho(t,y')dy' + \int_{\mathbb{R}^3\setminus B_1(y)} |y'-y|^{-2}\rho(t,y')dy' \\
&\le \|\rho(t)\|_{L^\infty}\int_{B_1(y)} |y'-y|^{-2}dy' + \int_{\mathbb{R}^3\setminus B_1(y)} \rho(t,y')dy' \lesssim \|\rho_0\|_{L^\infty} + 1.
\end{aligned}$$

Hence,

$$I_{2,1} \lesssim (1 + \|\rho_0\|_{L^\infty})\eta(t).$$

Summarizing and using (B2), we end up with

$$\tfrac{d}{dt}\eta(t) \lesssim (1 + \|\rho_0\|_{L^\infty} + L)\eta(t) + \left(Le^{2C_1 t}\left(\phi_N + R\right)\right)^p.$$

Applying Gronwall's Lemma results in

$$\mathcal{W}_p(t)^p \leq \eta(t) \leq (\eta(0) + c\left(L\left(\phi_N + R\right)\right)^p)e^{pCt} =: (\eta(0) + E_N^p)e^{pCt} \tag{2.5}$$

for some constants c independent of the actual particle configuration and C depending on C_1, ρ_0 and L. It is worth to notice that (2.5) is exactly the estimate on the Wasserstein distance claimed in the theorem, since $E_N \to 0$ by (A2).

Step 3: *Bounding $S_{2,0}(t)$:*

It was necessary to bound the Wasserstein distance first since we now rely on this estimate to utilize the bound on $S_{2,0}$ in Lemma 2.2. Combining this with (2.5) leads to

$$\begin{aligned} S_{2,0}(t) &\lesssim \|\rho(t)\|_\infty^{2/3} + \left(1 + N^{-2/3}d_{min}^{-2}\left(\|\rho(t)\|_\infty^{\frac{p}{3(3+p)}} + \|\rho(t)\|_\infty^{1/3}\right)\right)\mathcal{W}_p(t)^{\frac{p}{3+p}} \\ &\lesssim \|\rho_0\|_\infty^{2/3} + (1 + e^{cT})(\mathcal{W}_p(0)^p + E_N^p)e^{pCT} \end{aligned}$$

for $0 \leq t \leq T \wedge T^*(N)$ and some constant c depending only on C_1. According to $\mathcal{W}_p(\rho_N(0), \rho_0) \to 0$ and $E_N \to 0$, we can choose N large enough to enforce the right-hand side to be smaller than $\|\rho_0\|_{L^\infty}^{2/3}+2 =: \frac{1}{2}L$. This ensures that the strict version of (B2) holds true for all $0 \leq t \leq T \wedge T^*(N)$.

Step 4: *Bounding $d_{min}(t)$:*

For $0 \leq t \leq T \wedge T^*(N)$ and $i \neq j$, we obtain the estimate

$$\frac{d}{dt}d_{ij}^2(t) \geq -2|X_i(t) - X_j(t)||u_N(t, X_i(t)) - u_N(t, X_j(t))| \geq 2CS_{2,0}(t)|X_i(t) - X_j(t)|^2 \geq 2CLd_{ij}^2(t)$$

by appplying Lemma 2.3. Accordingly, by Gronwall's Lemma we conclude

$$d_{ij}(t) \geq d_{ij}(0)e^{-C_1 t}$$

with $C_1 = CL$. On the one hand, this justifies the bound on the minimal distance given in the statement of the theorem, while on the other hand, it proves that (B1) is strictly fulfilled for $0 \leq t \leq T \wedge T^*(N)$ with N large enough.

Step 5: *Conclusion:*

So far we realized that for N large enough the strict pendants of (B1) and (B2) are fulfilled for any $0 \leq t \leq T \wedge T^*(N)$. But since the involved quantities are continuous in time, we conclude the validity of (B1) and (B2) for all $0 \leq t \leq T \wedge T^*(N) + \varepsilon$ with some $\varepsilon > 0$ small. But to not contradict the definition of $T^*(N)$ it most hold $T < T^*(N)$. This ends the proof of the bootstrap argument.

We will not prove the convergence of the velocity fields here, since an analogous calculation will be performed later. □

Remark 2.1.

- By analyzing the previous proof and investigating, where the critical assumption (A1) is needed, one observes that the problematic part is to bound $S_{2,0}$ even initially. Using (1.1) the threshold scaling of d_{min} to prevent critical behaviour of $S_{2,0}$ is $N^{-1/3}$. This can be pushed for p large enough by using Lemma 2.2 up to $N^{-1/2}$ as $\mathcal{W}_p(\rho_N(0), \rho_0) \sim N^{-1/3}$. However, there is still a significant gap between this result and the expected scaling being $d_{min} \sim N^{-2/3}$.

- There are essentially three contributions in terms of $S_{2,0}$: (i) the bound of the kernel-related term $I_{2,1}$ in (2.4), (ii) the approximation error from (2.1) in Lemma 2.1, and (iii) the Lipschitz-type estimate for u_N.
  A way to overcome the $S_{2,0}$-dependence in the estimates on the particle distances is presented in [HS25] for the case of kernel-based interactions. The idea will be motivated at the beginning of the next chapter and subsequently an adaption of the method of reflections will be illustrated, which allows to utilize this idea to handle (ii) and (iii).
  The source (i) will be controlled by carrying out a more refined analysis in Lemma 3.12.

2.1 Limit Equations and Renormalization

Before we start discussing refinements of the existing tools, we want to emphasize, how results like those presented in Theorem 2.1 can be adjusted to obtain a macroscopic equation independent of N. On the one hand, if the particle velocity $(6\pi RN)^{-1}g$ converges to some limit velocity v_∞, one can easily utilize the well-posedness of the macroscopic equation in terms of the Wasserstein distance to identify a limiting object fulfilling the transport-Stokes equation with this limit velocity as an offset. On the other hand, in the case, where the particle velocity tends to infinity, such an approach is not tractable. However, we will investigate how to renormalize the problem to establish a macroscopic equation independent of N.

Finite Limit Velocity

To handle the case, where $\gamma_N = NR \to \gamma^* \in (0, \infty]$, we are going to apply the following stability result.

Lemma 2.4. *Let $v_1, v_2 \in \mathbb{R}^3$, consider $\tau_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ and $p \in [1, \infty]$. For $i = 1, 2$ let τ_i be the solution to*

$$\begin{cases} \partial_t \tau_i + (K * \tau_i + v_i) \cdot \nabla \tau_i = 0 & \textit{for } (t, x) \in [0, \infty) \times \mathbb{R}^3, \\ \tau_i(0) = \tau_0, \end{cases}$$

then there is a constant C depending on $\|\tau_0\|_{L^\infty}$ and p, such that

$$\mathcal{W}_p(\tau_1(t), \tau_2(t)) \le |v_1 - v_2| e^{Ct}.$$

Moreover, for $q < 3/2$ and $\Omega \Subset \mathbb{R}^3$, it holds

$$\int_\Omega |K * \tau_1(t) - K * \tau_2(t)|^q dx \le \operatorname{diam}(\Omega)^{3-2q} \mathcal{W}_q(\tau_1(t), \tau_2(t))^q.$$

Proof. The proof is a standard argument similar to the Wasserstein estimate in the proof of Theorem 2.1. It can be found in section A.1 in the appendix. □

Thus, if we denote by (ρ^*, v^*) the unique solution to

$$\begin{cases} \partial_t \rho + (v + v_\infty) \cdot \nabla \rho = 0 & \text{for } (t, x) \in [0, \infty) \times \mathbb{R}^3, \\ \rho(0) = \rho_0, \end{cases}$$

with $v_\infty = \lim_{N\to\infty} (6\pi\gamma_N)^{-1} g$, we conclude from Theorem 2.1 the following

Corollary 2.1. *Under the assumptions of Theorem 2.1 and $(6\pi\gamma_N)^{-1} g \to v_\infty$ there exists some $C > 0$ depending on ρ_0 and p, such that for all $T > 0 : \exists N(T) \in \mathbb{N} : \forall N \ge N(T)$:*

$$\mathcal{W}_p(\rho_N(t), \rho^*(t)) \le (\mathcal{W}_p(\rho_N(0), \rho_0) + E_N) e^{Ct} \quad \textit{for all } 0 \le t \le T$$

and some sequence $E_N \to 0$ for $N \to \infty$. In addition, for $q \le 3/2$ it holds $\|u_N(t) - v^*(t)\|_{L^q_{loc}} \to 0$ as $N \to \infty$.

Renormalizing the Equations

If the particle radius tends to zero faster than N^{-1}, i.e. $\gamma_N \to 0$, the previous result does not apply. Heuristically speaking, this is caused by the fact that the magnitude of the single particle velocity tends to infinity and so the self-interaction is dominating the interparticle interaction. In this senario, one might expect that by considering a reference frame moving with the same speed, the remaining dynamics approach a limit behaviour. We will see that performing this transformation summarizes all the N-dependencies in the microscopic model, while the Wasserstein distance and the local L^q-norms are preserved.

Given $N \in \mathbb{N}$ and $t \geq 0$, we consider the isometric diffeomorphism

$$\Psi_t : \mathbb{R}^3 \to \mathbb{R}^3, x \mapsto x - \tfrac{t}{6\pi RN} g.$$

For $Y_i(t) = \Psi_t(X_i(t))$ the dynamics reformulate as

$$\tfrac{d}{dt} Y_i(t) = u_N(t, X_i(t)) - \tfrac{1}{6\pi RN} g = \tilde{u}_N(t, Y_i(t)) - \tfrac{1}{6\pi RN} g$$

with $\tilde{u}_N(t, \cdot) = u_N(t, \Psi_t^{-1}(\cdot))$ solving the same system of equations as u_N just with particle positions $\{Y_i\}_i$ instead of $\{X_i\}_i$. Denote by $\tilde{\rho}_N$ the empirical distribution of $\{Y_i\}_i$, then the relation can be expressed via $\tilde{\rho}_N(t) = \Psi_{t\#}\rho_N(t)$. If we consider $\tilde{\rho}(t) := \Psi_{t\#}\rho(t)$, then the density function is given by $\tilde{\rho}(t, y) := \rho(t, \Psi_t^{-1}(y)) = \rho(t, y + \frac{t}{6\pi RN} g)$. Accordingly, one easily verifies that $\tilde{\rho}$ satisfies (at least in a distributional sense)

$$\begin{aligned}
\partial_t \tilde{\rho}(t, x) &= (\partial_t \rho)(t, \Psi_t^{-1}(x)) + \tfrac{1}{6\pi RN} g \cdot (\nabla \rho)(t, \Psi_t^{-1}(x)) \\
&= -(v_N(t, \Psi_t^{-1}(x)) + \tfrac{1}{6\pi RN} g) \cdot \nabla \tilde{\rho}(t, x) + \tfrac{1}{6\pi RN} g \cdot \nabla \tilde{\rho}(t, x) = -\tilde{v}(t, x) \cdot \nabla \tilde{\rho}(t, x),
\end{aligned}$$

with $\tilde{v}(t, x) = v_N(t, \Psi_t^{-1}(x))$. Alternatively, $\tilde{v}(t)$ is the unique solution in $\dot{H}^1_\sigma(\mathbb{R}^3)$ to

$$-\Delta v + \nabla p = g\tilde{\rho}(t) \quad \text{in } \mathbb{R}^3.$$

Hence, the macroscopic equation has now the form

$$\begin{cases} \partial_t \tilde{\rho} + (K * \tilde{\rho}) \cdot \tilde{\rho} = 0 & \text{in } [0, \infty) \times \mathbb{R}^3, \\ \tilde{\rho}(0) = \rho_0 & \text{in } \mathbb{R}^3, \end{cases}$$

which is free from any N dependence - as claimed before.

It remains to note that results like those in Theorem 2.1 are invariant under this renormalization. Indeed, since Ψ_t is an isometric diffeomorphism, the Wasserstein distances are not influenced by this transformation as

$$\int_{\mathbb{R}^3 \times \mathbb{R}^3} |x - y|^p d\pi(x, y) = \int_{\mathbb{R}^3 \times \mathbb{R}^3} |\Psi_t(x) - \Psi_t(y)|^p d\pi(x, y) = \int_{\mathbb{R}^3 \times \mathbb{R}^3} |x - y|^p d\Psi_{t\#}\pi(x, y)$$

and similiar for $p = \infty$. Hence, we obtain

$$\mathcal{W}_p(\tilde{\rho}_N(t), \tilde{\rho}(t)) = \mathcal{W}_p(\Psi_{t\#}\rho_N(t), \Psi_{t\#}\rho(t)) = \mathcal{W}_p(\rho_N(t), \rho(t))$$

and also for $q < 3/2$ and $\Omega \Subset \mathbb{R}^3$

$$\|\tilde{u_N}(t) - \tilde{v}(t)\|_{L^q(\Omega)} = \|u_N(t) - v(t)\|_{L^q(\Psi_t^{-1}(\Omega))} \leq C_{\Psi_t^{-1}(\Omega)} e_N(t).$$

Since $\mathrm{diam}(\Omega) = \mathrm{diam}(\Psi_t^{-1}(\Omega))$ it holds $C_\Omega = C_{\Psi_t^{-1}(\Omega)}$ and consequently the convergence of the vector fields transfers.

Remark 2.2.

- This renormalization unifies the considerations on the dilute regime stated in [JO04] with the mean-field results in the possibly non-dilute case. By performing the coordinate change one subtracts the self-interaction part from the driving velocity and focuses on the remaining interaction part. According to the previous analysis the corresponding macroscopic interaction effects satisfy an N-independent equation and thus are of order 1. Now, the dilute scenario corresponds to the case, where γ_N is sufficiently small. In this scenario, the self-interaction is of order γ_N^{-1} and thus highly dominates the particle transport.
  To increase rigour, we can consider the critical parameter Λ, which is used in [JO04] to distinguish, whether or not a configuration is in the dilute regime. In the setup of Theorem 2.1, where $d_{min} \sim N^{-1/3}$, it holds $\Lambda = d_{min}^{-1} RN^{2/3} \sim RN = \gamma_N$. Following the discussion in [JO04], the dilute regime is characterized by Λ being small, which goes hand in hand with γ_N being small and thus aligns nicely to the stated interpretation.

3 Towards a Modified Mean-Field Result

3.1 Main Idea

As emphasized in the previous chapter, the main obstacle for extending the current methodology to more natural scalings is the $S_{2,0}$-dependence of most of the relevant quantities. To overcome this problem - at least on a level of binary, kernel-based dynamics - two observations proved to be helpful.

First, given randomly placed particle centers, such that one pair of particles is excessively close to each other, it is quite unlikely to find a third particle having the same distance to the given pair. In consequence, one expects pairs of close particles to be significantly separated from each third.
To increase rigour, consider $X_1, \dots, X_N$ independently drawn from a uniform distribution on the unit cube $[0,1]^3 \subset \mathbb{R}^3$, then for $r_1, r_2 > 0$

$$\mathbb{P}\left(\exists i \neq j \neq k \neq i : d_{ij} \leq r_1, d_{ik} \leq r_2\right) \leq N^3 \mathbb{P}\left(d_{12} \leq r_1, d_{13} \leq r_2\right) \lesssim N^3 r_1^3 r_2^3 \lesssim (N r_1 r_2)^3.$$

If this probability does not vanish, then $N r_1 r_2 \nrightarrow 0$ must hold. So assuming $d_{ij} \sim r_1 = N^{-1/2-\varepsilon}$ implies $d_{ik} \gtrsim N^{-1/2+\varepsilon}$. Thus, the two distances scale on different levels.
Accordingly, one might consider particle configurations fulfilling the following separation property to the next-to-nearest neighbour: Given some threshold value $\delta_N > 0$, it holds for $i \neq j \neq k \neq i$

$$d_{ij} < \delta_N \quad \Rightarrow \quad d_{ik} \gg \delta_N \text{ and } \tfrac{1}{N} d_{ij}^{-1} d_{ik}^{-1} \leq C_N^{(4)} \to \infty. \tag{3.1}$$

At the moment, this observation seems useless as it does not exclude the influence of nearest particles, which was the critical point. But in fact, there is no attracting influence between nearest particles - being the second important observation. Indeed, the non-attractivity of the Stokes kernel is a straightforward consequence of its symmetry as for two given particles X_1, X_2, it applies

$$\tfrac{d}{dt}|X_1 - X_2|^2 = 2(X_1 - X_2)^T (K(X_1 - X_2) - K(X_2 - X_1)) = 0.$$

In fact, even in the case of N particles it holds

$$\tfrac{d}{dt}|X_i - X_j|^2 = 2(X_i - X_j)^T \left(\frac{1}{N}\sum_{k \neq i,j} K(X_i - X_k) - K(X_j - X_k)\right)$$

and so the distance between two particles with $d_{ij} < \delta_N$ is just influenced by the external field $u_\infty^{i,j}(x) = \sum_{k\neq i,j} \frac{1}{N} K(x - X_k)$, which fulfills

$$|u_\infty^{i,j}(X_i) - u_\infty^{i,j}(X_j)| \lesssim \frac{1}{N}\sum_{k\neq i,j}\left(|X_i - X_k|^{-2} + |X_j - X_k|^{-2}\right)|X_i - X_j| \lesssim 2 S_{2,\delta_N}|X_i - X_j|$$

with the truncated sum of inverse distances

$$S_{\beta,\delta} := \sup_i \tfrac{1}{N} \sum_{d_{ij} > \delta} d_{ij}^{-\beta}. \tag{3.2}$$

In conclusion, under assumptions similar to (3.1) one might exclude the influence of the nearest particle leading to dynamics of the particle distance satisfying

$$\tfrac{d}{dt} d_{ij}^2 \geq -2\left(S_{2,\delta_N} + C_N^{(4)}\right) d_{ij}^2.$$

The emergence of the $C_N^{(4)}$ term will be explained in more detail in the proof of Lemma 3.20. Heuristically speaking, it encodes the fact that any particle distance $d_{ij} < \delta_N$ only influences the evolution of distances of significant larger amplitude. Based on the presented ideas and assumptions, improved mean-field results for more general kernels sharing the non-attractivity property

$$(X_i - X_j)^T (K(X_i - X_j) - K(X_j - X_i)) \geq 0$$

were established in [HS25]. In some cases even propagation of chaos was verifiable, but unfortunately not in the case of the Stokes kernel. The reason for this is the fact that the likelihood of (3.1) being fulfilled will not tend to 1 for $N \to \infty$ and δ_N small enough. A rigourous discussion of this claim and similar statements on expected scaling properties can be found in section A.4 of the appendix. Due to dealing with the case of point particles, the application of Hauray's method in [HS25] is simplified in the sense that no approximative system has to be introduced and controlled. Consequently, it suffices to bound S_{2,δ_N} and prove the Lipschitz-type estimate by a concrete calculation. The main improvement is based on the following estimate.

Lemma 3.1 ([HS25, Lem. 2.1])**.** *Let $\beta \in (0,d), p \in [1,\infty]$. Furthermore, let $X \in (\mathbb{R}^d)^N$, and $\sigma_N = \frac{1}{N}\sum \delta_{X_i}$, as well as $\sigma \in L^\infty \cap \mathcal{P}(\mathbb{R}^d)$. Let $M, \delta > 0$ be such that*

$$\sup_i \#\{j \mid X_j \in B_\delta(X_i)\} \leq M.$$

Then, $S_{\beta,\delta}$ can be estimated by

$$S_{\beta,\delta} \lesssim \|\sigma\|_\infty^{\frac{\beta}{d}} + \left(\left(\frac{M^{\beta/d} \|\sigma\|_\infty^{\frac{(d-\beta)p}{d(d+p)}}}{N^{\beta/d}\delta^\beta} \right) + \left(\frac{M^{\beta/d}\|\sigma\|_\infty^{\frac{d-\beta}{d}}}{N^{\beta/d}\delta^\beta} \right) \right)^{\frac{\beta+p}{d+p}} \mathcal{W}_p(\sigma_N,\sigma)^{\frac{(d-\beta)p}{d+p}}. \tag{3.3}$$

In the case $p = \infty$, this means

$$S_{\beta,\delta} \lesssim \|\sigma\|_\infty^{\frac{\beta}{d}} + M^{\beta/d}\|\sigma\|_\infty^{\frac{d-\beta}{d}} N^{-\beta/d}\delta^{-\beta}\mathcal{W}_\infty(\sigma_N,\sigma)^{d-\beta}.$$

In order to apply the presented ideas in the case of spherical particles with positive radius, it is necessary to work on three aspects:

1. Proving that the interaction between two particles in a Stokes fluid is non-attractive.
2. Finding a kernel-based approximation with error bounded in terms of S_{2,δ_N} and $C_N^{(4)}$.
3. Finding a Lipschitz-type estimate in terms of S_{2,δ_N} and $C_N^{(4)}$.

We will work on these points chronologically. Starting from a quantitative analysis of the two-particle problem, we are going to utilize the two-particle solutions to modify the approximation based on the method of reflections. This modification will allow us to handle the last two points on the list. But before starting with this agenda, some facts about the method of reflections in general and the formulation in terms of orthogonal projections are revisited to introduce the necessary ideas and notation.

3.2 Method of Reflections

In the following work, it will be necessary on several occasions to approximate the fluid and particle velocities as explicitly as possible. In doing so, we will always resort to the method of reflections, which is used in many problems of microhydrodynamics – both for theoretical statements and numerical methods.

The method, which was introduced by Smoluchowski in [Smo11], is based on an intuitive idea: As the Stokes equation is a linear equation, one might ask, why the solution to the multi-particle system is not given by a superposition of single-particle solutions. Physically speaking, this approach would not include the fact that the presence of additional particles influence how two particles interact hydrodynamically. From the mathematical point of view, by summing up the single-particle solutions the condition on the fluid velocity to be constant inside all the particles is violated. Now, a natural approach to include a reaction to the disturbance flow generated by the surrounding particles is to search for a correction term R_i at particle i satisfying that $u_N^{(0)} - R_i$ is constant inside particle i and still fulfills Stokes equation in the outside. This can be seen as a reflection of the effect on particle i caused by the other particles. Now, repeating this for all particles and defining $u_N^{(1)} = u_N^{(0)} - \sum R_i$ one obtains a new approximation step including some of the interaction between the particles. Restarting the procedure from this point, a sequence of approximations can be generated and one might hope to approach the actual solution of the problem.

There are different implementations of this idea, some authors use rather explicit formulas to define the residual term, which will not eliminate the total error inside the particles, but reduces it up to some terms of higher order; see for example [GMP11]. Starting with the theory presented in [Luk89] there is also a way to state the method of reflections in terms of orthogonal projections. Further progress in this direction was done for example in [HV18] and for the case of sedimenting particles in [Hö18] and [Hö23]. In the subsequent discussion we will mostly follow the presentation in [Hö18].

Starting point for this approach is the observation that the multi-particle solution can be identified[2] as the unique minimizer of the energy

$$E(w) = \frac{1}{2}\int_{\mathbb{R}^3} |\nabla w|^2 dx - \sum_{i=1}^{N} \frac{1}{N} \fint_{\partial B_i} w \, d\mathcal{H}^2 \tag{3.4}$$

in the space

$$W = \{u \in \dot{H}^1_\sigma(\mathbb{R}^3) \mid u = \text{const. in } B_i \text{ for } i = 1, \dots, N\},$$

whereas the superposition of the single-particle solutions $u_N^{(0)} = \sum u_i$ is the minimizer of (3.4) in $\dot{H}^1_\sigma(\mathbb{R}^3)$. Accordingly - utilizing the connection between the weak formulation of Stokes equation and the inner product in $\dot{H}^1_\sigma(\mathbb{R}^3)$ - we can identify $u_N = P u_N^{(0)}$, where P is the orthogonal projection onto W in $\dot{H}^1_\sigma$. On the one side, this goes hand in hand with variational inequalities allowing us to use well-prepared competitors to establish bounds on $\|u_N - u_N^{(0)}\|_{\dot{H}^1}$. On the other side, we can restate the procedure behind the method of reflections in a more general setting. By identifying

$$W = \cap_{i=1}^N W_i \quad \text{with} \quad W_i = \{u \in \dot{H}^1_\sigma(\mathbb{R}^3) \mid u = \text{const. in } B_i\}$$

and since all the spaces are closed, we obtain $W^\perp = \overline{\sum W_i^\perp}$. Now, the idea is to rewrite $P = 1 - Q$ and approximate the projection onto $W^\perp$ via $Q \approx \sum_{i=1}^N Q_i$ with Q_i the projection onto $W_i^\perp$. In general, this approximation will differ from the real projection, but we hope the error to vanish by applying this idea iteratively, i.e. we wish

$$P u_N^{(0)} = \lim_{k\to\infty} (1 - \sum_{i=1}^{N} Q_i)^k u_N^{(0)} \tag{3.5}$$

to hold true.

In order to make this more plausible, let us investigate that in case of convergence the term on the right coincides with the claimed object. Denoting the limit by $u^\star$ we are going to check first that this is an element in W. We know,

$$(u^\star, u^\star)_{\dot{H}^1} = (u^\star, (1 - \sum_{i=1}^{N} Q_i) u^\star)_{\dot{H}^1} = (u^\star, u^\star)_{\dot{H}^1} - \sum_{i=1}^{N} (Q_i u^\star, Q_i u^\star)_{\dot{H}^1}$$

and so by positivity for all i it holds $(Q_i u^\star, Q_i u^\star)_{\dot{H}^1} = 0$ implying $u^\star \in W_i$ and leading to $u^\star \in W$. Moreover, for any $w \in W$ it holds $(1 - \sum_{i=1}^N Q_i) w = w$ and so

$$(u_N^{(0)} - u^\star, w)_{\dot{H}^1} = (u_N^{(0)}, w)_{\dot{H}^1} - \lim_{k\to\infty} (u_N^{(0)}, (1 - \sum_{i=1}^{N} Q_i)^k w)_{\dot{H}^1} = (u_N^{(0)}, w)_{\dot{H}^1} - (u_N^{(0)}, w)_{\dot{H}^1} = 0.$$

In summary, this implies $u^\star = P u_N^{(0)}$.

Using (3.5) as a guideline we define the intermediate steps of the method of reflections by

$$u_N^{(k)} = (1 - \sum_{i=1}^{N} Q_i) u_N^{(k-1)} = (1 - \sum_{i=1}^{N} Q_i)^k u_N^{(0)}.$$

[2]See [Hö18, 3.3] for further justification.

A rigourous formulation of this theory including detailed proofs can be found in [Hö18].

Here, we state the most important properties and inequalities used in the ongoing discussion without proofs. However, we will mimic some of the essential arguments in the later sections in order to establish slightly modified estimates. We start with recalling an *a priori* estimate on the $\dot{H}^1$-distance between the iterations and the final function.

Lemma 3.2. *For particle configurations $\{X_i\}_{i=1}^N$ with $d_{min} > 4R$, it holds*

$$\|u_N - u_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq CR^3 \sum_{i=1}^N \|\nabla u_N^{(k)}\|^2_{L^\infty(B_i)}. \tag{3.6}$$

Proof. The proof can be found in section A.2 of the appendix. □

Next, we introduce some of the properties of the projections P_i and Q_i.

Lemma 3.3 ([Hö18, Prop. 3.10])**.** *For $w \in \dot{H}^1_\sigma(\mathbb{R}^3)$ it holds*

$$P_i(w) = (w)_i \quad \text{on } B_i \tag{3.7}$$

with $(w)_i = \fint_{\partial B_i} w \, d\mathcal{H}^2$.

Proof. The proof is included in the appendix in section A.2. □

In addition, since $\dot{H}^1_{\sigma,0}(\mathbb{R}^3 \setminus B_i) := \overline{C_c^\infty(\mathbb{R}^3 \setminus B_i)}^{\|\cdot\|_{\dot{H}^1}} \cap \dot{H}^1_\sigma$ is a subset of W_i, we obtain for $w \in W_i^\perp$

$$\int_{\mathbb{R}^3 \setminus B_i} \nabla w : \nabla u \, dx = 0 \quad \forall u \in \dot{H}^1_{\sigma,0}(\mathbb{R}^3 \setminus B_i),$$

meaning that w is a weak solution to $-\Delta w + \nabla p = 0$ in $\mathbb{R}^3 \setminus B_i$. Accordingly, we can deduce an expected decay behaviour for $Q_i w \in W_i^\perp$ as stated below.

Lemma 3.4 ([Hö18, Prop. 3.12])**.** *Consider a particle configuration with $d_{min} \geq 4R$. Then, for $y \notin B_{2R}(X_j)$ it holds*

$$|Q_j u_N^{(k)}|(y) \leq C \frac{R^3}{|X_j - y|^2} \|\nabla u_N^{(k)}\|_{L^\infty(B_j)} \tag{3.8}$$

and

$$|\nabla Q_j u_N^{(k)}|(y) \leq C \frac{R^3}{|X_j - y|^3} \|\nabla u_N^{(k)}\|_{L^\infty(B_j)} \tag{3.9}$$

Proof. The proof is elaborated in detail in [Hö18, Prop. 3.12]. □

3.3 Two-Particle Problem

In this section we are going to analyze the weak solution for the two-particle problem given by

$$\begin{cases} -\Delta w + \nabla p = 0 & \text{in } \mathbb{R}^3 \setminus (B_1 \cup B_2), \\ \operatorname{div}(w) = 0 & \text{in } \mathbb{R}^3, \\ |w(x)| \to 0 & \text{as } |x| \to \infty, \\ \nabla w = 0 & \text{on } B_i \text{ for } i = 1,2, \\ \int_{\partial B_i} \sigma(w,p) \cdot n \, d\mathcal{H}^2 = \frac{1}{N} g & \text{for } i = 1,2, \end{cases} \tag{3.10}$$

with particle centers X_i and $B_i = B_R(X_i)$. This can be justified by two reasons: On the one hand, the solution will turn out to be an important ingredient to modify the method of reflections in the next section. On the other hand, we will use the method of reflections to gain insights to the solutions behaviour and so we familiarize with the typical arguments used in this context.

But first, we will use abstract arguments to prove that the real dynamics share the non-attractivity property. The proof will be based on the symmetry of the two-particle system.

Lemma 3.5. *For any two-particle configuration* $\{X_1, X_2\}$ *with* $d_{12} > R$ *it holds*

$$w(X_1) = w(X_2).$$

Proof. Given a solution w and an associated pressure p, we can consider

$$\tilde{w}(x) = w(X_2 + X_1 - x) \quad \text{and} \quad \tilde{p}(x) = -p(X_2 + X_1 - x).$$

Then $(\tilde{w}, \tilde{p})$ solves the same problem as (w, p) and by uniqueness of the solution we deduce

$$w(X_1) = \tilde{w}(X_1) = w(X_2).$$

□

In the next step, we want to employ the method of reflections to describe the solution for the two-particle problem quantitatively. Therefore, we denote by u_i the solution to the one-particle system with the same external force condition as before, i.e. $u_i = U(\cdot - X_i)$ with U given by (1.3). By Q_i we denote the projection onto $W_i^\perp$ in $\dot{H}^1(\mathbb{R}^3)$ as introduced in the previous section. In this setting, the iterations of the method of reflections read like

$$\begin{aligned} w^{(k)} &= (1 - Q_1 - Q_2) w^{(k-1)} \\ w^{(0)} &= u_1 + u_2. \end{aligned}$$

The estimates stated in the lemma below will be deduced by arguments borrowed from [Hö18, Prop. 3.12] and emphasize the usefulness of the method of reflections.

Lemma 3.6. *There exists a* $\theta_1 > 0$ *with the following property. Assume a two-particle configuration with* $d_{12} > 4R$ *and* $R^3 d_{12}^{-3} < \theta_1$*, then it holds*

1. *for* $x \notin B_{2R}(X_1) \cup B_{2R}(X_2)$
$$|w(x)| \lesssim \tfrac{1}{N}(|x - X_1|^{-1} + |x - X_2|^{-1}), \tag{3.11}$$
$$|w(x) - w^{(0)}(x)| \lesssim \tfrac{R}{N}(|x - X_1|^{-2} + |x - X_2|^{-2}), \tag{3.12}$$

2. *for* $x \in B_R(X_1)$
$$|w(x) - \tfrac{1}{6\pi RN} g| \lesssim \tfrac{1}{N}|X_1 - X_2|^{-1}, \tag{3.13}$$

3. *for* $x \in B_{2R}(X_1)$
$$|w(x) - w^{(0)}(x)| \lesssim \tfrac{1}{N}|X_1 - X_2|^{-1}, \tag{3.14}$$

4. *for* $x \notin B_{4R}(X_1) \cup B_{4R}(X_2)$
$$\|\nabla w\|_{L^\infty(B_R(x))} \lesssim \tfrac{1}{N}(|x - X_1|^{-2} + |x - X_2|^{-2}). \tag{3.15}$$

Remark 3.1.

- For $x \in B_{2R}(X_1)$ it holds $|x - X_1| \le 2R \le |X_1 - X_2|$ and so we conclude
$$|x - X_2| \le |X_1 - X_2| + |x - X_1| \le 2|X_1 - X_2|.$$
Accordingly, the x-dependent versions of (3.13) and (3.14) given by
$$|w(x) - \tfrac{1}{6\pi RN} g| \le C\tfrac{1}{N}|x - X_2|^{-1} \quad \text{and} \quad |w(x) - w^{(0)}(x)| \le C\tfrac{1}{N}|x - X_2|^{-1}$$
are fulfilled.

Proof. The proof will be based on the estimates introduced in Lemma 3.2 and 3.4. Following the definition of Q_1 we obtain for $y \in B_1$

$$\nabla w^{(k)}(y) = \nabla(w^{(k-1)} - Q_1 w^{(k-1)})(y) - \nabla Q_2 w^{(k-1)}(y) = -\nabla Q_2 w^{(k-1)}(y)$$

as $w^{(k-1)} - Q_1 w^{(k-1)}$ is constant on B_1. Thus, we observe by (3.9) and $|y - X_2| > \frac{1}{2}|X_1 - X_2|$

$$\|\nabla w^{(k)}\|_{L^\infty(B_1)} \le CR^3 d_{12}^{-3} \|\nabla w^{(k-1)}\|_{L^\infty(B_1 \cup B_2)}.$$

Iterating this estimate, we end up with

$$\|\nabla w^{(k)}\|_{L^\infty(B_1 \cup B_2)} \le (CR^3 d_{12}^{-3})^k \|\nabla w^{(0)}\|_{L^\infty(B_1 \cup B_2)} \le (CR^3 d_{12}^{-3})^k \tfrac{1}{N} d_{12}^{-2}, \tag{3.16}$$

where we used $\|\nabla w^{(0)}\|_{L^\infty(B_1 \cup B_2)} \lesssim \frac{1}{N} d_{12}^{-2}$.

Choosing $\theta_1 < \frac{1}{2C}$, we assume from now on $R^3 d_{12}^{-3} < \theta_1$. Before proving the stated estimates, we indicate that the method is convergent in this setting. Indeed, by (3.6) from Lemma 3.2 one deduces

$$\|w - w^{(k)}\|^2_{\dot H^1(\mathbb{R}^3)} \le CR^3 \sum_{i=1}^{2} \|\nabla w^{(k)}\|^2_{L^\infty(B_i)} \le CR^3 \left(2^{-k} \tfrac{1}{N} d_{12}^{-2}\right)^2 \to 0 \text{ for } k \to 0.$$

To obtain convergence in L^∞, we choose $x \in \mathbb{R}^3$ arbitrary and assume X_2 to be the particle nearest to x. In this scenario it holds for $R^3 d_{12}^{-3} < \theta_1$

$$\begin{aligned}
|w^{(k+1)}(x) - w^{(k)}(x)| &\le |Q_1 w^{(k)}(x)| + |Q_2 w^{(k)}(x)| \\
&\lesssim R^3 |x - X_2|^{-2} \|\nabla w^{(k)}\|_{L^\infty(B_1)} + \|w^{(k)} - (w^{(k)})_i\|_{L^\infty(B_2)} \\
&\lesssim (R^3 |X_i - X_j|^{-2} + R) \|\nabla w^{(k)}\|_{L^\infty(\cup_j B_j)} \\
&\lesssim R 2^{-k} \tfrac{1}{N} d_{12}^{-2},
\end{aligned}$$

where we used Lemma 1.1 and Lemma 3.3 to bound $|Q_i w^{(k)}(x)|$. By summing up these estimates we obtain uniform convergence of the sequence $(w^{(k)})_k$. Now, as we have ensured the convergence and identified the limit object, we can start proving the claimed estimates. First, let $x \notin B_{2R}(X_1) \cup B_{2R}(X_2)$, then by (3.9)

$$\begin{aligned}
|w^{(k+1)}(x) - w^{(k)}(x)| &\le |Q_1 w^{(k)}(x)| + |Q_2 w^{(k)}(x)| \\
&\le CR^3 \left(|x - X_1|^{-2} + |x - X_2|^{-2}\right) \|\nabla w^{(k)}\|_{L^\infty(B_1 \cup B_2)} \\
&\le CR^3 \left(|x - X_1|^{-2} + |x - X_2|^{-2}\right) 2^{-k} \tfrac{1}{N} d_{12}^{-2}
\end{aligned}$$

and thus by using $d_{12} > R$

$$|w(x) - w^{(0)}(x)| \le \sum_{k=1}^{\infty} 2^{-k} \tfrac{R}{N} \left(|x - X_1|^{-2} + |x - X_2|^{-2}\right) \le C\tfrac{R}{N} \left(|x - X_1|^{-2} + |x - X_2|^{-2}\right).$$

Moreover,

$$|w(x)| \le |w^{(0)}(x)| + |w(x) - w^{(0)}(x)| \le C\frac{1}{N}\left(|x - X_1|^{-1} + |x - X_2|^{-1}\right),$$

where we utilized $|x - X_i| > R$ and the estimate for the single-particle solution.

Next, for $x \in B_{2R}(X_1)$

$$\begin{aligned}
|w^{(k+1)}(x) - w^{(k)}(x)| &\le |Q_2 w^{(k)}(x)| + |Q_1 w^{(k)}(x)| \\
&\le CR^3|x - X_2|^{-2}\|\nabla w^{(k)}\|_{L^\infty(B_1\cup B_2)} + \|w^{(k)} - (w^{(k)})_1\|_{L^\infty(B_1)} \\
&\le C(R^3|X_1 - X_2|^{-2} + R)\|\nabla w^{(k)}\|_{L^\infty(B_1\cup B_2)} \\
&\le C\frac{R}{N}|X_1 - X_2|^{-2}2^{-k},
\end{aligned}$$

where we used Lemma 1.1 as well as Lemma 3.3 to handle the Q_1 term. Accordingly, we obtain

$$|w(x) - w^{(0)}(x)| \le \sum_{k=0}^{\infty} |w^{(k+1)}(x) - w^{(k)}(x)| \le C\frac{R}{N}|X_1 - X_2|^{-2},$$

which implies (3.14).

For $x \in B_R(X_1)$ we conclude by using $|x - X_2| \ge \frac{1}{2}|X_1 - X_2| > R$ and the previous that the following inequalities are satisfied

$$\begin{aligned}
|w(x) - \tfrac{1}{6\pi RN}g| &\le |w^{(0)}(x) - \tfrac{1}{6\pi RN}g| + |w(x) - w^{(0)}(x)| \\
&\lesssim \tfrac{1}{N}|x - X_2|^{-1} + \tfrac{1}{N}|X_1 - X_2|^{-1} \lesssim \tfrac{1}{N}|X_1 - X_2|^{-1}.
\end{aligned}$$

Finally, let $x \notin B_{4R}(X_1) \cup B_{4R}(X_2)$ and consider $y \in B_R(x)$ then

$$\nabla w^{(k+1)}(y) = \nabla w^{(k)}(y) - \nabla Q_1 w^{(k)}(y) - \nabla Q_2 w^{(k)}(y),$$

and thus by (3.9)

$$\begin{aligned}
\|\nabla w^{(k+1)}\|_{L^\infty(B_R(y))} &\le \|\nabla w^{(k)}\|_{L^\infty(B_R(y))} + CR^3\left(|x - X_1|^{-3} + |x - X_2|^{-3}\right)2^{-k}\tfrac{1}{N}d_{12}^{-2} \\
&\lesssim \|\nabla w^{(k)}\|_{L^\infty(B_R(y))} + \tfrac{1}{N}\left(|x - X_1|^{-2} + |x - X_2|^{-2}\right)2^{-k}.
\end{aligned}$$

Applying this estimate iteratively, we end up with

$$\begin{aligned}
\|\nabla w^{(k)}\|_{L^\infty(B_R(x))} &\lesssim \|\nabla w^{(0)}\|_{L^\infty(B_R(x))} + \tfrac{1}{N}\left(|x - X_1|^{-2} + |x - X_2|^{-2}\right)\sum_{n=0}^{k} 2^{-n} \\
&\lesssim \tfrac{1}{N}\left(|x - X_1|^{-2} + |x - X_2|^{-2}\right).
\end{aligned}$$

Accordingly, all the $w^{(k)}$ are Lipschitz on $B_R(x)$ with an uniform constant $C\frac{1}{N}\left(|x - X_1|^{-2} + |x - X_2|^{-2}\right)$ and since $w^{(k)} \to w$ pointwise also w is Lipschitz on this domain with at most twice of this constant. So $|\nabla w| \lesssim \frac{1}{N}\left(|x - X_1|^{-2} + |x - X_2|^{-2}\right)$ almost everywhere. □

3.4 Adaption of the Method of Reflections

As hinted in the previous section, we are going to modify the existing framework of the method of reflections for the case of sedimenting particles. The goal is to obtain suitable approximation bounds and incorporate the non-attractivity in the Lipschitz-type estimate regarding the solution function u_N.

Whereas the existing estimates in [Hö18] always include the full sum of the inverse distances, one expects under separation assumptions - like (A3) below - the influence between nearest particles to be negligible as it was illustrated in the two-particle case. The reason, why this effect is not visible at the moment, although the zeroth-order term shares most of the estimates established in the kernel case, is the following. Since the projections Q_j extinguish the gradient at particle X_j, the induced correction is of order $\|\nabla u_N^0\|_{L^\infty(B_j)}$ in the first step. However, between two particles X_i and X_j the gradient satisfies $\nabla u_i(X_j) \sim N^{-1} d_{ij}^{-2}$ and no cancellation effect between nearest particles are known. Accordingly, $\|\nabla u_N^0\|_{L^\infty(\cup B_i)} \sim S_{2,0}$ and in the worst case this dependence might be propagated through all approximation steps.

Our approach is to use a mixed-type zeroth approximation in the method of reflection consisting of single-particle solutions for dilute particles and two-particle solutions for pairs of quite close particles. We will see that this allows to overcome the aforementioned problem.

The Adjusted Initialization

Given some threshold value $\delta > 0$, we assume in the subsequent discussion particle configurations with $d_{min,1} > \delta$ and

$$d_{ij} \leq \delta \Rightarrow d_{ik} > 4\delta \quad \text{for } k \neq i \neq j \neq k, \tag{A3}$$

$$d_{ij} < \delta \Rightarrow N^{-1} d_{ik}^{-1} d_{ij}^{-1} < C^{(4)} \quad \text{for } k \neq i \neq j \neq k. \tag{A4}$$

Remark that (A3) ensures that there are only pairs of particles with $d_{ij} < \delta$, in particular it negates the existence of a chain X_i, X_j, X_k with $d_{ij} < \delta$ and $d_{jk} < \delta$ for $k \neq i \neq j \neq k$. Hence, recalling that i_{nn} refers to the index of the particle nearest to X_i we can define the sets

$$\mathcal{S}_\delta := \{i \mid d_{ii_{nn}} > \delta\}, \qquad \tilde{\mathcal{D}}_\delta = \{j \mid d_{jj_{nn}} < \delta\}$$

and

$$\mathcal{D}_\delta = \tilde{\mathcal{D}}_\delta / \sim \quad \text{with } j \sim i \Leftrightarrow d_{ij} < \delta.$$

The set $\mathcal{S}_\delta$ contains all the single particles, meaning that they are sufficiently separated from their nearest neighbour. On the contrary, the set $\mathcal{D}_\delta$ contains one representative for each pair of δ-close particles. In most cases the value δ is known from the context and thus we mainly suppress the additional index.

As before, let u_i denote the solution in W_i to the single-particle problem

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{on } \mathbb{R}^3 \setminus B_i, \\ \int_{\partial B_i} \sigma(u,p) \cdot n \, d\mathcal{H}^2 = \frac{1}{N} g, \end{cases}$$

and w_{ij} the two-particle solution in $W_i \cap W_j$ solving

$$\begin{cases} -\Delta w + \nabla p = 0 & \text{on } \mathbb{R}^3 \setminus (B_i \cup B_j), \\ \int_{\partial B_k} \sigma(w,p) \cdot n \, d\mathcal{H}^2 = \frac{1}{N} g & \text{for } k = i, j. \end{cases}$$

We consider now the modified zeroth-order approximation given by

$$w_N^{(0)} = \sum_{i \in \mathcal{S}} u_i + \sum_{i \in \mathcal{D}} w_{ii_{nn}},$$

i.e. we use the single particle solution for those particles in $\mathcal{S}$, whereas the flow around a pair of close particles is no longer approximated by solutions for the individual particles, but by the common flow around the two involved particles.

Remark 3.2.

- A distinction similar to $\mathcal{D}$ and $\mathcal{S}$ is performed in [GVH20]. There an improved derivation of Einstein's viscosity formula was presented and the possibly critical influence of the minimal distance needed to be controlled. However, in contrast to the method presented in this thesis they did not refine the analysis around pairs of close particles but excluded these from the approximation. This approach was applicable as they made use of energy estimates as well as duality arguments and considered the fixed time problem instead of the dynamical one[3].

[3] See [HS21] for a treatment of influence of Einstein's viscosity in the dynamical case.

Since the idea behind the method of reflections in terms of orthogonal projections states as $P = \lim_{k\to\infty}(1 - \sum_j Q_j)^k$ for P the projection onto

$$W = \{u \in \dot{H}^1_\sigma(\mathbb{R}^3) \mid u = \text{const. on } B_i \text{ for } i = 1, \dots, N\},$$

we first need to ensure $u_N = Pw_N^{(0)}$ for the desired solution u_N of the multi-particle system.

Lemma 3.7. *It holds $Pw_N^{(0)} = u_N$.*

Proof. We already know that $u_N = P(\sum_{i=1}^N u_i)$ and $w_{ij} = P_{ij}(u_i + u_j)$ with P_{ij} the projection onto

$$W_{ij} = \{u \in \dot{H}^1_\sigma(\mathbb{R}^3) \mid u = \text{const. on } B_k \text{ for } k = i, j\} = W_i \cap W_j.$$

It is worth to notice that $W \subset W_{ij}$ for all i, j. Now, pick any $\varphi \in W$, then

$$(w_N^{(0)} - u_N, \varphi)_{\dot{H}^1} = (\sum_{i=1}^N u_i - u_N, \varphi)_{\dot{H}^1} + \sum_{i\in D}(w_{ii_{nn}} - u_i - u_{i_{nn}}, \varphi)_{\dot{H}^1} = 0 + \sum_{i\in D} 0 = 0.$$

This already implies the statement by the characterization of orthogonal projections. □

As a consequence, we can be sure that in the case of convergence of the method, the limit will be u_N. Our first and in some way the most important observation is that using the adapted initialization allows to overcome the full $S_{2,0}$-dependence of $\|\nabla u_N^{(0)}\|_{L^\infty(\cup B_i)}$. This insight will play a key role in the later discussion.

Lemma 3.8. *If $R^3 d_{min}^{-3} < \theta_1$ with θ_1 from Lemma 3.6 and $d_{min} > 4R$, then it holds*

$$\|\nabla w_N^{(0)}\|_{L^\infty(\cup_i B_i)} \lesssim S_{2,\delta}.$$

Proof. On the one hand, we have $\nabla u_i = 0$ on B_i and $|\nabla u_i(x)| \lesssim \frac{1}{N}|x - X_i|^{-2} \lesssim \frac{1}{N} d_{ij}^{-2}$ for $x \in B_j$. On the other hand, the two-particle solutions satisfy $\nabla w_{ii_{nn}} = 0$ on B_i and $B_{i_{nn}}$ and for $j \neq i, i_{nn}$

$$|\nabla w_{ii_{nn}}| \lesssim \tfrac{1}{N}(d_{ji}^{-2} + d_{ji_{nn}}^{-2}) \quad \text{on } B_j$$

by (3.15) from Lemma 3.6, which is applicable due to the assumption on the particle configuration. Combining these, we end up with

$$\begin{aligned}\|\nabla w_N^{(0)}\|_{L^\infty(B_i)} &\le \sum_{k\in S, k\neq i} \|\nabla u_k\|_{L^\infty(B_i)} + \sum_{k\in D, k\nsim i} \|\nabla w_{kk_{nn}}\|_{L^\infty(B_i)} \\ &\lesssim \frac{1}{N}\sum_{j: d_{ij} > 4\delta} d_{ij}^{-2} \lesssim S_{2,\delta} \quad \forall i = 1, \dots, N.\end{aligned}$$

□

The previous lemma indicates that by incorporating the two-particle solutions one can overcome the d_{min}-influence in the estimate, which originally was caused by the non-vanishing gradients between nearest particles. Since this was the main obstacle for propagating the relevant estimates of the zeroth approximation through the method, we might be optimistic about possible improvements of the estimates for u_N. We will start with revisiting the general approximation bound and prove the following adaption of [Hö18, Prop. 3.12].

Proposition 3.1. *There exists a constant $\theta_2 > 0$ with the following property. Assume $\phi_N S_{3,0} < \theta_2$ and $d_{min} > 4R$, then*

$$\|\nabla w_N^{(k)}\|_{L^\infty(B_i)} \le S_{2,\delta} 2^{-k} \quad \text{and} \quad w_N^{(k)} \to u_N \text{ in } \dot{H}^1(\mathbb{R}^3) \text{ and } L^\infty(\mathbb{R}^3).$$

In addition, we have

$$\|u_N - w_N^{(0)}\|_{L^\infty(\mathbb{R}^3)} \lesssim S_{2,\delta}(S_{2,0}\phi_N + R). \tag{3.17}$$

Proof. We adapt the notation and set $w_N^{(k)} = (1 - \sum_{j=1}^N Q_j) w_N^{(k-1)}$. If we assume $\theta_2 < \theta_1$, we deduce by (3.6) from Lemma 3.2

$$\|u_N - w_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq CR^3 \sum_{i=1}^N \|\nabla w_N^{(k)}\|^2_{L^\infty(B_i)} \leq CR^3 N \|\nabla w_N^{(k)}\|^2_{L^\infty(\cup B_i)}.$$

By (3.9) it holds

$$\|\nabla w_N^{(k)}\|_{L^\infty(B_i)} \leq C \sum_{j \neq i} R^3 d_{ij}^{-3} \|\nabla w_N^{(k-1)}\|_{L^\infty(\cup B_k)}$$

and thus by Lemma 3.8 we observe

$$\|\nabla w_N^{(k)}\|_{L^\infty(B_i)} \leq (C\phi_N S_{3,0})^k S_{2,\delta}.$$

So for $\phi_N S_{3,0} < \theta_2 < \frac{2}{C}$ we end up with

$$\|u_N - w_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq C\phi_N (2^{-k} S_{2,\delta})^2. \tag{3.18}$$

This proves convergence in $\dot{H}^1(\mathbb{R}^3)$ for $k \to \infty$. To obtain convergence in L^∞, we choose $x \in \mathbb{R}^3$ arbitrary and let X_i be the particle nearest to x, then we notice for $\phi_N S_{3,0} < \theta_2$

$$\begin{aligned}
|w_N^{(k+1)}(x) - w_N^{(k)}(x)| &\leq |\sum_{j=1}^N Q_j w_N^{(k)}(x)| \\
&\leq \sum_{j \neq i} |Q_j w_N^{(k)}(x)| + |Q_i w_N^{(k)}(x)| \\
&\leq \sum_{j \neq i} CR^3 |x - X_j|^{-2} \|\nabla w_N^{(k)}\|_{L^\infty(\cup_i B_i)} + \|w_N^{(k)} - (w_N^{(k)})_i\|_{L^\infty(B_i)} \\
&\leq (\sum_{j \neq i} CR^3 |X_i - X_j|^{-2} + CR) \|\nabla w_N^{(k)}\|_{L^\infty(\cup_j B_j)} \\
&\lesssim (S_{2,0}\phi_N + R) 2^{-k} S_{2,\delta},
\end{aligned} \tag{3.19}$$

where we used again Lemma 1.1 and Lemma 3.3 to bound $|Q_i w_N^{(k)}(x)|$. By summing up these estimates, we obtain convergence in L^∞ and the claimed estimate. □

The remainder of this section is devoted to improve also the Lipschitz-type estimate - the missing statement. Again the proof will be a straightforward adaption of the proof in [Hö18, Lem. 3.16].

As a first step, we establish the desired statement for $w_N^{(0)}$.

Lemma 3.9. *Assume $d_{min} > 4R$ and $S_{3,0}\phi_N < \theta_2$. Then, for all $i \neq j$ and all $h \in \bar{B}_R(0) \subset \mathbb{R}^3$ it holds*

$$|w_N^{(0)}(X_i + h) - w_N^{(0)}(X_j + h)| \lesssim (S_{2,\delta} + C^{(4)}) |X_i - X_j|. \tag{3.20}$$

Proof. Before starting, we remark $|\nabla u_i(x)| \lesssim \frac{1}{N} |x - X_i|^{-2}$ for $x \notin B_{2R}(X_i)$ and by Lemma 3.15

$$|\nabla w_{ii_{nn}}(x)| \lesssim \frac{1}{N}(|x - X_i|^{-2} + |x - X_{i_{nn}}|^{-2}) \quad \text{for } x \notin B_{4R}(X_i) \cup B_{4R}(X_{i_{nn}}).$$

For $i \nsim k \nsim j$ and $h \in B_R(0)$ we can find an ellipsoid arc $s : [0,1] \to \mathbb{R}^3$ connecting $X_i + h$ and $X_j + h$ and satisfying for $l = k, k_{nn}$

$$|s(t) - X_l| \geq \frac{1}{2} \min\{|X_a - X_b| \mid a \in \{i,j\}, b \in \{k, k_{nn}\}\} \tag{3.21}$$

$$|\tfrac{d}{dt} s(t)| \leq C|X_i - X_j|. \tag{3.22}$$

with a universal constant C. Using this curve we can estimate

$$
\begin{aligned}
|w_{kk_{nn}}(X_i+h)-w_{kk_{nn}}(X_j+h)| &= |\int_0^1 \nabla w_{kk_{nn}}(s(t))\tfrac{d}{dt}s(t)dt| \\
&\lesssim \tfrac{1}{N}\int_0^1 |s(t)-X_k|^{-2}+|s(t)-X_{k_{nn}}|^{-2}dt|X_i-X_j| \lesssim \tfrac{1}{N}(d_{ik}^{-2}+d_{ik_{nn}}^{-2}+d_{jk}^{-2}+d_{jk_{nn}}^{-2})d_{ij}.
\end{aligned}
$$

Analogously, for $i\neq k\neq j$ it holds $|u_k(X_i+h)-u_k(X_j+h)|\lesssim \frac{1}{N}(d_{ik}^{-2}+d_{jk}^{-2})d_{ij}$ and by Lemma 3.5 we have $|w_{ij}(X_i+h)-w_{ij}(X_j+h)|=0$. For the one-particle solutions we can use symmetry to obtain

$$
|u_j(X_i+h)-u_i(X_j+h)| = |u_j(X_i+h)-u_j(X_i-h)| \leq C\tfrac{1}{N}d_{ij}^{-2}d_{ij}.
$$

Having these estimates at hand, we can start the actual proof. Denote by $x_i = X_i+h$ and $x_j = X_j+h$, we need to distinct several cases:

Case 1: $i\in\mathcal{S}, j\in\mathcal{S}$:

$$
\begin{aligned}
|w_N^{(0)}(x_i)-w_N^{(0)}(x_j)| &\leq |u_i(x_j)-u_j(x_i)| + \sum_{k\in\mathcal{S},k\neq i,j}|u_k(x_i)-u_k(x_j)| + \sum_{k\in\mathcal{D}}|w_{kk_{nn}}(x_i)-w_{kk_{nn}}(x_j)| \\
&\lesssim \tfrac{1}{N}d_{ij}^{-2}d_{ij} + \sum_{k\in\mathcal{S},k\neq i,j}\tfrac{1}{N}(d_{ik}^{-2}+d_{jk}^{-2})d_{ij} + \sum_{k\in\mathcal{D}}\tfrac{1}{N}(d_{ik}^{-2}+d_{ik_{nn}}^{-2}+d_{jk}^{-2}+d_{jk_{nn}}^{-2})d_{ij} \\
&\lesssim S_{2,\delta}d_{ij}
\end{aligned}
$$

Case 2: $i\in\mathcal{D}, j\in\mathcal{S}$:

In this case, $d_{ij}>\delta$ and we obtain with the help of Lemma 3.6

$$
\begin{aligned}
|w_N^{(0)}(x_i)-w_N^{(0)}(x_j)| &\leq |u_j(x_i)| + \sum_{k\in\mathcal{S},k\neq j}|u_k(x_i)-u_k(x_j)| \\
&\qquad + |w_{ii_{nn}}(x_j)| + |w_{ii_{nn}}(x_i)-u_j(x_j)| + \sum_{k\in\mathcal{D},k\neq i}|w_{kk_{nn}}(x_i)-w_{kk_{nn}}(x_j)| \\
&\lesssim \tfrac{1}{N}d_{ij}^{-2}d_{ij} + \tfrac{1}{N}\sum_{k\in\mathcal{S}}(d_{ik}^{-2}+d_{jk}^{-2})d_{ij} \\
&\qquad + \tfrac{1}{N}(d_{ij}^{-1}+d_{ji_{nn}}^{-1})d_{ij}^{-1}d_{ij} + \tfrac{1}{N}d_{ii_{nn}}^{-1}d_{ij}^{-1}d_{ij} \\
&\qquad + \tfrac{1}{N}\sum_{k\in\mathcal{D},k\neq i}(d_{ik}^{-2}+d_{ik_{nn}}^{-2}+d_{jk}^{-2}+d_{jk_{nn}}^{-2})d_{ij} \\
&\lesssim (S_{2,\delta}+C^{(4)})d_{ij}
\end{aligned}
$$

Case 3: $i\in\mathcal{D}, j\in\mathcal{D}$ and $d_{ij}<\delta$:

$$
\begin{aligned}
|w_N^{(0)}(x_i)-w_N^{(0)}(x_j)| &\leq \sum_{k\in\mathcal{S}}|u_k(x_i)-u_k(x_j)| + \sum_{k\in\mathcal{D},k\nsim i}|w_{kk_{nn}}(x_i)-w_{kk_{nn}}(x_j)| \\
&\lesssim \sum_{k\in\mathcal{S}}\tfrac{1}{N}(d_{ik}^{-2}+d_{jk}^{-2})d_{ij} + \sum_{k\in\mathcal{D},k\nsim i}\tfrac{1}{N}(d_{ik}^{-2}+d_{ik_{nn}}^{-2}+d_{jk}^{-2}+d_{jk_{nn}}^{-2})d_{ij} \\
&\leq S_{2,\delta}d_{ij}
\end{aligned}
$$

Case 4: $i \in \mathcal{D}, j \in \mathcal{D}$ and $d_{ij} > \delta$:

$$
\begin{aligned}
|w_N^{(0)}(x_i) - w_N^{(0)}(x_j)| &\leq \sum_{k\in\mathcal{S}} |u_k(x_i) - u_k(x_j)| + \sum_{k\in\mathcal{D},k\neq i,j} |w_{kk_{nn}}(x_i) - w_{kk_{nn}}(x_j)| \\
&\quad + |w_{ii_{nn}}(x_i) - w_{jj_{nn}}(x_j)| + |w_{jj_n}(x_i)| + |w_{ii_n}(x_j)| \\
&\lesssim \frac{1}{N}\sum_{k\in\mathcal{S}}(d_{ik}^{-2} + d_{jk}^{-2})d_{ij} + \frac{1}{N}\sum_{k\in\mathcal{D},k\neq i,j}(d_{ik}^{-2} + d_{ik_{nn}}^{-2} + d_{jk}^{-2} + d_{jk_{nn}}^{-2})d_{ij} \\
&\quad + |w_{ii_{nn}}(x_i) - \frac{1}{6\pi RN} g| + |\frac{1}{6\pi RN} g - w_{jj_{nn}}(x_j)| \\
&\quad + \frac{1}{N}(d_{ij_{nn}}^{-1} + d_{ij}^{-1}) + \frac{1}{N}(d_{ji_{nn}}^{-1} + d_{ji}^{-1}) \\
&\lesssim S_{2,\delta} d_{ij} + \frac{1}{N} d_{ii_{nn}}^{-1} d_{ij}^{-1} d_{ij} + \frac{1}{N} d_{jj_{nn}}^{-1} d_{ij}^{-1} d_{ij} + \frac{1}{N}(d_{ij_{nn}}^{-1} + d_{ij}^{-1} + d_{ji_{nn}}^{-1} + d_{ji}^{-1}) d_{ij}^{-1} d_{ij} \\
&\lesssim (S_{2,\delta} + C^{(4)}) d_{ij},
\end{aligned}
$$

by using $|w_{ii_{nn}}(x_i) - \frac{1}{6\pi RN} g| \lesssim \frac{1}{N} d_{ii_{nn}}^{-1} d_{ij}^{-1} d_{ij}$. □

The final step consists of lifting the estimate via the method of reflections to conclude the validity also for u_N. At this stage it is important that the Lipschitz estimate for $w_N^{(0)}$ and estimate on $\nabla w_N^{(0)}$ are compatible. Indeed, the same estimate as in the previous lemma is provable also with the classical initialization using the same case distinctions. However, if one tries to propagate this via the method of reflections the full $S_{2,0}$-dependence would appear already in the first reflection step due to the $\|\nabla u_N^{(0)}\|_{L^\infty(\cup B_i)}$-dependence of the projection operators.

Proposition 3.2. *There exists $\theta_\star < \min\{\theta_1, \theta_2\}$ with the following property. Assume a particle configuration with $d_{min} > 4R$, $\phi_N S_{3,0} < \theta_\star$ and* (A3) - (A4)*, then for all i, j and $k \in \mathbb{N}$ we have*

$$
|w_N^{(k)}(X_i) - w_N^{(k)}(X_j)| \lesssim (S_{2,\delta} + C^{(4)})|X_i - X_j|.
$$

In particular,

$$
|u_N(X_i) - u_N(X_j)| \lesssim (S_{2,\delta} + C^{(4)})|X_i - X_j|.
$$

Proof. To simplify the notation, set $\beta := S_{2,\delta} + C^{(4)}$ and assume $\phi_N S_{3,0} < \theta < \min\{\theta_1, \theta_2\}$. We are going to prove the following estimate

$$
|w_N^{(k)}(X_i + h) - w_N^{(k)}(X_j + h)| \leq \beta \sum_{n=0}^{k} (C\theta)^n |X_i - X_j| \quad \text{for } i \neq j, h \in B_R(0)
$$

by induction. The case $k = 0$ is the statement of Lemma 3.9. Next, assume the statement is true up to some fixed $k > 0$, then for $x_i = X_i + h$

$$
w_N^{(k+1)}(x_i) = w_N^{(k)}(x_i) - \sum_{l=1}^{N} Q_l w_N^{(k)}(x_i) = (w_N^{(k)})_i - \sum_{l\neq i} Q_l w_N^{(k)}(x_i).
$$

Hence,

$$
|w_N^{(k+1)}(x_i) - w_N^{(k+1)}(x_j)| \leq |(w_N^{(k)})_i - (w_N^{(k)})_j| + \sum_{l\neq i,j} |Q_l w_N^{(k)}(x_i) - Q_l w_N^{(k)}(x_j)| + |Q_j w_N^{(k)}(x_i)| + |Q_i w_N^{(k)}(x_j)|.
$$

By induction hypothesis it holds

$$
|(w_N^{(k)})_i - (w_N^{(k)})_j| \leq \beta \sum_{n=0}^{k} (C\theta)^n |X_i - X_j|.
$$

Moreover, by using again appropriate arcs connecting x_i and x_j, we obtain by (3.9)

$$
|Q_l w_N^{(k)}(x_i) - Q_l w_N^{(k)}(x_j)| \leq C|X_i - X_j| R^3 (|X_i - X_l|^{-3} + |X_j - X_l|^{-3}) \|\nabla w_N^{(k)}\|_{L^\infty(\cup B_i)}
$$

and thus

$$\sum_{l\neq i,j} |Q_l w_N^{(k)}(x_i) - Q_l w_N^{(k)}(x_j)| \leq C|X_i - X_j|\theta(C\theta)^k \|\nabla w_N^{(0)}\|_{L^\infty(\cup B_i)} \leq |X_i - X_j|(C\theta)^{k+1}\beta$$

by Lemma 3.9. Finally,

$$\begin{aligned}|Q_j w_N^{(k)}(x_i)| &\leq CR^3|x_i - X_j|^{-2}\|\nabla w_N^{(k)}\|_{L^\infty(\cup B_i)} \\ &\leq C\phi_N \tfrac{1}{N}|X_i - X_j|^{-3}(C\theta)^k\beta|X_i - X_j| \leq |X_i - X_j|(C\theta)^{k+1}\beta.\end{aligned}$$

After summing up the estimates, we obtain the claimed relation also for $k+1$. The claims follow by considering $\theta_\star$ with $C\theta_\star < \frac{1}{2}$ with $h = 0$ and utilizing the previous L^∞-convergence result. □

Remark 3.3.

- With this statement at hand, we have fulfilled the tasks set out at the beginning and gathered all the necessary resources to extend the application of Hauray's method: Proposition 3.1 quantifies the approximation error, Proposition 3.2 provides the Lipschitz-type estimate and from Lemma 3.1 we borrow the necessary estimate for the inverse distances.
- The presented approach is not limited to the case of a mixture between one- and two-particle solution, but can be extended to mixtures including solution of k-particle problems for arbitrary, but fixed k. However, the notational effort might increase rapidly and also to quantify the behaviour of a k-particle solution will get more complicated. But at least for three particles, this approach is still tractable.

3.5 Mean-Field Result

Building on the results of the previous subsections, we now present a mean-field result that extends the statements in Theorem 2.1 and is similar to the point particle result in [HS25].

In order set a foundation for the subsequent formulation of the result, we first introduce necessary notation. Given $\delta > 0$ and particle positions $\{X_i\}_{i=1}^N$, we denote the set of δ-close particles by

$$D_\delta = \{X_i \mid d_{ii_{nn}} < \delta\} = \{X_i, X_{i_{nn}} \mid i \in \mathcal{D}_\delta\}.$$

Moreover, we set $\rho_N^0 = \rho_N(0)$. Using this notation, we can describe the assumptions, which are strongly based on those in [HS25]. In detail, given $p \in [1,\infty]$ and $\rho_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$, we assume sequences of particle configurations $\{X_i\}_{i=1}^N$ and $\delta_N > 0$ such that

$$\mathcal{W}_p(\rho_N^0, \rho_0) \to 0, \tag{A1}$$

$$\begin{aligned}(N^{-2/3}\delta_N^{-2} + 1)[\mathcal{W}_p(\rho_N^0, \rho_0)^{p/(3+p)} + (N^{-p}\rho_N^0[D_{\delta_N}]d_{min}^{-p}(0))^{1/(3+p)} \\ + (S_{2,\delta}(0)(\phi_N S_{2,0}(0) + R))^{p/(3+p)}] \to 0,\end{aligned} \tag{A2}$$

$$\forall i : \forall j \neq i \neq k \neq j : \quad d_{ij}(0) < \delta_N \Rightarrow d_{ik}(0) \gg \delta_N, \tag{A3}$$

$$\forall i : \forall j \neq i \neq k \neq j : \quad d_{ij}(0) < \delta_N \Rightarrow \tfrac{1}{N}d_{ij}^{-1}(0)d_{ik}^{-1}(0) \leq C_N^{(4)} \to 0 \text{ as } N \to \infty, \tag{A4}$$

$$d_{min}(0)^{-3}R^3\log(N) \to 0. \tag{A5}$$

Remark 3.4. In the case $p = \infty$ assumption (A2) must be understood as

$$(N^{-2/3}\delta_N^{-2} + 1)[\mathcal{W}_\infty(\rho_N^0, \rho_0) + N^{-1}d_{min}^{-1}(0) + S_{2,\delta}(0)(\phi_N S_{2,0}(0) + R)] \to 0.$$

Exploiting these assumptions, the main result states as follows.

Theorem 3.1. *Let $\rho_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$ and $p \in [1,\infty]$ and assume* (A1)-(A5) *to be satisfied. Then there is a constant $C = C(\|\rho_0\|_{L^1\cap L^\infty}, p)$, s.t. for all $T > 0$ there is $N^\star(T) \in \mathbb{N}$ with the following property: For $0 \le t \le T$ and $N > N^\star(T)$*

$$\mathcal{W}_p(\rho_N(t), \rho(t)) \le (\mathcal{W}_p(\rho_N^0, \rho_0) + CE_N)e^{Ct}$$

and

$$d_{ij}(t) \ge d_{ij}(0)e^{-Ct},$$

where

$$E_N = \left(\rho_N^0[D_{\delta_N}]\left(N^{-1}d_{min}(0)^{-1} + N^{-2/3}\right)^p + \left(S_{2,\delta_N}(0)(\phi_N S_{2,0}(0) + R)\right)^p\right)^{1/p}$$

and $E_N \to 0$. Moreover, for $q < 3/2$ and $\Omega \Subset \mathbb{R}^3$ it holds $\|u_N(t) - v_N(t)\|_{L^q(\Omega)} = C_\Omega e_N(t)$ as $N \to \infty$. Here, the constant C_Ω is just depending on $\mathrm{diam}(\Omega)$ *and the error term $e_N(t)$ is independent of Ω.*

Remark 3.5. In the case $p = \infty$ it holds $E_N = N^{-1}d_{min}^{-1}(0) + S_{2,\delta}(0)(\phi_N S_{2,0}(0) + R)$.

In the proof it will be necessary to use both types of known approximations, i.e. the standard single-particle approximation $u_N^{(0)} = \sum_{i=1}^N u_i$ and the mixed-type approximation $w_N^{(0)} = \sum_{i\in S} u_i + \sum_{i\in D} w_{ii_{nn}}$. But why does the classical approximation reappear? The reason is its explicit expression in terms of the Oseen tensor, which makes it compatible with Hauray's method. This important property is no longer valid, when considering the mixed-type approximation. In summary, the mixed-type approximation is needed due to the improved global approximation errors, whereas $u_N^{(0)}$ compares easier to the macroscopic velocity v_N. However, one might object that inserting $w_N^{(0)}$ as an intermediate step cannot improve the total approximation by $u_N^{(0)}$. This objection is generally correct, but it should be noted that by using $w_N^{(0)}$, the generally larger approximation error by $u_N^{(0)}$ can be limited to smaller areas, as indicated by the following lemma.

Lemma 3.10. *Assume* (A1)-(A5) *are satisfied, as well as $4R < d_{min}$ and $R^3 d_{min}^{-3} < \theta_\star$, then*

$$|u_N^{(0)}(X_k) - w_N^{(0)}(X_k)| \lesssim (\tfrac{1}{N}d_{min}^{-1}\mathbb{1}_{k\in D_{\delta_N}} + RS_{2,\delta_N}).$$

Proof. By Lemma 3.6 we obtain

$$|w_{ii_{nn}}(X_k) - u_i(X_k) - u_{i_{nn}}(X_k)| \lesssim \begin{cases} \frac{R}{N}(d_{ik}^{-2} + d_{i_{nn}k}^{-2}) & \text{if } k \nsim i, \\ \frac{1}{N}d_{ii_{nn}}^{-1} & \text{if } i \sim k. \end{cases}$$

Thus,

$$\begin{aligned} |w_N^{(0)}(X_k) - u_N^{(0)}(X_k)| &= |\sum_{i\in D_{\delta_N}} w_{ii_{nn}}(X_k) - u_i(X_k) - u_{i_{nn}}(X_k)| \\ &\le \sum_{i\in D_{\delta_N}} |w_{ii_{nn}}(X_k) - u_i(X_k) - u_{i_{nn}}(X_k)| \lesssim (\tfrac{1}{N}d_{min}^{-1}\mathbb{1}_{k\in D_{\delta_N}} + RS_{2,\delta_N}). \end{aligned}$$

□

To slim the proof of the main theorem, some estimates will be outsourced in the following two lemmas.

Lemma 3.11. *If* (A3) *is valid, then it holds*

$$\tfrac{R^2}{6}|\Delta K * \rho_N(X_k)| \lesssim \tfrac{1}{N}d_{min}^{-1}\mathbb{1}_{k\in D_{\delta_N}} + RS_{2,\delta_N}.$$

Proof. We recall $|\Delta K(x)| \lesssim |x|^{-3}$ and conclude

$$\begin{aligned}
\tfrac{R^2}{6}|\Delta K * \rho_N(X_k)| &= \frac{R^2}{6N}|\sum_{i\neq k} \Delta K(X_k - X_i)| \\
&\lesssim \sum_{i\neq k} \tfrac{R^2}{N}|X_k - X_i|^{-3} \\
&\lesssim \tfrac{R^2}{N}|X_k - X_{k_{nn}}|^{-3}\mathbb{1}_{k\in\mathcal{D}_{\delta_N}} + \sum_{d_{ik}>\delta_N} \tfrac{R^2}{N}|X_k - X_i|^{-3} \\
&\lesssim \tfrac{1}{N} d_{min}^{-1}\mathbb{1}_{k\in\mathcal{D}_{\delta_N}} + RS_{2,\delta_N},
\end{aligned}$$

where we used $R < d_{ik}$ for $i \neq k$. □

Finally, we state how the $S_{2,0}$-dependence introduced by source (i) from Remark 2.1 can be handle using a more careful analysis.

Lemma 3.12. *Using the notation from Theorem 3.1 assume that* (A3) *is fullfilled for some $\delta > 0$. Denote by π a coupling between ρ_N and ρ and set $\eta = \int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^p d\pi(x,y)$, then for $p < \infty$*

$$\int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^{p-1}|K * \rho_N(x) - K * \rho(y)|\, d\pi(x,y)$$
$$\leq C(S_{2,\delta}+1)\eta + \rho_N[D_\delta](N^{-1}d_{min}^{-1} + N^{-2/3})^p \quad (3.23)$$

for some constant C depending on $\|\rho\|_{L^\infty}$. For $p = \infty$ we consider $\eta = \pi - \mathrm{esssup}|x-y|$ and obtain

$$\pi - \mathrm{esssup}|K * \rho_N(x) - K * \rho(y)| \leq C(S_{2,\delta}+1)\eta + N^{-1}d_{min}^{-1} + N^{-2/3} \quad (3.24)$$

for C depending on $\|\rho\|_{L^\infty}$.

Proof. This estimate is borrowed from the proof of [HS25, Thm. 1.1]. The calculations are similar to those used to bound I_2 when proving Theorem 2.1 and are presented in section A.3 in the appendix. □

With these tools at hand, we can start the actual proof. Its structure will highly follow the general steps motivated in section 2.

Proof of Theorem 3.1:

The proof is organized into two parts: First, we derive the statements about the Wasserstein and the interparticle distances. The second part is devoted to the convergence of the vector fields. In the proof, we restrict to the case $p < \infty$, but the proof can easily be adapted to the case $p = \infty$ using the corresponding formulations given in Remarks 3.4 and 3.5 as well as Lemma 3.12. Further insights on how to handle the calculations in the case $p = \infty$ are available in [HS21].

PART 1: The bootstrap argument:

Step 0: *Defining the bootstrap setup:*

We consider $\theta_\star$ from Lemma 3.2 and set $L = \|\rho_0\|_\infty^{2/3} + 1$. Moreover, we define $\delta_N(t) := \frac{1}{2}\delta_N e^{-C_1 t}$ for a constant C_1 independent of N and t, which will be specified later. In the following $\mathcal{S}$ and $\mathcal{D}$ implicitely refer to the choice $\delta = \delta_N(t)$. We denote by $S_{k,\beta}(t)$ the value of $S_{k,\beta}$ related to the particle configuration $\{X_i(t)\}_i$ at time t.

In this setting, we define $T^*(N)$ as the maximal time, s.t.

$$S_{3,0}(t)\phi_N \leq \theta_\star \quad \text{and} \quad d_{min}(t) > 4R, \quad \text{(BT1)}$$

$$S_{2,\delta_N(t)}(t) \leq L, \quad \text{(BT2)}$$

$$d_{ij}(t) \geq \tfrac{1}{2}d_{ij}(0)e^{-C_1 t}, \quad \text{(BT3)}$$

hold true for all $t \leq T^*(N)$.

Our goal is now to utilize a continuity argument in order to prove that given $T \geq 0$ there is some $N(T)$ such that $T^*(N) > T$ for all $N \geq N(T)$. Again, it will turn out that the bounds on the Wasserstein and interparticle distances will be necessary side products. Before starting the proof, we introduce $C_N^{(4)}(t) := C_N^{(4)} e^{2C_1 t}$ and remark that for any $t \leq T^*(N)$ assumption (A4) holds with the new constant $C_N^{(4)}(t)$.

Step 1: *Bounding $S_{3,0}(t)\phi_N$:*

Concerning (BT1) for $t \leq T \wedge T^*(N)$, we obtain by (BT3)

$$S_{3,0}(t)\phi_N \lesssim (d_{min}(t)^{-3}R^3 + d_{min,1}(t)^{-3}R^3\log(N)) \\ \lesssim d_{min}(t)^{-3}R^3\log(N) \lesssim e^{3C_1T}d_{min}(0)^{-3}R^3\log(N) \to 0$$

and

$$Rd_{min}(t)^{-1} \leq Rd_{min}(0)^{-1}e^{C_1T} \to 0,$$

where the convergence follows in both cases from (A5). Consequently, for $N > N_1$ we can ensure the strict version of (BT1) for all $t \leq T \wedge T^*(N)$.

Step 2: *Differential inequality for the Wasserstein distance:*

According to Remark 1.1 the measures $\rho_N(t)$ and $\rho(t)$ can be represented by means of the associated flow maps as

$$\rho_N(t) = Y(t,0)_\# \rho_N(0) \quad \text{and} \quad \rho(t) = Z(t,0)_\# \rho_0$$

with

$$\begin{cases} \frac{d}{dt}Y(t,s,x) = u_N(t, Y(t,s,x)), \\ Y(s,s,x) = x, \end{cases} \qquad \begin{cases} \frac{d}{dt}Z(t,s,x) = v_N(t, Z(t,s,x)) + \frac{1}{6\pi RN}g, \\ Z(s,s,x) = x. \end{cases}$$

Thus, starting with an optimal transport plan $\gamma \in \Gamma(\rho_N(0), \rho_0)$, we can consider the pushforward measure $\gamma_t := (Y(t,0), Z(t,0))_\#\gamma$, which is an admissible coupling between $\rho_N(t)$ and $\rho(t)$. Hence,

$$\mathcal{W}_p(t)^p := \mathcal{W}_p(\rho_N(t), \rho(t))^p \leq \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^p d\gamma_t(y,z) =: \eta(t).$$

Now for $t \leq T \wedge T^*(N)$,

$$\begin{aligned} \frac{d}{dt}\eta(t) &\leq p\int_{\mathbb{R}^3\times\mathbb{R}^3} |Y(t,0,y) - Z(t,0,z)|^{p-1} \\ &\qquad \cdot |u_N(t, Y(t,0,y)) - v_N(t, Z(t,0,z)) - \tfrac{1}{6\pi RN}g| d\gamma(y,z) \\ &= p\int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|u_N(t,y) - v_N(t,z) - \tfrac{1}{6\pi RN}g| d\gamma_t(y,z) \\ &\lesssim \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|u_N(t,y) - w_N^{(0)}(t,y)| d\gamma_t(y,z) \\ &\quad + \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|w_N^{(0)}(t,y) - u_N^{(0)}(t,y)| d\gamma_t(y,z) \\ &\qquad + \int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1}|u_N^{(0)}(t,y) - v_N(t,z) - \tfrac{1}{6\pi RN}g| d\gamma_t(y,z) \\ &=: I_1 + I_2 + I_3. \end{aligned}$$

By (BT1) we can apply (3.17) from Prop. 3.1 to get

$$\begin{aligned} I_1 &\leq C\int_{\mathbb{R}^3\times\mathbb{R}^3} |y-z|^{p-1} S_{2,\delta_N(t)}(t)(\phi_N S_{2,0}(t) + R) d\gamma_t(y,z) \\ &\leq C\left(\eta(t) + \left(S_{2,\delta_N(t)}(t)\left(\phi_N S_{2,0}(t) + R\right)\right)^p\right), \end{aligned} \tag{3.25}$$

after using Young's inequality.
By exploiting Lemma 3.10 the second term I_2 satisfies

$$I_2 \leq C\left(\eta(t) + \rho_N(t)[D_{\delta_N(t)}]\left(\tfrac{1}{N}d_{min}^{-1}(t)\right)^p + \left(RS_{2,\delta_N(t)}(t)\right)^p\right). \tag{3.26}$$

Here it is appropiate to emphasize that the local error between $w_N^{(0)}$ and $u_N^{(0)}$ leads to an error term, which is already known from Lemma 3.12.

The bound on I_3 will be based on the kernel estimates introduced in [HS25]. Remarking $v_N(t,z) = [K * \rho(t)](z)$ and $\operatorname{supp}(\gamma_t) \subset \{X_i(t)\}_{i=1}^N \times \mathbb{R}^3$, one observes

$$|u_N^0(t,y) - v_N(t,z) - \tfrac{1}{6\pi RN}| \leq |[K * \rho_N(s)](y) - [K * \rho(s)](z)| + \tfrac{R^2}{6}|[\Delta K * \rho_N(s)](y)|\ \gamma_t - \text{a.e.}$$

and by Lemma 3.11 and Lemma 3.12 one ends up with

$$I_3 \leq C\left(\left(S_{2,\delta_N(t)}(t)+1\right)\eta(t) + \rho_N(t)[D_{\delta_N(t)}]\left(N^{-1}d_{min}(t)^{-1} + N^{-2/3}\right)^p + \left(RS_{2,\delta_N(t)}(t)\right)^p\right). \tag{3.27}$$

Combining (3.25), (3.26) and (3.27) leads to

$$\begin{aligned}\frac{d}{dt}\eta(t) \lesssim{}& \left(S_{2,\delta_N(t)}(t)+1\right)\eta(t)\\ &+ \rho_N(t)[D_{\delta_N(t)}]\left(N^{-1}d_{min}(t)^{-1} + N^{-2/3}\right)^p + \left(S_{2,\delta_N(t)}(t)\left(S_{2,0}(t)\phi_N + R\right)\right)^p.\end{aligned}$$

By definition of $\delta_N(t)$ and (BT3), we deduce $X_i(t) \in D_{\delta_N(t)} \Rightarrow X_i(0) \in D_{\delta_N}$ and thus $\rho_N(t)[D_{\delta_N(t)}] \leq \rho_N(0)[D_{\delta_N}]$. Using this as well as (BT2) and (BT3) one concludes

$$\frac{d}{dt}\eta(t) \leq C(L+1)\eta(t) + \underbrace{\left(\rho_N^0[D_{\delta_N}]\left(N^{-1}d_{min}(0)^{-1} + N^{-2/3}\right)^p + \left(S_{2,\delta_N}(0)\left(S_{2,0}(0)\phi_N + R\right)\right)^p\right)e^{Ct}}_{=:E_N^p}$$

for a constant C just depending on C_1 and p.

In consequence, Gronwall's Lemma implies

$$\mathcal{W}_p(t)^p \leq \eta(t) \leq (\eta(0) + CE_N^p)e^{C_2t} = (\mathcal{W}_p(0)^p + CE_N^p)e^{C_2t} \tag{3.28}$$

with a constant C_2 depending only on L, C_1 and p. It is noteworthy that the aforementioned estimate is the Wasserstein bound formulated in Theorem 3.1.

Step 3: *Bounding $S_{2,\delta_N(t)}(t)$:*

As we already identified $\rho(t)$ as the solution of a transport equation associated to a Lipschitz vector field, it holds $\|\rho(t)\|_\infty \leq \|\rho_0\|_\infty$. Using Lemma 3.1 and (3.28) we deduce

$$\begin{aligned}S_{2,\delta_N(t)}(t) &\leq \|\rho_0\|_\infty^{2/3} + \left(\frac{2^{5/3}\|\rho_0\|_\infty^{p/(3(3+p))}}{N^{2/3}\delta_N^2} + \left(\frac{2^{5/3}\|\rho_0\|_\infty^{1/3}}{N^{2/3}\delta_N^2}\right)^{\frac{2+p}{3+p}}\right)e^{CT}(\mathcal{W}_p(0)^p + E_N^p)^{1/(3+p)}\\ &\leq \|\rho_0\|_\infty^{2/3} + C(N^{-2/3}\delta_N^{-2}+1)e^{CT}(\mathcal{W}_p(0)^p + E_N^p)^{1/(3+p)}\end{aligned}$$

by applying Young's inequality after noticing $\frac{2+p}{3+p} < 1$. Here the constant C can be chosen independent of N and T. Now, by (A2) the second term tends to zero and so there is a N_2 s.t. for all $N > N_2$ and $t \leq T \wedge T^*(N)$ it holds

$$S_{2,\delta_N(t)}(t) \leq \|\rho_0\|_\infty^{2/3} + \tfrac{1}{2} < L.$$

This is the strict version of (BT2).

Step 4: *Bounding $d_{ij}(t)$*

By (BT1) the improved Lipschitz-type estimate from Proposition 3.2 can be applied and we deduce

$$\begin{aligned}\frac{d}{dt}d_{ij}(t)^2 &= 2(X_i(t)-X_j(t))\cdot(u_N(t,X_i(t))-u_N(t,X_j(t)))\\ &\geq -2(CS_{2,\delta_N(t)}(t)+C_4^{(N)}(t))d_{ij}(t)^2\\ &\geq -2(CL+C_N^{(4)}(0)e^{2C_1T})d_{ij}(t)^2 \geq -2(CL+1)d_{ij}(t)^2\end{aligned}$$

for $N > N_3$ by assumption (A4). After setting $C_1 = CL+1$, Gronwall's Lemma implies

$$d_{ij}(t) \geq d_{ij}(0)e^{-C_1t} > \tfrac{1}{2}d_{ij}(0)e^{-C_1s}$$

and so the strict version of (BT3) is fulfilled for any $t \leq T \wedge T^\star(N)$ with $N > N_3$.

Step 5: *Conclusion:*

We proved that for $N \geq \max\{N_1, N_2, N_3\}$ the strict versions of (BT1) - (BT3) are fulfilled for all $t \leq T \wedge T^*(N)$. Analogously to the proof of Theorem 2.1 we can use the continuity of the involved terms to conclude $T < T^*(N)$.

PART 2: Convergence of the vector fields:

Assume $q < 3/2$ and pick $\Omega \Subset \mathbb{R}^3$. We begin with a few statements that will prove to be helpful later on. First, for $\pi \in \Gamma(\rho_N(t),\rho(t))$ optimal, it holds

$$\begin{aligned}\int_\Omega |[K*\rho_N(t)](x)-[K*\rho(t)](x)|^q dx &= \int_\Omega \Big|\int_{\mathbb{R}^3\times\mathbb{R}^3} K(x-y)-K(x-z)d\pi(y,z)\Big|^q dx\\ &\leq \int_\Omega\int_{\mathbb{R}^3\times\mathbb{R}^3} |K(x-y)-K(x-z)|^q d\pi(y,z)dx\\ &\lesssim \int_\Omega\int_{\mathbb{R}^3\times\mathbb{R}^3} \left(|x-y|^{-2q}+|x-y|^{-2q}\right)|y-z|^q d\pi(y,z)dx\\ &= \int_{\mathbb{R}^3\times\mathbb{R}^3}\int_\Omega (|x-y|^{-2q}+|x-z|^{-2q})dx|y-z|^q d\pi(y,z)\\ &\leq \operatorname{diam}(\Omega)^{3-2q}\mathcal{W}_q(t)^q. \end{aligned}\tag{3.29}$$

Moreover, we remark that

$$\begin{aligned}\int_\Omega \operatorname{dist}(x,\{X_i\}_i)^{-q}dx &\leq \int_\Omega \sum_{i=1}^N |x-X_i|^{-q}\mathbb{1}_{B_{\frac{d_{min}}{2}}(X_i)}(x)+\left(\frac{d_{min}}{2}\right)^{-q}dx\\ &\lesssim Nd_{min}^{3-q}+|\Omega|\left(\frac{d_{min}}{2}\right)^{-q}.\end{aligned}\tag{3.30}$$

Now, we will use again the same mixture of approximations to estimate $u_N(t,x)-v_N(t,x)$

$$\begin{aligned}|u_N(t,x)-v_N(t,x)| &\leq |u_N(t,x)-w_N^{(0)}(t,x)|+|w_N^{(0)}(t,x)-u_N^{(0)}(t,x)|+|u_N^{(0)}(t,x)-v_N(t,x)|\\ &=: I_1+I_2+I_3.\end{aligned}$$

Assuming $S_{3,0}(t)\phi_N \leq \theta_\star$ and $d_{min}(t) > 4R$ (which is possible by the previous step for N large enough) it holds

$$I_1 \lesssim S_{2,\delta_N(t)}(t)(S_{2,0}(t)\phi_N+R).$$

Given $x \in \mathbb{R}^3$ we consider $k \in \mathcal{D}$ the index of the particle nearest to x in $D_{\delta_N(t)}$ and obtain analogously to Lemma 3.10

$$
\begin{aligned}
I_2 = |\sum_{i\in\mathcal{D}} w_{ii_{nn}}(x) - u_i(x) - u_{i_{nn}}(x)| \\
&\lesssim \tfrac{1}{N}\operatorname{dist}(x,\{X_i\}_i)^{-1} + \tfrac{R}{N}\sum_{j\nsim k, j\in\mathcal{D}} (|X_j - X_k|^{-2} + |X_{j_{nn}} - X_k|^{-2}) \\
&\lesssim \tfrac{1}{N}\operatorname{dist}(x,\{X_i\}_i)^{-1} + RS_{2,\delta_N(t)}(t).
\end{aligned}
$$

To bound I_3 we distinguish two cases. On the one hand, for $x \in B_R(X_k)$

$$
\begin{aligned}
|u_N^{(0)}(t,x) - v_N(t,x)| &\le |\tfrac{1}{6\pi RN} g| + |\tfrac{1}{N}K(X_k - x)| \\
&\quad + \tfrac{R^2}{6N}\sum_{i\neq k} |\Delta K(x - X_i)| + |[K * \rho_N(t)](x) - [K * \rho(t)](x)| \\
&\lesssim \tfrac{1}{NR} + \tfrac{1}{N}\operatorname{dist}(x,\{X_i\}_i)^{-1} \\
&\quad + \tfrac{1}{N}\sum_{i\neq k} R^2|x - X_i|^{-3} + |[K * \rho_N(t)](x) - [K * \rho(t)](x)| \\
&\lesssim \tfrac{1}{N}\operatorname{dist}(x,\{X_i\}_i)^{-1} + RS_{2,\delta_N(t)}(t) + |[K * \rho_N(t)](x) - [K * \rho(t)](x)|,
\end{aligned}
$$

where we absorbed the term according to the case $i = k_{nn}$ into the first summand and used $R > |x - X_k|$ as well as $2|x - X_i| > |X_k - X_i|$.

On the other hand, for $x \notin \cup_i B_R(X_i)$, let k be the index of the particle nearest to x, then

$$
\begin{aligned}
|u_N^{(0)}(t,x) - u(t,x)| &\le |[K * \rho_N(t)](x) - [K * \rho(t)](x)| + \tfrac{R^2}{6}|[\Delta K * \rho_N(t)](x)| \\
&\lesssim |[K * \rho_N(t)](x) - [K * \rho(t)](x)| + \tfrac{R^2}{N}\operatorname{dist}(x,\{X_i\}_i)^{-3} + \tfrac{R^2}{3N}\sum_{i\neq k,k_{nn}} |x - X_i|^{-3} \\
&\lesssim |[K * \rho_N(t)](x) - [K * \rho(t)](x)| + \tfrac{1}{N}\operatorname{dist}(x,\{X_i\}_i)^{-1} + RS_{2,\delta_N(t)}(t),
\end{aligned}
$$

by using $R < \operatorname{dist}(x,\{X_i\}_i)$, $|X_k - X_i| < 2|x - X_i|$ and again absorbing the term related to k_{nn} into the term involving $\operatorname{dist}(x,\{X_i\}_i)^{-1}$. Hence, combining the termwise bounds and using (3.29) and (3.30), we end up with

$$
\begin{aligned}
\|u_N(t,\cdot) - u(t,\cdot)\|_{L^q(\Omega)} &\lesssim S_{2,\delta_N(t)}(t)(S_{2,0}(t)\phi_N + R)|\Omega|^{1/q} + \operatorname{diam}(\Omega)^{\frac{3-2q}{q}}\mathcal{W}_q(t) \\
&\quad + \tfrac{1}{N}N^{1/q}d_{min}(t)^{\frac{3-q}{q}} + \tfrac{|\Omega|}{N}d_{min}(t)^{-1} \\
&\le C_\Omega e^{Ct}\bigg(S_{2,\delta_N}(0)(\phi_N S_{2,0}(0) + R) + \tfrac{1}{N}N^{1/q}d_{min}(0)^{\frac{3-q}{q}} \\
&\quad \tfrac{1}{N}d_{min}(0)^{-1} + \mathcal{W}_q(0) + B(0)^{1/q}\bigg) \to 0
\end{aligned}
$$

as $N \to \infty$ by assumption (A2). We remark that the constant C_Ω is just depending on diam(Ω) as claimed before. □

3.6 Further Remarks and Limitations

The presented mean-field result extends the result in Theorem 2.1 and published in [HS21] to more natural scalings of the particle distances. As pointed out in [HS25] this result applies to particle configurations with $d_{min} \sim N^{-2/3+\varepsilon}$ and $d_{min,1} \sim N^{-1/2+\varepsilon}$ for ε small. This scaling is already near to the natural case, where $d_{min} \sim N^{-2/3}$ and $d_{min,1} \sim N^{-1/2}$.

However, to speculate about further refinements of the existing technique, which might even include propagation of chaos, we need to identify the bottlenecks among the assumptions (A1) - (A5). Here, (A2) and (A3) are the two assumptions that seem most critical.

Regarding (A2) one obtains for $\delta_N \sim N^{-1/2}$ that the term

$$N^{-2/3}\delta_N^{-2}\mathcal{W}_p(\rho_N,\rho)^{p/(3+p)} \sim N^{-1/3}\mathcal{W}_p(\rho_N,\rho)^{p/(3+p)}$$

does not converge to zero for any $p \in [1,\infty]$ due to the scaling of the Wasserstein distance. Indeed, according to [HS21, (1.11)] it holds $\mathcal{W}_p(\rho_N,\rho) \gtrsim N^{-1/3}$. This might be solved by redefining δ_N. While at the moment δ_N acts as threshold differentiating between dilute and close particle pairs, one might extend this to triples of particles by reformulating assumption (A3) to

$$\forall i \in \{1,\dots,N\} : \#\{j \neq i \mid d_{ij} \leq \delta_N\} \leq 2. \tag{A3'}$$

In this scenario one expects $\delta_N \sim N^{-4/9-\varepsilon}$, so that at least the considered term will no longer cause problems for p large.

However, this already sheds light on the important insight, which we will gain from considering assumption (A4): the necessity to understand the three-particle problem in more detail.

But one thing at a time, let us first investigate what is critical about (A4). As it was already mentioned in the introduction, for independent uniformly distributed particle centers, one would expect the existence of $C_N^{(4)} \in O(1)$, but not the assumed convergence to zero. To be precise, one expects the existence of particles with $d_{ij} \sim N^{-2/3}$ and $d_{ik} \sim N^{-1/3}$.

Besides the technical necessity of (A4) for proving the previous results, there is also a physical interpretation regarding this assumption. To see this, we need to analyze, where the $C^{(4)}$-dependencies arise in the proofs. Considering for example the verification of the Lipschitz-type estimate in Lemma 3.20, we observe that the $C^{(4)}$-terms are not caused by interactions between nearest particles, but originate from the interplay of two particles, where one is member of a pair of close particles not including the other. Thus, assumption (A4) encodes a non-criticality assumption for three-particle constellations.

Indeed, let us emphasize on this heuristically. Given particle positions X_1, X_2 and X_3 with $d_{12} \ll d_{13}$, then X_1 and X_2 form a pair of close particles travelling with some velocity of magnitude $\sim N^{-1}d_{12}^{-1}$ with respect to particle X_3. Remaining in this simplification, one expects the pair of particles to travel the distance d_{13} in a time scaling like

$$Nd_{12}d_{13} \geq 1/C_4^{(N)}.$$

Consequently, in real world – refering to the case $C_4^{(N)} > c > 0$ – a pair of fast particles might catch up the third in $O(1)$ and then induce drastic changes in the particle constellation. In contrast, the assumption (A4) ensures that for fixed T and N large enough, the two particles are not able to reduce the distance to the third significantly, and so exclude drastic changes in the particle constellation.

These observations suggest that further refinement of the assumptions can only be achieved with a better understanding of the three-particle case.

4 Insights into Three-Particle Configurations

As motivated at the end of last section, the natural continuation of the two-particle considerations is to take into account the special behaviour of three particles. However, in contrast to the non-attractivity in the two-particle case there is no obvious property that can be utilized to express a potential preservation of distances in three-particle constellations. In order to simplify the considerations, we will mainly work with the point-particle approximation. In this setting, the dynamics are given by

$$\frac{d}{dt}X_i = \tfrac{1}{6\pi RN}g + \frac{1}{N}\sum_{j\neq i} K(X_i - X_j).$$

There is a variety of questions arising when thinking on how to include the behaviour of three sedimenting particles to weaken the separation assumptions:

- How to modify the assumptions for the mean-field limit? Which additional separation assumptions might be proposed?
- Is there some preservation of the minimal distance in the case of three particles? If not, can we identify a sufficiently large subclass, for which the minimal distance can be controlled?
- Is the behaviour sensitive to the influence of an external field? And which type of external disturbance might occur?
- Is there a way to extend results from the point-particle case to the case of spherical particles with $R > 0$?

In the following, we are going to illuminate some of these questions. However, the results are often not answering the questions, but emphasize on possible difficulties.

4.1 Modified Mean-Field Result

The goal of this section is to modify the conditions imposed on the particle configuration. We start with formulating our assumptions on the initial particle configuration.

As before, we consider some threshold values $\delta_N > 0$ differentiating between close particles and distinct ones. According to the discussions on the limitations of Theorem 3.1, we will allow not only pairs, but triples of particles inside a ball of radius δ_N, expressed by

$$\forall i \in \{1,\dots,N\}: \quad \#\{j\neq i \mid d_{ij} < \delta_N\} \leq 2 \quad \text{and} \quad k \notin \{j \neq i \mid d_{ij} < \delta_N\} \Rightarrow d_{ik} > 4\delta_N. \tag{A3'}$$

As not all critical constellations inducing a failure of (A4) are triples inside a δ_N-ball, we need to impose a condition to catch these exceptions. To formulate this, we denote by $i_{nnn} \in \operatorname{argmin}\{d_{ij} \mid j \notin \{i, i_{nn}\}\}$ the index of the next-to-nearest neighbour and assume the following modification of (A4)

$$\exists C_N^{(4)} \to 0 \; s.t. \; d_{ij} \leq \delta_N \;\text{ and} \tfrac{1}{N} d_{ij}^{-1}\, d_{ik}^{-1} \; > C_N^{(4)} \\ \Rightarrow k \in \{i, i_{nn}, i_{nnn}\} \text{ and } \{i, i_{nn}, i_{nnn}\} = \{j, j_{nn}, j_{nnn}\}. \tag{A4'}$$

In summary, there are two kind of critical three-particle constellations, which we want to handle specifically: (i) three particles in a ball of radius δ_N and (ii) three particles with distances not fulfilling the condition $\frac{1}{N}d_{ij}^{-1}d_{ik}^{-1} \leq C_N^{(4)}$. The reason for (i) is to extend the validity of assumption (A2), whereas configurations of group (ii) are critical by not fulfilling (A4). Consequently, we define

$$\mathcal{E}_1 = \{\{i,j,k\} \mid i \neq j \neq k \neq i \text{ and } X_i, X_j \in B_{\delta_N}(X_k)\} \\ \text{and} \quad \mathcal{E}_2 = \{\{i,j,k\} \mid i \neq j \neq k \neq i \text{ and } N^{-1}d_{ij}^{-1}d_{ik}^{-1} > C_4^{(N)}\}$$

and call $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$ the set of exceptional particle trios. The associated set of particles is denoted by

$$G_{\delta_N, C_N^{(4)}} = \{X_i \mid i \in e \text{ for some } e \in \mathcal{E}\}.$$

Remark 4.1.

- As the minimal distance to the third-nearest particle scales like $N^{-4/9}$, a natural choice for δ_N is given by $N^{-4/9-\varepsilon}$ for some $\varepsilon > 0$ small.
- Under assumptions (A3') and (A4') we obtain for $i \neq j \neq k \neq i$ with $\frac{1}{N} d_{ij}^{-1} d_{ik}^{-1} > C_N^{(4)}$ that $\{i, i_{nn}, i_{nnn}\} = \{i, j, k\}$. This follows from $j \in \{i, i_{nn}, i_{nnn}\}$ by (A3').
- Assumption (A4') is still unrealistic, as one expects the existence of configurations with $d_{ij} \sim N^{-2/3}$ and more than two additional particles with distance to X_i of order $N^{-1/3}$. However, all the other critical cases are included.

After defining the larger class of exceptional particle constellations, we need to consider how to control these special configurations. Whereas this was rather easy in the case of two particles by using the non-attractivity, it is more involved in the three-particle case. In fact, there is no explicit symmetry leading to a preservation of the minimal distance and in the multi-particle setting the three-particle constellations are influenced by an additional external disturbance field. In order to formalize the desired stability of the three-particle behaviour, we define the following.

Definition 4.1. *Given $T > 0$ and $Y \in (\mathbb{R}^3)^3$, we define for $L > 0$ the set of disturbance flows*

$$\mathcal{U}(Y, L) := \{u \in C(\mathbb{R}^3, \mathbb{R}^3) \mid \operatorname{div}(u) = 0 \text{ on } \mathbb{R}^3, |u(Y_i) - u(Y_k)| \leq L|Y_i - Y_k| \text{ for } i, k = 1, 2, 3\},$$

where $\operatorname{div}(u)$ *has to be understood in the weak sense. Moreover, we set $\mathcal{F} = \{f : [0, T] \times \mathbb{R}^3 \to \mathbb{R}^3 \mid f(\cdot, x) \text{ continuous}\}$ and given $N > 0$, $f : [0, \infty) \to (0, 1]$ continuous and decreasing, we define for $T > 0$*

$$\mathcal{M}(N, f, T, L) := \{X \in (\mathbb{R}^3)^3 \mid d^u_{min}(t) \geq f(t) d_{min}(0) \;\; \forall 0 \leq t \leq T \;\; \forall u \in \mathcal{F}, u(t) \in \mathcal{U}(X^u(t), L)\}$$

with X^u following the perturbed flow, i.e. for $i \neq k \neq j \neq i$

$$\begin{cases} \frac{d}{dt} X_i^u(t) = \frac{1}{N} \left(K(X_i(t) - X_j(t)) + K(X_i(t) - X_k(t)) \right) + u(t, X_i(t)), \\ X_i^u(0) = X_i, \end{cases}$$

for $i = 1, 2, 3$ and d^u_{min} the minimial particle distance in configuration X^u. We implicitly assume that the dynamics are well-defined until T for all $X \in \mathcal{M}(N, f, T, L)$, otherwise $X \notin \mathcal{M}(N, f, T, L)$.

Remark 4.2.

- Since we are just interested in the values of any disturbance field u at the points Y_1, Y_2 and Y_3 it is sufficient to consider affine linear fields, i.e. one can replace $\mathcal{U}(Y, L)$ by

$$\hat{\mathcal{U}}(Y, L) = \{u(x) = Ax + v \mid v \in \mathbb{R}^3, A \in \mathbb{R}^{3\times 3}, \operatorname{tr}(A) = 0, \text{ and } |A(X_i - X_j)| \leq L|X_i - X_j|\}.$$

Using this definition, we would hope the following to hold true:
There exists $f : [0, \infty) \to (0, 1]$ continuous and decreasing and $L > 0$, s.t.

$$\forall T > 0 \; \exists N(T) \in \mathbb{N} : \forall N > N(T) : \\ i, j, k \in \mathcal{E} \text{ for } i \neq k \neq j \neq i \;\Rightarrow (X_i, X_k, X_j) \in \mathcal{M}(N, f, T, L). \quad (\mathrm{A_L})$$

Assuming that this set is non-empty for a specific choice of L one obtains the following mean-field result.

Theorem 4.1. *Assume $\rho_0 \in L^\infty \cap \mathcal{P}(\mathbb{R}^3)$, $p \in [1, \infty]$ and consider a sequence of particle configurations and threshold values δ_N, such that there exists f as above with* ($\mathrm{A_L}$) *satisified for $L = 2(\|\rho_0\|_{L^\infty}^{2/3} + 1)$, as well as* (A3') *and* (A4'). *Moreover, we assume*

$$\mathcal{W}_p(\rho_N^0, \rho_0) \to 0, \tag{A1'}$$

$$(N^{-2/3}\delta_N^{-2} + 1)\left(\mathcal{W}_p(\rho_N^0, \rho_0)^{\frac{p}{3+p}} + \left(N^{-p}\rho_N^0[G_{\delta_N, C_N^{(4)}}]d_{min}(0)^{-p}\right)^{\frac{1}{3+p}}\right) \to 0. \tag{A2'}$$

Then, there exists some $C = C(p, \|\rho_0\|_{L^\infty}) > 0$, s.t. for all $T > 0$ $\exists N^*(T) > 0 :\ \forall N > N^*(T)$:

$$\mathcal{W}_p(\rho_N(t), \rho(t)) \leq \left(\mathcal{W}_p(\rho_N^0, \rho_0) + CE_N F(t)\right) e^{Ct} \quad \forall 0 \leq t \leq T$$

and

$$d_{min}(t) \geq \left(f(t) \wedge e^{-Ct}\right) d_{min}(0) \quad \forall 0 \leq t \leq T$$

with $E_N \to 0$ and $F(t) = \int_0^t f(s)^{-p} ds$.

Remark 4.3.

- In the case $p = \infty$ assumption (A2') reduces to

$$(N^{-2/3}\delta_N^{-2} + 1)\left(\mathcal{W}_\infty(\rho_N^0, \rho_0) + N^{-1}d_{min}(0)^{-1}\right) \to 0$$

- If f has the form $f(t) = ae^{-At}$ for $a, A > 0$, then the bound for the Wasserstein distance reduces to

$$\mathcal{W}_p(\rho_N(t), \rho(t)) \leq \left(\mathcal{W}_p(\rho_N^0, \rho_0) + \tilde{E}_N\right) e^{\tilde{C}t} \quad \forall\, 0 \leq t \leq T$$

  with slightly modified error term $\tilde{E}_N$ and constant $\tilde{C}$.

Proof. The proof follows the general structure of the previous proofs. We will therefore only clarify the setup and outline the individual steps. Let $\delta_N(t) = \frac{1}{2}\delta_N e^{-Ct}$ and introduce $C_N^{(4)}(t) := C_N^{(4)} e^{2C_1 t}$, then we define $T^*(N)$ to be the maximal time such that for all $0 \leq t \leq T^*(N)$ the following are satisfied

$$S_{2,\delta_N(t)}(t) \leq L = \|\rho_0\|_{L^\infty}^{2/3} + 1, \tag{BT1'}$$

$$d_{ij}(t) \geq \tfrac{1}{2}e^{-Ct}d_{ij}(0) \quad \text{for } i, j \text{ with } \{i, j\} \not\subseteq e\ \forall e \in \mathcal{E}, \tag{BT2'}$$

$$d_{ij}(t) \geq f(t)d_{ijk}(0) \text{ for } \{i, j, k\} \in \mathcal{E}, \tag{$\mathrm{BT_{new}}$}$$

where $d_{ijk} := \min\{d_{ij}, d_{ik}, d_{jk}\}$. Again, after fixing $T > 0$ arbitrary, the proof is based on a continuity argument inducing that $T^*(N) > T$ for N large enough. The necessary bounds on the Wasserstein distance and S_{2,δ_N} are the same as before and follow from Lemma 3.1 and 3.12. Here it is worth to mention that the stated estimates are easily adapted to the case of three close particles instead of two - only a constant factor is introduced by this. The function F arises when applying Gronwall's Lemma. Indeed, if one considers $\eta(t)$ as in the previous proofs, then one obtains

$$\frac{d}{dt}\eta(t) \lesssim (S_{2,\delta_N(t)}(t) + 1)\eta(t) + \rho_N(t)[G_{\delta_N(t), C_N^{(4)}(t)}](N^{-1}d_{min}(t)^{-1} + N^{-2/3})^p.$$

Next, we remark that $\rho_N(t)[G_{\delta_N(t), C_N^{(4)}(t)}] \leq \rho_N^0[G_{\delta_N, C_N^{(4)}}]$ by definition of $\delta_N(t)$ and assumption (A3'). Next, by using (BT2') and ($\mathrm{BT_{new}}$) we end up with

$$\frac{d}{dt}\eta(t) \lesssim (L + 1)\eta(t) + \rho_N^0[G_{\delta_N, C_N^{(4)}}](N^{-1}d_{min}(0)^{-1} + N^{-2/3})^p g(t)^{-p}$$

with $g(t) = \frac{1}{2}f(t) \wedge e^{-Ct}$. Integrating this inequality implies the stated estimate on the Wasserstein distance. The bounds on the minimal distance are essentially the same as in the proof of [HS25, Thm. 1.1] or

Lemma 3.9 in most of the cases. The only novel case is the one, when considering two particles being part of an exceptional trio. Up to renumeration, we assume these particles are X_1, X_2 and X_3. Then,

$$\frac{d}{dt}X_1(t) = \frac{1}{N}\left(K(X_1 - X_2) + K(X_1 - X_3)\right) + \underbrace{\frac{1}{N}\sum_{k=4}^{N} K(X_1 - X_k)}_{=:u_\infty(t,X_1)}.$$

In this scenario, $u_\infty(t,\cdot)$ is divergence-free and satisfies for $0 \le t \le T \wedge T^*(N)$

$$|u_\infty(t,X_1) - u_\infty(t,X_1)| \le 2S_{2,\delta_N(t)}|X_1(t) - X_2(t)| \le L|X_1(t) - X_2(t)|,$$

in other words $u_\infty(t,\cdot) \in \mathcal{U}((X_1(t), X_2(t), X_3(t)), L)$. By assumption ($\mathrm{A_L}$), we conclude $d_{12}(t) \ge f(t)d_{123}(0)$. Combining all these steps allows to finish the proof. □

4.2 Behaviour of Three Particles

As pointed out in the previous discussion, understanding the behaviour of trios of sedimenting particles is essential for extending the results to more general settings. Accordingly, this section is devoted to the study of this behaviour. We start with the particles' movement in the undisturbed scenario, where we identify both non-critical and critical particle constellations with regards to the preservation of distance. Based on this, we illustrate that the additional influence of an external field might destroy the essential preservation of the minimal distance. Finally, we end with a discussion of some invariant and why this invariant is too weak to obtain good control of the distances. However, one needs to dement any notion of sufficiency for these results, as they will not combine to a result justifying that there exists a large set of constellations fulfilling ($\mathrm{A_L}$).

In the remainder of this chapter, we will mostly consider the point particle approximation given by

$$\dot{X}_i = \tfrac{1}{6\pi RN}g + \tfrac{1}{N}\left(K(X_i - X_j) - K(X_i - X_k)\right) \qquad \text{for } i \ne j \ne k \ne i$$

and we set $g = (0,0,-1)$. In this scenario, it is quite natural to describe the behaviour of the particles with respect to a fixed reference particle, say X_1. By defining $y = X_2 - X_1$ and $z = X_3 - X_1$, we can make use of the symmetry of the kernel and obtain

$$\begin{aligned}\dot{y} &= \dot{X}_2 - \dot{X}_1\\ &= \tfrac{1}{N}\left(K(X_2 - X_1) + K(X_2 - X_3) - K(X_1 - X_2) - K(X_1 - X_3)\right)\\ &= \tfrac{1}{N}\left(K(z - y) - K(z)\right)\end{aligned}$$

and thus the dynamics simplify to

$$\begin{cases} \dot{y} = \frac{1}{N}(K(z-y) - K(z)),\\ \dot{z} = \frac{1}{N}(K(z-y) - K(y)).\end{cases} \tag{4.1}$$

4.2.1 Undisturbed Case

We start our discussion with a result justifying some kind of preservation of the minimal distance for special initial constellations. The idea is to relate the vertical velocity of a falling pair of close particles with the horizontal velocity and use this to show that necessary horizontal movement already implies excessive vertical displacement.

In order to describe this, we consider $\Pi : \mathbb{R}^3 \to \mathbb{R}^3$ the orthogonal projection onto the x_1x_2-plane. Then the result states as follows.

Proposition 4.1. *Given $a > 5$, $c \in (0,1)$ and consider a sequence of three-particle configurations with mass $\frac{1}{N}$ satisfying*

$$\frac{|z_0 - y_0|}{|z_0|} \to 0 \quad \textit{and} \quad \frac{|z_0 - y_0|}{|y_0|} \to 0, \tag{4.2}$$

$$\tfrac{1}{N}\left(|y_0|^{-2} + |z_0|^{-2}\right) \to 0, \tag{4.3}$$

$$|\Pi(y_0)| > b|y_0| \quad \textit{and} \quad |\Pi(z_0)| > b|z_0|, \tag{4.4}$$

with $b \in (0,1)$ fulfilling

$$2\left(\frac{a-2}{a+1}\right)(b-c) - (1-b^2)^{1/2} > 0. \tag{4.5}$$

Then, for all $T > 0$ there exists $N(T) \in \mathbb{N}$, s.t. for $N > N(T)$

$$d_{min}(t) \geq \tfrac{c}{2} d_{min}(0) \quad \textit{for all } 0 \leq t \leq T.$$

Remark 4.4.

- Equation (4.5) encodes that z and y are initially placed outside some truncated cone. In this case, the third particle at the origin will be sufficiently separated from every possible trajectory of the pair formed by z and y.

Proof. Again, the proof will be based on a bootstrap and continuity argument. We start with defining the setup. Given N we set $T^*(N)$ to be the maximal time such that

$$a|z(t) - y(t)| \leq |z(t)| \quad \text{and} \quad a|y(t) - z(t)| \leq |y(t)| \tag{b1}$$

$$|y(t)| \geq \tfrac{c}{2}|y_0|, \quad |z(t)| \geq \tfrac{c}{2}|z_0| \quad \text{and} \quad |z(t) - y(t)| \geq \tfrac{c}{2}|z_0 - y_0| \tag{b2}$$

for all $0 \leq t \leq T^*(N)$.

Consider now $T > 0$ arbitrary, but fixed, then for $t \leq T \wedge T^*(N)$, it holds

$$\frac{d}{dt}|y - z|^2(t) = \frac{2}{N}\left(y(t) - z(t)\right)^T \left(K(y) - K(z)\right) \begin{cases} \lesssim \frac{1}{N}\left(|y(t)|^{-2} + |z(t)|^{-2}\right)|y(t) - z(t)|^2, \\ \gtrsim \frac{-1}{N}\left(|y(t)|^{-2} + |z(t)|^{-2}\right)|y(t) - z(t)|^2. \end{cases}$$

Exploiting (b2) for y and z, we conclude by Gronwall's Lemma

$$\tfrac{c}{2}|y_0 - z_0| < |y_0 - z_0| e^{-\frac{C}{c^2 N}\left(|y_0|^{-2} + |z_0|^{-2}\right)T} \leq |y(t) - z(t)| \leq |y_0 - z_0| e^{C\frac{C}{c^2 N}\left(|y_0|^{-2} + |z_0|^{-2}\right)T} < 2c^{-1}|y_0 - z_0|$$

for N large enough by assumption (4.3). Here, the constant C is generic and independent of the actual particle configuration. Moreover, whenever (b2) holds true, then for N large enough, we have (using the foregoing)

$$a|z(t) - y(t)| \leq a2c^{-1}\,|y_0 - z_0| \leq 2ac^{-1}\,\frac{|y_0 - z_0|}{|y_0|}|y_0| \leq 4ac^{-2}\,\frac{|y_0 - z_0|}{|y_0|}|y(t)| < |y(t)|$$

as the prefactor tends to zero by (4.2). Summarizing, the only critical part might be proving that

$$|y(t)| > \tfrac{c}{2}|y_0| \quad \text{for all } 0 \leq t \leq T \wedge T^*(N)$$

and N large enough (and the same for z). To do so, we argue by contradiction and assume there is $t_0 \leq T \wedge T^*(N)$ with $|y(t_0)| = \frac{c}{2}|y_0|$. Then, for some $\lambda \in [0,1]$ it holds $|g^T y(t_0)| = (1-\lambda^2)^{1/2}\frac{c}{2}|y_0|$ and $|\Pi(y(t_0))| = \lambda\frac{c}{2}|y_0|$. In this case, we already know that y has performed an horizontal change of at least $(b-\lambda\frac{c}{2})|y_0|$. But we will see that this already enforces a vertical movement of y resulting in $y(t_0)\notin B_{\frac{c}{2}|y_0|}(0)$. First, we remark that the velocity in g-direction satisfies

$$\begin{aligned}\frac{d}{dt}g^T y(t) &= \frac{1}{8\pi N}\left(\frac{1}{|y(t)-z(t)|}+\frac{(g^T(y(t)-z(t)))^2}{|y(t)-z(t)|^3}-\frac{1}{|z(t)|}-\frac{(g^T z(t))^2}{|z(t)|^3}\right)\\ &\geq \frac{1}{8\pi N}\left(\frac{1}{|y(t)-z(t)|}-\frac{2}{|z(t)|}\right)>0\end{aligned}$$

as $a>2$. On the other hand, the horizontal velocity is given by

$$\begin{aligned}\Pi(\tfrac{d}{dt}y(t)) &= \frac{1}{8\pi N}\left(K(z(t)-y(t))-g^T K(z(t)-y(t))g-\left(K(z(t))-g^T K(z(t))g\right)\right)\\ &= \frac{1}{8\pi N}\left(\frac{g^T(z(t)-y(t))\left((z(t)-y(t))-g^T(z(t)-y(t))g\right)}{|z(t)-y(t)|^3}-\frac{g^T z(t)\left(z(t)-g^T z(t)g\right)}{|z(t)|^3}\right).\end{aligned}$$

Noticing that for $w\in\mathbb{R}^3$ and ϕ the angle between w and g it holds

$$|g^T w(w-g^T wg)|\leq|g^T w||\Pi(w)|=|\sin(\phi)||w||\cos(\phi)||w|\leq\tfrac{1}{2}|w|^2,$$

we deduce

$$|\Pi(\tfrac{d}{dt}y(t))|\ \leq\frac{1}{16\pi N}\left(\frac{1}{|y(t)-z(t)|}+\frac{1}{|z(t)|}\right).$$

In consequence, in the regime enforced by (b1) and (b2), one obtains

$$\frac{|\Pi(\frac{d}{dt}y(t))|}{\frac{d}{dt}g^T y(t)}\leq\frac{1}{2}\left(\frac{|z(t)|+|y(t)-z(t)|}{|z(t)|-2|y(t)-z(t)|}\right)\leq\frac{1}{2}\left(\frac{(1+a^{-1})|z(t)|}{(1-2a^{-1})|z(t)|}\right)=\frac{1}{2}\left(\frac{a+1}{a-2}\right).$$

Based on this, we conclude

$$\begin{aligned}(b-\lambda\tfrac{c}{2})|y_0|\leq\Big|\int_0^{t_0}\Pi(\tfrac{d}{dt}y(s))ds\Big|&\leq\int_0^{t_0}|\Pi(\tfrac{d}{dt}y(s))|ds\\ &\leq\frac{1}{2}\left(\frac{a+1}{a-2}\right)\int_0^{t_0}\tfrac{d}{dt}g^T y(s)ds=\frac{1}{2}\left(\frac{a+1}{a-2}\right)(g^T y(t_0)-g^T y_0)\end{aligned}$$

or equivalently using $|g^T y_0|\leq(1-b^2)^{1/2}|y_0|$ from (4.4)

$$\left(2\left(\frac{a-2}{a+1}\right)(b-\lambda\tfrac{c}{2})-(1-b^2)^{1/2}\right)|y_0|\leq 2\left(\frac{a-2}{a+1}\right)(b-\lambda\tfrac{c}{2})|y_0|+g^T y_0\leq g^T y(t_0). \tag{4.6}$$

By (4.5) and $a>5$, we deduce

$$2\left(\frac{a-2}{a+1}\right)(b-\lambda\tfrac{c}{2})-(1-b^2)^{1/2}>2\left(\frac{a-2}{a+1}\right)(2-\lambda)\tfrac{c}{2}>(2-\lambda)\tfrac{c}{2}\geq(1-\lambda^2)^{1/2}\tfrac{c}{2}.$$

But then we obtain by (4.6)

$$g^T y(t_0)>(1-\lambda^2)^{1/2}\tfrac{c}{2}|y_0|=|g^T y(t_0)|.$$

This contradiction implies $|y(t)|>\frac{c}{2}|y_0|$ for all $0\leq t\leq T\wedge T^*(N)$. Now, a standard continuity argument allows us to deduce $T<T^*(N)$ for N large enough. □

The previous proposition imposes some kind of separation assumption on the particle constellation and the trajectories. By ensuring that the third particle is far away from any possible trajectory of the remaining two particles, we prevent any significant change in the particles roles and behaviours, analogously to assumption (A4) but in a less coarse manner. If the pair of close particles is placed outside the truncated cone, it will stay a pair of close particles on the relevant time scale.

However, if we place the pair of particles inside the cone, more interesting phenomena might arise. In this case, the two fast particles might catch up to the third and several effects are imaginable: (i) the roles of the particles mix and a new pair of close particles is formed and escapes to infinity, (ii) the particles perform a periodic movement with repeating change of roles (see [CLLS88]), or (iii) the particles might collide in finite time. Here, scenario (iii) would be the most critical. In the ongoing discussion, we demonstrate that there are constellations with such a behaviour.

Collision in the symmetric scenario

In order to make the calculations analytically tractable, we will restrict to the case of three particles arranged in a vertical plane symmetrically around $g = (0, 0, -1)$. We will see that no equilibrium state exists and either the pair of particles escapes to infinity or the particles collide in finite time. For this, it will prove to be helpful to express the relative positions in spherical coordinates for some $\phi \in (0, \pi)$ and $r \geq 0$, meaning

$$y = (0, -r\sin(\phi), r\cos(\phi)) \quad \text{and} \quad z = (0, r\sin(\phi), r\cos(\phi)).$$

Accordingly, after performing a time scaling to remove the prefactor $\frac{1}{8\pi}$ the system (4.1) can be expressed by

$$\dot{y} = \frac{1}{N}\begin{pmatrix} 0 \\ \frac{1}{r}\cos(\phi)\sin(\phi) \\ \frac{1}{2r\sin(\phi)}(-1 + 2\sin(\phi) + 2\sin(\phi)\cos(\phi)^2) \end{pmatrix},$$

$$\dot{z} = \frac{1}{N}\begin{pmatrix} 0 \\ -\frac{1}{r}\cos(\phi)\sin(\phi) \\ \frac{1}{2r\sin(\phi)}(-1 + 2\sin(\phi) + 2\sin(\phi)\cos(\phi)^2) \end{pmatrix}.$$

Consequently, we are interested in the temporal behaviour of r^2 and ϕ. By using $r^2 = z^T z$, we derive

$$\begin{aligned} \frac{d}{dt}r^2 &= 2z^T\left(\tfrac{d}{dt}z\right) \\ &= \tfrac{2}{N}\left[-\cos(\phi)\sin(\phi)^2 + \tfrac{\cos(\phi)}{2\sin(\phi)}(-1 + 2\sin(\phi) + 2\sin(\phi)\cos(\phi)^2)\right] \\ &= \tfrac{1}{N\sin(\phi)}\left[-2\cos(\phi)\sin(\phi)^3 - \cos(\phi) + 2\sin(\phi)\cos(\phi) + 2\sin(\phi)\cos(\phi)^3\right] \\ &= \tfrac{\cos(\phi)}{N\sin(\phi)}\left[-1 + 4\sin(\phi)\cos(\phi)^2\right] \\ &= \tfrac{\cos(\phi)}{N\sin(\phi)}\left[-1 + 4\sin(\phi) - 4\sin(\phi)^3\right]. \end{aligned} \tag{4.7}$$

Moreover, by $\phi = \operatorname{arccotan}(\frac{z_3}{z_2})$ it holds

$$\begin{aligned} \frac{d}{dt}\phi &= -\tfrac{z_2^2}{z_2^2+z_3^2}\left[\tfrac{\dot{z}_3 z_2 - \dot{z}_2 z_3}{z_2^2}\right] \\ &= \tfrac{-1}{Nr^2}\left[(-\tfrac{1}{2} + \sin(\phi) + \sin(\phi)\cos(\phi)^2) + \cos(\phi)^2\sin(\phi)\right] \\ &= \tfrac{1}{2Nr^2}\left[1 - 2\sin(\phi) - 4\sin(\phi)\cos(\phi)^2\right] \\ &= \tfrac{1}{2Nr^2}\left[1 - 6\sin(\phi) + 4\sin(\phi)^3\right]. \end{aligned} \tag{4.8}$$

For the further analysis, we introduce the notation

$$F(T) = -1 + 4T - 4T^3, \qquad f(\phi) = F(\sin(\phi)),$$
$$G(T) = 1 - 6T + 4T^3, \qquad g(\phi) = G(\sin(\phi)).$$

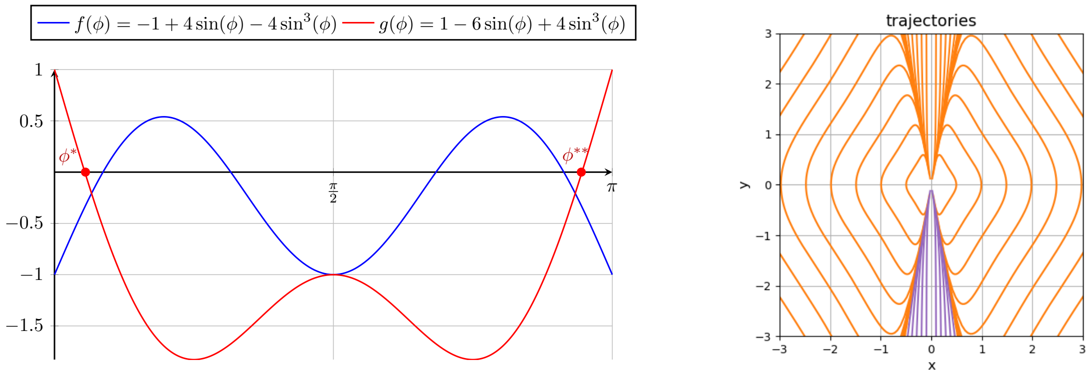

Figure 1: LEFT: plot of the functions f and g on the interval $[0, \pi]$, RIGHT: plot of the trajectories of the two particles around the third. The two particles follow symmetrically the drawn lines. The right particle is travelling counter-clockwise. The violet cone is indicating the trajectories in the case of the close pair escaping to infinity.

According to this, given $r_0 > 0, \phi_0 \in (0, \pi)$ we are interested in solutions to the dynamical system

$$\begin{cases} \frac{d}{dt}\phi(t) = \frac{1}{2Nr^2(t)} g(\phi(t)), \\ \frac{d}{dt}r^2(t) = \frac{\cos(\phi(t))}{N\sin(\phi(t))} f(\phi(t)), \\ r(0) = r_0, \phi(0) = \phi_0, \end{cases}$$

for $0 \le t < T_c := \sup\{t > 0 \mid r^2(t) > 0\}$ with T_c the first collison time, when the system becomes ill-posed.

Combining (4.7) and (4.8) and observing that $\frac{d}{dt}\phi(t) \neq 0$, whenever $g(\phi_0) \neq 0$, we obtain a differential relation between r^2 and ϕ along trajectories, given by

$$\frac{dr^2}{d\phi} = 2\frac{\cos(\phi)f(\phi)}{\sin(\phi)g(\phi)}r^2. \tag{4.9}$$

It is worth noticing that the previous equation suggests that the trajectories are independent of the current mass of the particles, just the travelling velocity depends on the mass.

Before starting the analysis of the dynamics of ϕ and r^2, we mention a last important property. Since G has only one zero $x^* \approx 0.169938$ in $[-1, 1]$, there exist only two steady angles $\phi^* = \arcsin(x^*) \in [0, \frac{\pi}{2}]$ and $\phi^{**} = \pi - \phi^*$. We recommend having a look at the plots of f and g in Figure 1 as the ongoing analysis is highly depending on these functions. We start with observing the following monotonicity of the angle.

Lemma 4.1. *Given some initial angle ϕ_0, the following monotonicity statements hold true:*

1. *If $0 < \phi_0 \le \phi^*$, then $\frac{d}{dt}\phi(t) \ge 0$ for $0 < t < T_c$.*
2. *If $\phi^* \le \phi_0 \le \phi^{**}$, then $\frac{d}{dt}\phi(t) \le 0$ for $0 < t < T_c$.*

Proof. Observing that

$$g(\phi) \quad \begin{cases} \ge 0 & \text{for } \phi \in (0, \phi^*], \\ \le 0 & \text{for } \phi \in [\phi^*, \phi^{**}], \end{cases}$$

and that $\phi(t)$ will never cross ϕ^*, we conclude the claim as $\frac{d}{dt}\phi(t) = \frac{1}{2Nr^2}g(\phi)$. □

Lemma 4.2. *There exist an $\varepsilon^* > 0$ and $a^* > 0$, such that*

$$g(\phi^* + \varepsilon^*) > -1 \quad \textit{and} \quad \frac{d}{dt}r^2(t) \le -\frac{1}{N}a^* \quad \forall t \ge 0,$$

whenever $\phi_0 \in (0, \phi^* + \varepsilon^*]$.

Proof. Consider $h(\phi) = \frac{\cos(\phi)}{\sin(\phi)} f(\phi)$. The whole statement is based on the observation that $h(\phi^*) < 0$. Setting $a^* = -\frac{1}{2} h(\phi^*)$ and using the continuity of h, we deduce the existence of $\varepsilon^* > 0$ with $h(\phi^* + \delta) < -a^*$ for all $\delta \in (0, \varepsilon^*)$. Together with $h(\phi) \leq h(\phi^*)$ for $\phi \in (0, \phi^*)$ and $\frac{d}{dt} r^2(t) = \frac{1}{N} h(\phi)$, the claim follows by applying Lemma 4.1. □

Corollary 4.1. *Assume $\phi_0 \in (0, \phi^* + \varepsilon)$, then*

$$r^2(t) \leq r_0^2 - \tfrac{1}{N} a^* t \quad \forall t \in (0, T_c).$$

Moreover, $T_c \leq N \frac{r_0^2}{a^*}$.

Next, we want to investigate whether or not the region $(0, \phi^* + \varepsilon^*)$ can be reached in finite time.

Lemma 4.3. *There exist constants $b^*, c^* > 0$ depending only on ε^*, such that for all $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$*

$$\min\{t > 0 \mid \phi(t) \in (0, \phi^* + \varepsilon]\} \leq b^* N r_0^2$$

and

$$r^2(t) \leq c^* r_0^2.$$

Proof. In the spirit of Lemma 4.1 we only need to consider $\phi_0 \in (\phi^* + \varepsilon^*, \phi^{**} - \varepsilon^*)$. To utilize the ODE for ϕ in (4.8), we need to upperbound $r^2(t)$ first. To do so, we make use of the relation between r^2 and ϕ given by (4.9). As we just need to consider the time interval $[0, T]$ with $T = \min\{t > 0 \mid \phi(t) \in [0, \phi^* + \varepsilon^*]\}$ we are interested in bounding the terms in (4.9) only for $\phi \in (\phi^* + \varepsilon^*, \phi^{**} - \varepsilon^*)$. On this interval, we have

$$|g(\phi)| \geq |g(\phi^* + \varepsilon^*)| > 0, \quad \sin(\phi) > \sin(\phi^*) > 0, \quad |f(\phi)| \leq 9$$

and thus we can find a constant $c > 0$ with

$$-cr^2 \leq \tfrac{d}{d\phi} r^2 = 2r^2 \frac{\cos(\phi) f(\phi)}{\sin(\phi) g(\phi)} \leq cr^2.$$

By Gronwall's Lemma, we obtain

$$r^2(t) = r^2(\phi(t)) \leq r_0^2 e^{c|\phi(t) - \phi_0|} \leq r_0^2 e^{c|\phi^{**} - \phi^*|} =: c^* r_0^2.$$

Now, for $t \in [0, T]$ it holds $g(\phi(t)) < g(\phi^* + \varepsilon) < 0$ and so

$$\frac{d}{dt}\phi(t) = \tfrac{1}{2Nr^2(t)} g(\phi(t)) \leq \tfrac{1}{2Nc^* r_0^2} g(\phi^*) =: -\tfrac{1}{Nr_0^2} b.$$

Consequently, for $t \in [0, T]$

$$\phi(t) \leq \phi_0 - b \tfrac{t}{Nr_0^2}$$

and so

$$T \leq \frac{\phi_0 - \phi^* - \varepsilon^*}{b} N r_0^2 \leq \frac{\phi^{**} - \phi^* - 2\varepsilon^*}{b} N r_0^2 =: b^* N r_0^2,$$

which proves the claim. □

Proposition 4.2. *Given a sequence of three-particle configurations such that $r_0 = O(N^{-1/2})$ and $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$. Then, the first collision time is bounded uniformly.*

Proof. By assumption, we know that $r_0^2 < cN^{-1}$ for some $c > 0$. Now, for any $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$, by Lemma 4.3 the angle $\phi(t)$ reaches $[0, \phi^* + \varepsilon^*]$ for some $t < b^* N r_0^2$ and $r^2(t) < c^* r_0^2$.
From this point on, the time span till first collision is bounded by $N \frac{r^2(t)}{a^*}$ according to Corollary 4.1. Thus, the overall time until the first collision is bounded by

$$b^* N r_0^2 + N \tfrac{r^2(t)}{a^*} \leq (b^* + \tfrac{c^*}{a^*}) c N^{-1} N \leq (b^* + \tfrac{c^*}{a^*}) c.$$

□

The previous proof does not directly extend to the case, where $d_{min,1}(0) = r(0) \in \omega(N^{-1/2})$, but still $N d_{min}(0) d_{min,1}(0) \sim 1$. To apply the previous result, one needs to ensure that for some fixed constant C a state satisfying $r(t) < C N^{-1/2}$ will be reached before some $T > 0$ with T independent of N. This will be part of the following lemma.

Lemma 4.4. *Given a sequence of three-particle configurations satisfying $d_{min}(0)d_{min,1}(0) \sim N^{-1}$ and $d_{min}(0)/d_{min,1}(0) \to 0$ as well as $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$. Then there exists a constant $C_* > 0$ and $T^* > 0$ such that*

$$\min\{t > 0 \mid r(t) \leq C_* d_{min}(0)\} < T^*.$$

Proof. Due to $d_{min}(0)/d_{min,1}(0) \to 0$ and $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$, we can assume $\phi_0 \in (0, \phi^*)$. To include the finitely many exceptions, one simply applies the previous result in these cases and takes the maximum over all generated time points. By assumption there is $\lambda > 0$ with

$$\lambda^{-1} d_{min}(0) d_{min,1}(0) < N^{-1} < \lambda d_{min}(0) d_{min,1}(0).$$

Set $I = [\frac{1}{2}\phi^*, \phi^*]$ and $T = \min\{t \geq 0 \mid \phi(t) \in I\}$. For $t < T$ it holds

$$\frac{d}{dt}\phi(t) = \frac{g(\phi(t))}{2Nr^2(t)} \geq \frac{g(\frac{1}{2}\phi^*)}{2Nr^2(t)} =: \frac{c}{Nr^2(t)} \tag{4.10}$$

and for $0 \leq \phi \leq \frac{1}{2}\phi^*$

$$\tfrac{d}{d\phi} r^2 = 2r^2 \frac{\cos(\phi)}{\sin(\phi)} \underbrace{\frac{f(\phi)}{g(\phi)}}_{\leq -1} \leq -2r^2 \frac{\cos(\phi)}{\sin(\phi)}.$$

Hence,

$$\log(r^2(t)) - \log(r_0^2) \leq -2\left(\log(\sin(\phi(t))) - \log(\sin(\phi_0))\right)$$

and by using $\sin(\phi_0) = \frac{d_{min}(0)}{2r_0}$, we end up with

$$r^2(t) \leq r_0^2 \left(\frac{\sin(\phi_0)}{\sin(\phi(t))}\right)^2 \leq \tfrac{1}{4} d_{min}^2(0) \sin(\phi(t))^{-2}. \tag{4.11}$$

Inserting (4.11) into (4.10) and using $\sin(\phi) \geq \frac{1}{2}\phi$ on $[0, \frac{1}{2}\phi^*]$, we obtain

$$\frac{d}{dt}\phi(t) \geq \tfrac{c}{N d_{min}^2(0)} \sin(\phi(t))^2 \geq \tfrac{c}{4N d_{min}^2(0)} \phi^2(t).$$

This implies

$$\phi_0^{-1} - \phi(t)^{-1} \ \geq \tfrac{c}{4N d_{min}^2(0)} t \quad \text{for } t \leq T,$$

or equivalently

$$\phi(t) \geq (\phi_0^{-1} - \tfrac{c}{4N d_{min}^2(0)} t)^{-1}.$$

Observing that $\phi_0^{-1} \leq C \frac{r_0}{d_{min}(0)}$, we obtain

$$0 \leq \phi_0^{-1} - \tfrac{c}{4N d_{min}^2(0)} t \leq C \tfrac{r_0}{d_{min}(0)} - \lambda d_{min}(0) r_0 \tfrac{c}{4 d_{min}^2(0)} t \leq \tfrac{r_0}{d_{min}(0)} (C - \tfrac{\lambda c}{4} t),$$

which leads to $T < \frac{4C}{\lambda c} =: T^*$. By (4.11), we conclude for T

$$r(T) \leq \frac{1}{2\sin(\frac{1}{2}\phi^*)} d_{min}(0) =: C_* d_{min}(0).$$

This proves the lemma. □

Corollary 4.2. *Given a sequence of three-particle configurations fulfilling* $N d_{min}(0) d_{min,1}(0) \sim 1$, $d_{min}(0)/d_{min,1}(0) \to 0$ *and* $\phi_0 \in (0, \phi^{**} - \varepsilon^*)$, *then the first collision time is bounded uniformly.*

Proof. Under the assumptions it holds $d_{min}(0) = O(N^{-1/2})$. By Lemma 4.4 after finite time, one reaches a state, for which Proposition 4.2 can be applied and so the claim follows. □

Remark 4.5.

- An analysis of the movement of three sedimenting spheres placed in an isosceles triangle with horizontal base can be found in [CLLS88] and [Hoc64]. There it was proven that the horizontal equilateral triangle forms a locally stable steady state. Furthermore, it was shown that a periodic movement of the particles can be observed, if the base particles are close enough and not too seperated from the third particle. In this case, the base particles overtake the third particle and are then separated from each other by the flow below the third particle. The third particle catches up with the base particles again and the distance between the base particles decreases due to the flow above the third particle. From this point on the movement repeats. In addition, in [CLLS88] it is mentioned that numerical studies suggest these oscillations to be rather stable under small perturbations destroying the symmetry.

- Combining these results with the foregoing discussion we can conclude that in the symmetric case there are at least three possible behaviours: (i) the pair of close base particles escapes to infinity, (ii) the particles perform a periodic movement, and (iii) the base particles collide in finite time with the third particle. Here, (iii) can be interpreted as a degeneration of the periodic movement in case (ii).

4.2.2 Disturbed Case

Recapitulating the previous results, we must admit that the dynamics of three particles cannot be characterized easily. Even in the undisturbed case there are configurations, whose symmetry prevent any essential preservation of the minimal distance. However, one might argue that these symmetric constellations are highly unlikely and thus excluding them by some additional assumption would not imply a significant loss of generality. But as we need stability even under the disturbance of some external field - like formulated in Theorem 4.1 - the class of critical situations that can be transformed into such a symmetric constellation might be even larger.

Let us ignore for a moment the influence of gravity and assume that the particles are purely driven by an external field $u(t) \in \hat{\mathcal{U}}(X(t), L)$. In this case, the particle system is fully controllable in the sense that given some continuous trajectory, there exists an external field producing this trajectory. Indeed, for any $v_1, v_2, v_3 \in \mathbb{R}^3$ we find some $\lambda > 0$ such that $\lambda |v_i - v_j| \leq L |X_i - X_j|$. Now, there exists an affine measure preserving linear map A with $A(X_i) = \lambda v_i$ and so we can enforce particle X_i to move in direction v_i.

This controllability might get lost by the strong interaction between the particles in critical constellations. Since $|K(y-z) - K(y)| \leq \frac{1}{N}(|z-y|^{-1}|z|^{-1} + |y|^{-1}|z|^{-1})|z| \sim |z|$ in this scenario, it is in general not possible to completely eliminate the internal dynamics by the external field. Accordingly, the task of describing the set of well-behaved constellations becomes more subtle and we are not in the position to prove any complete characterization.

However, we are going to illustrate that there exist constellations, which are essentially distance preserving in the undisturbed setting, but can be transformed into a critical symmetric constellation in finite time.

Proposition 4.3. *Consider a sequence of three-particle configurations with mass $\frac{1}{N}$ satisfying*

$$|z_0| = |y_0| = d_{min,1} \quad \text{and} \quad g^T(z_0 - y_0) = 0, \tag{4.12}$$
$$\lambda < N|z_0||z_0 - y_0| < \Lambda \quad \text{for some } \lambda, \Lambda > 0, \tag{4.13}$$
$$\frac{1}{N}|z_0|^{-2} \to 0, \tag{4.14}$$
$$-g^T z_0 \geq c|z_0|, \tag{4.15}$$

where c fulfills $\frac{9}{80}\lambda c^2 > (1-c^2)^{1/2}$.
Then, there exists some $T > 0$ s.t. for N large enough there is $A \in \mathbb{R}^{3\times 3}$ with $\|A\| \leq 1$ and $\operatorname{tr}(A) = 0$ s.t. for some $0 \leq t_0 < T$ it holds $-g^T z^A(t_0) > 0$ and $|z^A(t_0) - y^A(t_0)| \geq \frac{9}{10}|z_0 - y_0|$ as well as $|z^A(t_0)| = |y^A(t_0)| \geq \frac{1}{2}c|z_0|$ and $z(t_0), y(t_0)$ lie in a plane perpendicular to the x_1x_2-plane. Here, z^A and y^A refer to the dynamics disturbed by $u_\infty(x) = Ax$, i.e.

$$\begin{cases} \frac{d}{dt}y^A = \frac{1}{N}\left(K(z^A - y^A) - K(z^A)\right) + Ay^A, \\ \frac{d}{dt}z^A = \frac{1}{N}\left(K(z^A - y^A) - K(y^A)\right) + Az^A. \end{cases}$$

Remark 4.6.

- (4.12) ensures that the particles form an isosceles triangle with horizontal base, (4.15) implies that the base is inside some cone above the particle at the origin. By (4.13) and (4.14) we consider non-equilateral triangle that are critical in context of (A4).
- The values $\frac{9}{80}$ and $\frac{9}{10}$ are chosen more or less arbitrary to slighten the notation.
- If one is interested in checking, whether there are constellations fulfilling the assumptions of Prop. 4.1 and the foregoing proposition, one needs to pay attention on the different definitions of c in this context. Setting $a = 8$ and $c = \frac{1}{2}b$ in Prop. 4.1, one obtains the condition $\frac{2}{3}b > (1-b^2)^{1/2}$. The value c from Prop. 4.3 corresponds to $(1-b^2)^{1/2}$. Hence, to satisfy the assumptions of both propositions the following is needed

$$\frac{2}{3}b > \left(1 - b^2\right)^{1/2}, \tag{4.16}$$
$$\frac{9\lambda}{80}(1 - b^2) > b. \tag{4.17}$$

  Remarking that (4.17) is satisfied at least for all $b \in \left(0, (1 - \frac{80}{9\lambda})^{1/2}\right)$ and (4.16) for $b > 3/(13)^{1/2}$, we can choose λ large enough, such that both conditions are fulfilled. In consequence, there are configurations, which preserve the minimal distance essentially in the undisturbed scenario, but can be transformed via a linear disturbance flow in finite time into a constellation with finite collision time.

Proof. We start with constructing the disturbance field. Recalling that $\Pi : \mathbb{R}^3 \to \mathbb{R}^3$ denotes the projection onto the x_1x_2-plane, we define

$$v := \frac{\Pi(\frac{z_0+y_0}{2})}{|\Pi(\frac{z_0+y_0}{2})|}$$

and consider $A = -vg^T$. This matrix fulfills $\|A\| \leq 1$ and $\operatorname{tr}(A) = -g^T v = 0$. In the following, we will omit the superscript A and just consider $z(t) = z^A(t)$ as well as $y(t) = y^A(t)$. Before beginning with the main work, we remark that the mirror symmetry of the particle constellation is preserved during time by the dynamics. Accordingly, the middle point between z and y is always contained in the plane spanned by v and g. In the assumed scenario, the vertical movement satisfies

$$g^T(\tfrac{d}{dt}y(t)) = \frac{1}{N}\left(\frac{1}{|z(t)-y(t)|} - \frac{1}{|z(t)|} - \frac{(g^T z(t))^2}{|z(t)|^3}\right) \leq \frac{1}{N|y(t)-z(t)|} \tag{4.18}$$

and for the horizontal movement in direction v it holds

$$v^T(\tfrac{d}{dt}y(t)) = -g^T y(t) - \frac{1}{N}\frac{(g^T z(t))^2}{|z(t)|^3} \geq -g^T y(t) - \frac{1}{N|z(t)|}. \tag{4.19}$$

Set $T^*(N) = \sup\{T \geq 0 \mid -g^T y(t) \geq \frac{c}{2}|y_0| \text{ for all } 0 \leq t \leq T\}$ and observe that for $t \leq \min\{\frac{9}{20}c, T^*(N)\}$ it holds

$$\frac{d}{dt}|z(t)-y(t)|^2 \begin{cases} \leq \frac{8}{Nc^2}\left(|y_0|^{-2}+|z_0|^{-2}\right)|z(t)-y(t)|^2, \\ \geq -\frac{8}{Nc^2}\left(|y_0|^{-2}+|z_0|^{-2}\right)|z(t)-y(t)|^2 \end{cases}$$

since $-g^T y(t) \leq |y(t)|$.

So by Gronwall's Lemma, we deduce for $N \geq N_1$ large enough

$$\tfrac{10}{9}|y_0 - z_0| \geq |y_0 - z_0| e^{\frac{16}{Nc^2}|y_0|^{-2}\frac{9}{20}c} \geq |y(t)-z(t)| \geq |y_0 - z_0| e^{-\frac{16}{Nc^2}|y_0|^{-2}\frac{9}{20}c} \geq \tfrac{9}{10}|y_0 - z_0|. \tag{4.20}$$

Next, we want to lower bound $T^*(N)$ for $N \geq N_1$. Therefore, we $-g^T y(t_1) = \frac{c}{2}|y_0|$ for some $t_1 \leq \min\{\frac{9}{20}c, T^*(N)\}$, then the vertical component of y has to change by $\frac{c}{2}|y_0|$. Exploiting the upper bound on the vertical velocity (4.18) and (4.20) we obtain

$$\tfrac{10}{9N|y_0-z_0|}t_1 \geq \tfrac{c}{2}|y_0| \quad \Rightarrow \quad t_1 \geq \tfrac{9}{20}N|y_0||y_0-z_0|c \geq \tfrac{9}{20}\lambda c.$$

Thus, $T^*(N) \geq t_1 \geq \frac{9}{20}\lambda c$. On the other hand, for $t \leq T^*(N)$ we obtain by (4.19)

$$v^T\left(\frac{d}{dt}y(t)\right) \geq -g^T y(t) - \tfrac{|y(t)-z(t)|}{N|z(t)||y(t)-z(t)|} \geq -g^T y(t) - \frac{\frac{10}{9}|z_0-y_0|}{\frac{c}{2}|z_0|\frac{9}{10}|z_0-y_0|N} \geq \tfrac{c}{2}|y_0| - \tfrac{200}{81\lambda c}|z_0-y_0| \geq \tfrac{c}{4}|y_0|$$

for $N > N_2 > N_1$ large enough.

Accordingly, denoting by t_0 the first time satisfying $v^T y(t_0) = 0$ it holds $t_0 \leq \frac{4(1-c^2)^{1/2}}{c} =: t_*$, if $t_* < T^*(N)$. This condition is fulfilled, if

$$\tfrac{9}{20}\lambda c > \tfrac{4(1-c^2)^{1/2}}{c} \;\Leftrightarrow\; \tfrac{9}{80}\lambda c^2 > (1-c^2)^{1/2}$$

Summarizing, after choosing $T = \frac{9}{20}\lambda c$ the claimed statements follow for $N > N_2$. $\square$

4.2.3 Preservation of Triangle Area

Using invariants to describe the behaviour of dynamical systems can be a powerful tool as illustrated in the two-particle scenario. But whereas for two particles the distance between the particles was a helpful invariant, no comparably strong property is known in the three-particle case. However, as it is mentioned in different papers, e.g. [Hoc64] or [CLLS88], the area of the triangle formed by the three particles projected onto the plane perpendicular to g is constant in time, i.e.

$$\frac{d}{dt}g \cdot [(X_2 - X_1) \times (X_3 - X_1)] = 0. \tag{4.21}$$

Accordingly, one is tempted to use this invariant to establish new results on preservation of the minimal distance in the three-particle case, using for example

$$A_\Delta := \tfrac{1}{2}|g \cdot [(X_2 - X_1)\times(X_3 - X_1)]| \leq \frac{1}{2}|X_2 - X_1||X_3 - X_1|.$$

Thus, one would obtain the differential inequality

$$\frac{d}{dt}|X_i - X_j|^2 \geq -4\frac{1}{NA_\Delta}|X_i - X_j|^2,$$

which results directly in an exponential bound for the particle distances and hence excludes collisions in finite time.

At a first glance, such a result seems to give cause for optimism. But there are two main reasons, why we did not include this invariant in the previous discussion.

On the one hand, due to its lower-dimensional character this invariant behaves rather badly in the limit $N \to \infty$. Indeed, for uniformly distributed points one expects the minimal triangle area to scale like $N^{-3/2}$ and thus the constant in the exponential would tend to infinity for N large.

On the other hand, we will illustrate that this invariant is not lifted from the point-particle case to the actual dynamics. In fact, this area preservation is valid for a large class of kernels. To be precise, one easily checks that any kernel of the form $H(x) = f(x)g + k(x)x$ with $f, k : \mathbb{R}^3 \to \mathbb{R}$ and $k(-x) = -k(x)$ will lead to dynamics preserving the projected triangle area. However, as the higher order approximations in the method of reflections do not satisify this structural assumption, the validity of (4.21) will be lost.

To make this argument rigourous, we need to expand the approximation in the method of reflections up to the first reflection and so we are forced to include rotations of the particles into our considerations. The necessary theoretical framework for this case can be found in [NS20] and [Hö23], and we will just reformulate the results for the case of a fixed number of particles with variable radius and distances. In most cases, we will use the same notation as in the previous discussion, but the involved projections P and Q_i are now related to

$$W_i = \{u \in \dot{H}^1_\sigma \mid eu = 0 \text{ in } B_i\} \quad \text{and} \quad W = \cap W_i.$$

As the first order approximation is no longer explicit, we need to introduce several levels of approximation for the real fluid velocity u:

- the abstract approximation $u^{(1)} = (1 - \sum_{i=1}^3 Q_i)u^{(0)}$ in terms of orthogonal projections.
- the semi-explicit approximation $u^{(1)}_{se}$ using just the leading order dipole part of Q_i given in terms of

$$D[S](x) = \begin{cases} Sx & \text{in } B_R, \\ \frac{5}{2}R^3 \frac{xx^T Sx}{|x|^5} + \frac{R^5 Sx}{|x|^5} - \frac{5}{2}R^5 \frac{xx^T Sx}{|x|^7} & \text{else} \end{cases}$$

  with $S = \fint_{B_R(X_i)} eu^{(0)} dx$.
- the explicit approximation $u^{(1)}_e$, where we use the dipole approximation with $S = eu^{(0)}(X_i)$.

By bounding the associated approximation errors, we will be able to use the explicit approximation to deduce the existence of a particle configuration, for which the right-hand side in (4.21) has a sign. We will focus on the scaling behaviour of estimates in both the particle distances and the radius R. To measure the scaling in the particle distance, we consider fixed positions $Y_1, Y_2, Y_3 \in \mathbb{R}^3$ with $Y_1 = 0$ and $Y_2 \nparallel Y_3$ and set for $r > 0$ the actual particle positions $X_i = rY_i$. Furthermore, we assume that R and r are chosen in a way satisfying $B_R(X_i) \cap B_R(X_j) = \emptyset$ for $i \neq j$.

Lemma 4.5. *There exists a constant $C > 0$ independent of r and R, s.t.*

$$\|eu^{(0)}\|_{L^\infty(\cup B_i)} \lesssim r^{-2}.$$

Proof. The single-particle solution satisfies

$$|\nabla u_i(x)| \lesssim |X_k - X_i|^{-2} \lesssim r^{-2} \quad \text{for } x \in B_k, k \neq i.$$

By summing over the three different balls, one obtains the claimed upper bound. □

Lemma 4.6. *For $R^3 r^{-3}$ small enough, it holds*

$$\|u - u^{(1)}\|_{L^\infty} \lesssim (r^{-2}R^3 + R)(R^3 r^{-3})\|eu^0\|_{L^\infty(\cup B_i)}.$$

Proof. The proof is based on an analogous analysis as for the two-particle problem just using the symmetric gradient instead of the gradient itself. We point out which estimates change in the analysis in [NS20, Thm. 4.6] due to the fixed particle number. Denoting by $u^{(k)}$ the k-th approximation step, we get analogously to Lemma 3.4

$$\|eu^{(k+1)}\|_{L^\infty(\cup B_i)} \le C_1 R^3 r^{-3} \|eu^{(k)}\|_{L^\infty(\cup B_i)}$$

and similar arguments as in Prop. 3.1 imply convergence in $\dot{H}^1$ and L^∞ if $C_1 R^3 r^{-3} < 1$. We will even assume $C_1 R^3 r^{-3} < 1/2$ to simplify the later calculations. In addition, one deduces similar to the proof of (3.19)

$$\begin{aligned}\|u^{(k+1)} - u^{(k)}\|_{L^\infty} &\le C r^{-2} R^3 \|\nabla u^{(k)}\|_{L^\infty(\cup B_i)} + CR \|\nabla u^{(k)}\|_{L^\infty(\cup B_i)} \\ &\le C(r^{-2}R^3 + CR)(C_1 R^3 r^{-3})^k \|eu^{(0)}\|_{L^\infty(\cup B_i)},\end{aligned}$$

and thus

$$\begin{aligned}\|u - u^{(1)}\|_{L^\infty} &\le C(r^{-2}R^3 + CR) \sum_{k=1}^{\infty} (C_1 R^3 r^{-3})^k \|eu^{(0)}\|_{L^\infty(\cup B_i)} \\ &\le C(r^{-2}R^3 + CR) 2 C_1 R^3 r^{-3} \|eu^{(0)}\|_{L^\infty(\cup B_i)}.\end{aligned}$$

□

Remark 4.7.

- Using the scaling properties of $\|eu^{(0)}\|_{L^\infty(\cup B_i)}$ one ends up with

$$\|u - u^{(1)}\|_{L^\infty} \lesssim (r^{-2}R^3 + R) R^3 r^{-5} \lesssim R^4 r^{-5} \tag{4.22}$$

  for R small enough.

To simplify the ongoing discussion, we introduce the notation of the dipole operator. Denote by Sym(3) the space of symmetric 3×3-matrices, then we consider

$$D : \mathrm{Sym}(3) \to \dot{H}^1_\sigma \quad \text{given by}$$
$$D[S](x) := \begin{cases} Sx & \text{in } B_R(0), \\ \frac{5}{2} R^3 \frac{x x^T S x}{|x|^5} + R^5 \frac{Sx}{|x|^5} - \frac{5}{2} R^5 \frac{x x^T S x}{|x|^7} & \text{for } x \notin B_R(0), \end{cases}$$

which is the solution operator associated to the problem

$$\begin{cases} -\Delta u + \nabla p = 0 & \text{in } B_R(0), \\ \operatorname{div} u = 0 & \text{in } \mathbb{R}^3, \\ |u| \to 0 & \text{as } |x| \to 0, \\ eu(x) = S & \text{in } B_R(0). \end{cases}$$

This operator clearly satisfies for $S_1, S_2 \in \mathrm{Sym}(3)$, R small and r large

$$|D[S_1](x) - D[S_2](x)| \lesssim R^3 |x|^{-2} \|S_1 - S_2\|. \tag{4.23}$$

According to the more detailed analysis of the projection operators Q_i in [Hö23], $Q_i u$ can be decomposed into two parts, where the leading order part is characterized as a simple dipole and the remaining is denoted as a quadrupole part. The leading order term can be expressed in terms of D, as it is stated in the following lemma borrowed from [HS21, Lem. 2.2].

Lemma 4.7. *Assume $B_{2R}(X_i) \cap B_{2R}(X_j) = \emptyset$, then*

$$|Q_j(u_k)(X_i) - D[\fint_{B_j} eu_k dx](X_i - X_j)| \lesssim \frac{R^{5/2}}{|X_i - X_j|^3} \|eu_k\|_{L^2(B_j)} \lesssim R^4 r^{-5}. \tag{4.24}$$

As the matrix $S = \fint_{B_j} eu_k dx$ cannot be computed easily, we will use the evaluation at the midpoint instead, i.e. $\tilde{S} = eu_k(X_j)$. The produced error can be bounded as follows.

Lemma 4.8. *For R, r^{-1} small enough, it holds*

$$\| \fint_{B_j} eu_k - eu_k(X_j) dx\| \lesssim R r^{-3}.$$

In particular,

$$|D[\fint_{B_j} eu_k dx - eu_k(X_j)]|(x) \lesssim R^4 r^{-3}\ |x|^{-2}. \tag{4.25}$$

Proof. By Taylor's theorem, it holds for $x \in B_R(X_j)$

$$\begin{aligned} \|eu_k(x) - eu_k(X_j)\| &\leq \|\nabla eu_k(X_j)\| |x - X_j| + O(|x - X_j|^2) \\ &\lesssim |X_k - X_j|^{-3} R + O(R^2). \end{aligned}$$

Using (4.23) implies (4.25). □

Summarizing the previous lemmata, we see that the approximation error is of order R^4 for R small. Denoting the explicit approximation by

$$u_e^{(1)}(X_i) = \sum_{j=1}^{3} u_j(X_i) - \sum_{j=1}^{3} \sum_{k \neq j} D[eu_k(X_j)](X_i - X_j),$$

one obtains

Proposition 4.4. *For R small enough, it holds*

$$|u(X_i) - u_e^{(1)}(X_i)| \lesssim R^4 r^{-5}. \tag{4.26}$$

Proof. For R small enough we notice

$$\begin{aligned} |u(X_i) - u_e^{(1)}(X_i)| \leq{}& |u(X_i) - u^{(0)}(X_i) + \sum_{j=1}^{3} \sum_{k \neq j} Q_j(u_k)(X_i)| \\ &+ \sum_{j=1}^{3} \sum_{k \neq j} |Q_j(u_k)(X_i) - D[\fint_{B_j} eu_k dx](X_i - X_j)| \\ &+ \sum_{j=1}^{3} \sum_{k \neq j} |D[\fint_{B_j} eu_k dx] - D[eu_k(X_j)]|(X_i) \\ \lesssim{}& |u(X_i) - u^{(1)}(X_i)| + R^4 r^{-5}\ + R^4 r^{-5} \lesssim R^4 r^{-5} \end{aligned}$$

by (4.22), (4.24) and (4.25). □

Accordingly, if we can prove the existence of a particle configuration for which the leading order term of the explicit dynamics scales like R^3 with a fixed sign, then we can choose R small enough, such that the actual dynamics have the same sign. As we are only interested in the effect of the difference between $u^{(0)}$ and $u_e^{(1)}$ we denote $d(X_i) = u^{(0)} - u_e^{(1)}(X_i) = \sum_{j=1}^{3} \sum_{k \neq j} D[eu_k(X_j)](X_i - X_j)$.

Proposition 4.5. *Assume $X_1, X_2, X_3 \in \mathbb{R}^3$ such that*

$$g \cdot \left[\left(d(X_2) - d(X_1)\right) \times \left(X_3 - X_1\right)\right] + g \cdot \left[\left(X_2 - X_1\right) \times \left(d(X_3) - d(X_1)\right)\right] = AR^3 + O(R^4)$$

for some $A > 0$, then there exists some R^* small enough, such that

$$\frac{d}{dt}\left(g \cdot [(X_2 - X_1) \times (X_3 - X_1)]\right) \neq 0.$$

Proof. For R small enough, it holds for some $C > 0$

$$\begin{aligned}
&\frac{d}{dt} g \cdot [(X_2 - X_1) \times (X_3 - X_1)] \\
&\quad = g \cdot \left[(u(X_2) - u(X_1)) \times (X_3 - X_1) + (X_2 - X_1) \times (u(X_3) - u(X_1))\right] \\
&\quad = g \cdot \left[(u(X_2) - u_e^{(1)}(X_2) - u(X_1) + u_e^{(1)}(X_1)) \times (X_3 - X_1)\right. \\
&\qquad\qquad \left. + (X_2 - X_1) \times (u(X_3) - u_e^{(1)}(X_3) - u(X_1) + u_e^{(1)}(X_1))\right] \\
&\qquad + g \cdot \left[(u_e^{(1)}(X_2) - u_e^{(1)}(X_1)) \times (X_3 - X_1) + (X_2 - X_1) \times (u_e^{(1)}(X_3) - u_e^{(1)}(X_1))\right] \\
&\quad \leq CR^4 - g \cdot \left[\left(d(X_2) - d(X_1)\right) \times \left(X_3 - X_1\right)\right] - g \cdot \left[\left(X_2 - X_1\right) \times \left(d(X_3) - d(X_1)\right)\right] \\
&\quad = CR^4 - AR^3 < 0,
\end{aligned}$$

where we used $g \cdot \left[(u^{(0)}(X_2) - u^{(0)}(X_1)) \times (X_3 - X_1) + (X_2 - X_1) \times (u^{(0)}(X_3) - u^{(0)}(X_1))\right] = 0.$ □

Accordingly, we are left with the task of finding a particle configuration fulfilling the conditions of the previous proposition.

To do so, we first remark that we only need to consider the leading order term of $D[eu_k(X_j)](X_i - X_j)$. Using

$$\begin{aligned}
eu_k(X_j) = &\frac{g^T(X_k - X_j)}{|X_k - X_j|^3} \operatorname{Id} - 3\frac{g^T(X_k - X_j)(X_k - X_j)(X_k - X_j)^T}{|X_k - X_j|^5} \\
&+ \tfrac{R^2}{6}\left[\frac{(X_k - X_j)g^T + g(X_k - X_j)^T}{|X_k - X_j|^5} + \frac{(X_k - X_j)^T g}{|X_k - X_j|^5}\operatorname{Id}\right. \\
&\left. - 5\frac{g^T(X_k - X_j)(X_k - X_j)(X_k - X_j)^T}{|X_k - X_j|^7}\right]
\end{aligned}$$

we can express

$$\begin{aligned}
&D[eu_k(X_j)](X_i - X_j) \\
&\quad = \tfrac{5}{2}R^3\left[\frac{g^T(X_k - X_j)}{|X_i - X_j|^3|X_j - X_k|^3} - 3\frac{\left((X_i - X_j)^T(X_k - X_j)\right)^2 g^T(X_k - X_j)}{|X_i - X_j|^5|X_j - X_k|^5}\right](X_i - X_j) + O(R^4) \\
&\quad =: \tfrac{5}{2}R^3(C_{ijk})(X_i - X_j) + O(R^4),
\end{aligned}$$

where $C_{iik} = C_{ikk} := 0$. Remarking that $C_{iki} = -C_{kik}$ the two relevant terms simplify to

$$\begin{aligned}
&g \cdot \left[(d(X_2) - d(X_1)) \times (X_3 - X_1)\right] \\
&\qquad = \tfrac{5}{2}R^3\left(C_{213} + C_{231} + C_{232} + C_{123}\right)\; g \cdot \left[(X_2 - X_1) \times (X_3 - X_1)\right] + O(R^4)
\end{aligned}$$

and similarly

$$\begin{aligned}
&g \cdot \left[(X_2 - X_1) \times (d(X_3) - d(X_1))\right] \\
&\qquad = \tfrac{5}{2}R^3(C_{312} + C_{321} + C_{231} + C_{213} + C_{312} + C_{132})\; g \cdot [(X_2 - X_1) \times (X_3 - X_1)] + O(R^4).
\end{aligned}$$

Hence, the term of interest is given by

$$\tfrac{5}{2}R^3(C_{123} + C_{321} + C_{231} + C_{213} + C_{312} + C_{132}) g \cdot [(X_2 - X_1) \times (X_3 - X_1)].$$

Lemma 4.9. *For* $X_1 = (0,0,0)$, $X_2 = (1,0,1)$ *and* $X_3 = (3,2,1)$ *it holds*

$$C_{123} + C_{321} + C_{231} + C_{213} + C_{312} + C_{132} > 0.$$

Proof. It holds

$$\begin{aligned} C_{123} &= 0, & C_{132} &= 0, & C_{231} &= \tfrac{395}{351232}\sqrt{7}, \\ C_{213} &= \tfrac{-1}{686}\sqrt{7}, & C_{312} &= \tfrac{5}{2744}\sqrt{7}, & C_{321} &= \tfrac{1}{256}, \end{aligned}$$

and hence

$$C_{123} + C_{321} + C_{231} + C_{213} + C_{312} + C_{132} = \tfrac{523}{351232}\sqrt{7} + \tfrac{1}{256} \approx 0.01 > 0$$

□

Combining Lemma 4.9 with Proposition 4.5 implies the non-preservation of the triangle area in the case of spherical particles with $R > 0$.

Remark 4.8.

- This illustrates that even if one can utilize this invariant in the point-particle case by a more advanced analysis, difficulties will arise when trying to lift the result to the scenario of proper particles. Thus, this result is a warning that not all properties of the approximation will be observable in the actual system.

5 Conclusion

The main finding of this thesis is the observation that the idea presented in [HS25] can be extended to the case of the proper particle dynamics. Based on an adapted initialization modified estimates for the method of reflections were derived, which allow to deduce a mean-field result incorporating more general particle configurations as in [HS21]. However, the presented results still remain suboptimal as the natural scaling behaviour is not included yet.

In the second part of this thesis, one possible approach to further extend the results is illustrated. Building on a black box statement about the behaviour of triples of sedimenting particles, we outlined the proof of a more general result. Although discussing several insights on this behaviour, the introduced results remain on a rudimentary level. In fact, most literature emphasizes that describing the dynamics of a three-body system is significantly more complex than the two-body case, as a wider variety of phenomena may occur. Moreover, it was illustrated that one promising invariant inherits several downsides and thus might not be the desired counterpart to the preservation of distance in the two-particle case. Regarding the three-particle scenario a variety of questions remain unanswered and might be elaborated on in future research.

On the one hand, all the presented examples of configurations with unpreferred behaviour are characterized by a high level of symmetry, which causes important cancellations leading to the observed behaviour. As these symmetric constellations are rather unlikely to occur, it is an interesting question, whether one can quantify the ratio of such ill-behaved constellations, proving that in most of the cases the absence of symmetry and cancellation leads to configurations, where the minimal distance is essentially preserved.

A second thought-provoking aspect is the question, which external disturbance fields can be produced by the random placement of particles. At the moment we assumed all bounded measure-preserving flows to be possible, but maybe the actual set is smaller due to some expected cancellations.

Finally, even if appropiate results are available in the point-particle case, it remains an interesting task to investigate, how to modify the method of reflections to incorporate this behaviour. In addition, as illustrated by the example of the triangle-area, one cannot expect every characteristic of the point-particle approximation to be lifted to the case of proper particles.

In summary, even though this thesis extends existing results, there is still place for possible improvement. To eventually close the gap between the assumed and the expected particle initialization, one needs to further investigate the behaviour of three sedimenting particles. This might be a key to overcome some of the current problems in applying Hauray's method.

A Appendix

In this section proofs and statements are included, which are not directly relevant in the previous discussion. The appendix is structured as follows: The first part is devoted to a well-posedness result needed to handle the case of finite limit velocity in subsection 2.1. In the second subsection we include an overview of necessary inequalities used to prove the convergence of the method of reflections. After discussing the proof of Lemma 3.12, we end with introducing a collection of probabilistic results about the expected scaling behaviour of several quantities. We present lower and upper bounds using the Chen-Stein method for Poisson approximation. The collected results bring rigour to the remarks in the foregoing text.

A.1 Macroscopic Well-Posedness

Lemma A.1. *Let $v_1, v_2 \in \mathbb{R}^3$, consider $\tau_0 \in L^\infty \cap \mathcal{P}$ and $p \in [1, \infty]$. For $i = 1, 2$ let τ_i be the solution to*

$$\begin{cases} \partial_t \tau_i + (K * \tau_i + v_i) \cdot \nabla \tau_i = 0 & \text{for } (t,x) \in [0,\infty) \times \mathbb{R}^3, \\ \tau_i(0) = \tau_0, \end{cases}$$

then there is a constant C depending on $\|\tau_0\|_{L^\infty}$, s.t.

$$\mathcal{W}_p(\tau_1(t), \tau_2(t)) \le |v_1 - v_2| e^{Ct}.$$

Moreover, for $q < 3/2$ and $\Omega \Subset \mathbb{R}^3$, it holds

$$\int_\Omega |K * \tau_1(t) - K * \tau_2(t)|^q dx \le \operatorname{diam}(\Omega)^{3-2q} \mathcal{W}_q(\tau_1(t), \tau_2(t))^q.$$

Proof. Let $Y_i(t, s, \cdot)$ denote the flow map according to $K * \tau_i(t) + v_i$ and consider some optimal transport plan γ_0 between $\tau_1(0)$ and $\tau_2(0)$. We set $\gamma_t = (Y_1(t,0), Y_2(t,0))_\# \gamma_0$ and observe that this is an admissible coupling between $\tau_1(t)$ and $\tau_2(t)$, so

$$\mathcal{W}_p(t)^p := \mathcal{W}_p(\tau_1(t), \tau_2(t))^p \le \int_{\mathbb{R}^3 \times \mathbb{R}^3} |y_1 - y_2|^p d\gamma_t(y_1, y_2) =: \eta(t).$$

Then,

$$\begin{aligned} \frac{d}{dt}\eta(t) \le p \int_{\mathbb{R}^3} & |[K * \tau_1(t)](Y_1(t,0,y_1)) + v_1 - [K * \rho_2(t)](Y_2(t,0,y_2)) - v_2| \\ & |Y_1(t,0,y_1) - Y_2(t,0,y_2)|^{p-1} d\gamma_0(y_1,y_2) \\ \lesssim \int_{\mathbb{R}^3} & \left(|[K * \tau_1(t)](y_1) - [K * \tau_2(t)](y_2)| + |v_1 - v_2| \right) |y_1 - y_2|^{p-1} d\gamma_t(y_1, y_2). \end{aligned}$$

Observing

$$\begin{aligned} |[K * \tau_1(t)](y_1) - [K * \tau_2(t)](y_2)| &\le \int_{\mathbb{R}^3 \times \mathbb{R}^3} |K(y_1 - y_1') - K(y_2 - y_2')| d\gamma_t(y_1', y_2') \\ &\lesssim \int_{\mathbb{R}^3 \times \mathbb{R}^3} \left(|y_1 - y_1'|^{-2} + |y_2 - y_2'|^{-2} \right) \left(|y_1 - y_2| + |y_1' - y_2'| \right) d\gamma_t(y_1', y_2'), \end{aligned}$$

we can bound $\frac{d}{dt}\eta(t)$ analogously to the term $I_{2,2}$ in the proof of Theorem 2.1 via

$$\frac{d}{dt}\eta(t) \lesssim (1 + \|\tau_0\|_\infty)\eta(t) + |v_1 - v_2|^p.$$

By Gronwall's Lemma, we realize

$$\mathcal{W}_p(\tau_1(t), \tau_2(t))^p \le \eta(t) \le C(\tau_0 + |v_1 - v_2|^p) e^{Ct}$$

with C just depending on p and $\|\tau_0\|_{L^\infty}$. Since $\eta(0) = 0$ the claimed estimate on the Wasserstein distance follows.

To deduce the bound on the vector fields, we consider some optimal coupling π between $\tau_1(t)$ and $\tau_2(t)$. Then, it holds

$$\begin{aligned}
\int_\Omega |[K * \tau_1(t)](z) - [K * \tau_2(t)](z)|^q dz &= \int_\Omega | \int_{\mathbb{R}^3\times\mathbb{R}^3} K(z-x) - K(z-y) d\pi(x,y)|^q dz \\
&\leq \int_\Omega \int_{\mathbb{R}^3\times\mathbb{R}^3} |K(z-x) - K(z-y)|^q d\pi(x,y) dz \\
&\lesssim \int_\Omega \int_{\mathbb{R}^3\times\mathbb{R}^3} (|z-x|^{-2q} + |z-y|^{-2q})|y-x|^q d\pi(x,y) dz \\
&= \int_{\mathbb{R}^3\times\mathbb{R}^3} \int_\Omega (|z-x|^{-2q} + |z-y|^{-2q}) dz |y-x|^q d\pi(x,y) \\
&\leq \operatorname{diam}(\Omega)^{3-2q} \mathcal{W}_q(\rho_1(t), \rho_2(t))^q.
\end{aligned}$$

□

A.2 Method of Reflections

In order to prove the necessary inequalities used to deduce the convergence of the method of reflections, we need an auxiliary result about extensions of divergence-free vector fields.

Lemma A.2 ([Hö18, Lem. 3.5]). *For $r > 0$ and $x \in \mathbb{R}^3$, let $H_r = \{u \in \dot{H}^1_\sigma(B_r(x)) \mid \int_{B_r(x)} u dx = 0\}$. Then, for all $r > 0$ there exists an extension operator $E_r : H_r \to \dot{H}^1_{\sigma,0}(B_{2r}(x))$ with $\|\nabla E_r(u)\|_{L^2(B_{2r}(x))} \leq C\|\nabla u\|_{L^2(B_r(x))}$ for all $u \in H_r$. Moreover, the constant $C > 0$ is independent of r.*

Proof. The proof can be found in [Hö18, Lem. 3.5], it is based on a classical result on extension operators and the validity of a Poincaré inequality on H_1. □

This combined with the identification of u_N as the orthogonal projection, allows to prove the following bounds:

Lemma A.3. *For particle configurations $\{X_i\}_{i=1}^N$ with $d_{min} > 4R$, it holds*

$$\|u_N - u_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq CR^3 \sum_{i=1}^N \|\nabla u_N^{(k)}\|^2_{L^\infty(B_i)}.$$

Proof. We define $w_i = u_N^{(k)} - (u_N^{(k)})_i$ on B_i and define $\tilde{w}_i$ as the extension produced by the extension operator described in Lemma A.2. Setting $w = u_N^{(k)} - \sum \tilde{w}_i$, we remark that w is a possible competitor for the projection of $u_N^{(k)}$ on W. Accordingly, we obtain

$$\|u_N - u_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq \|w - u_N^{(k)}\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq \sum_{i=1}^N \|\tilde{w}_i\|^2_{\dot{H}^1(\mathbb{R}^3)} \leq C \sum_{i=1}^N \|\nabla w_i\|^2_{L^2(B_i)} \leq CR^3 \sum_{i=1}^N \|\nabla w_i\|^2_{L^\infty(B_i)}.$$

□

Finally, we recall some properties of the projections Q_i and P_i.

Lemma A.4. *For $w \in \dot{H}^1_\sigma(\mathbb{R}^3)$ it holds*

$$P_i(w) = (w)_i \quad \text{on } B_i$$

with $(w)_i = \fint_{\partial B_i} w \, d\mathcal{H}^2$.

Proof. Let $w \in \dot{H}^1_\sigma$ and consider for arbitrary $\psi_0 \in \mathbb{R}^3$ the solution ψ to

$$\begin{cases} -\Delta\psi + \nabla p = 0 & \text{in } \mathbb{R}^3 \setminus B_i, \\ \psi = \psi_0 & \text{on } B_i. \end{cases}$$

in $\dot{H}^1_\sigma$. This solves

$$-\Delta\psi + \nabla p = \frac{3}{2R}\psi_0 \, d\mathcal{H}^2 \restriction_{\partial B_i} .$$

Thus, since $\psi \in W_i^\perp$, we obtain

$$0 = (w - P_i(w), \psi)_{\dot{H}^1} = \frac{3}{2R}\psi_0 \int_{\partial B_i} (w - P_i(w)) \, d\mathcal{H}^2.$$

Varying ψ_0 we end up with

$$\int_{\partial B_i} P_i(w) \, d\mathcal{H}^2 = \int_{\partial B_i} w \, d\mathcal{H}^2,$$

which is the desired result. □

A.3 Proof of Lemma 3.12

We recall the statement of Lemma 3.12.

Lemma A.5. *Using the notation from Theorem 3.1 assume that* (A3) *is fullfilled for some* $\delta > 0$. *Denote by* π *a coupling between* ρ_N *and* ρ *and set* $\eta = \int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^p d\pi(x,y)$, *then*

$$\int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^{p-1} |K * \rho_N(x) - K * \rho(y)| \, d\pi(x,y) \le C(S_{2,\delta}+1)\eta + \rho_N[D_\delta](N^{-1}d_{min}^{-1} + N^{-2/3})^p$$

for some constant C *depending on* $\|\rho\|_{L^\infty}$. *For* $p = \infty$ *we consider* $\eta = \pi - \text{esssup}|x-y|$ *and obtain*

$$\pi - \text{esssup}|K * \rho_N(x) - K * \rho(y)| \le C(S_{2,\delta}+1)\eta + N^{-1}d_{min}^{-1} + N^{-2/3} \tag{A.1}$$

for C *depending on* $\|\rho\|_{L^\infty}$.

Proof. We present the proof in the case $p < \infty$ We define $D = D_\delta = \{X_i \mid d_{ii_{nn}} < \delta\}$ and $G = G_\delta = \{X_i \mid d_{ii_{nn}} > \delta\}$. Then, we can rewrite and split the integral

$$\begin{aligned}
&\int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^{p-1} \, |K * \rho_N(x) - K * \rho(y)| \, d\pi(x,y) \\
&\qquad \le \int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^{p-1} \int_{\mathbb{R}^3\times\mathbb{R}^3} |K(x-z) - K(y-w)| \, d\pi(z,w) d\pi(x,y) \\
&\qquad = \int_{G\times\mathbb{R}^3} |x-y|^{p-1} \int_{\mathbb{R}^3\times\mathbb{R}^3} |K(x-z) - K(y-w)| d\pi(z,w) d\pi(x,y) \\
&\qquad\quad + \int_{D\times\mathbb{R}^3} |x-y|^{p-1} \int_{B_\delta(x)^C\times\mathbb{R}^3} |K(x-z) - K(y-w)| d\pi(z,w) d\pi(x,y) \\
&\qquad\quad + \int_{D\times\mathbb{R}^3} |x-y|^{p-1} \int_{B_\delta(x)\times\mathbb{R}^3} |K(x-z) - K(y-w)| d\pi(z,w) d\pi(x,y) \\
&=: I_1 + I_2 + I_3.
\end{aligned}$$

To bound the different terms we recall that for $(x,y) \in \text{supp}(\pi)$ it holds

$$|K(x-z) - K(y-w)| \lesssim \left(\frac{\mathbb{1}_{\{x\neq z\}}}{|x-z|^2} + \frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2} \right) (|x-y| + |z-w|). \tag{A.2}$$

In addition, we remark that

$$
\begin{aligned}
\sup_{y\in\mathbb{R}^3}\int_{\mathbb{R}^3\times\mathbb{R}^3}\mathbb{1}_{\{y\neq w\}}|y-w|^{-2}\,d\pi(z,w) &= \sup_{y\in\mathbb{R}^3}\int_{\mathbb{R}^3}\mathbb{1}_{\{y\neq w\}}|y-w|^{-2}d\rho(w)\\
&\lesssim \sup_{y\in\mathbb{R}^3}\int_{B_1(y)}\mathbb{1}_{\{y\neq w\}}|y-w|^{-2}\,\|\rho\|_{L^\infty}\,dw+\int_{B_1(y)^C}|y-w|^{-2}d\rho(w)\lesssim\|\rho\|_{L^\infty}\;+1 \quad (A.3)
\end{aligned}
$$

as well as

$$
\sup_{x\in G}\int_{\mathbb{R}^3\times\mathbb{R}^3}\mathbb{1}_{\{x\neq z\}}|x-z|^{-2}d\pi(z,w)=\sup_{x\in G}\int_{\mathbb{R}^3}\mathbb{1}_{\{x\neq z\}}|x-z|^{-2}d\rho_N(z)\leq S_{2,\delta} \qquad (A.4)
$$

and

$$
\sup_{x\in D}\int_{B_\delta(x)^C\times\mathbb{R}^3}|x-z|^{-2}d\pi(z,w)\leq S_{2,\delta}. \qquad (A.5)
$$

Using (A.2) and Young's inequality, we obtain

$$
\begin{aligned}
I_1 &\lesssim \int_{G\times\mathbb{R}^3}\int_{\mathbb{R}^3\times\mathbb{R}^3}|x-y|^{p-1}\left(\frac{\mathbb{1}_{\{x\neq z\}}}{|x-z|^2}+\frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2}\right)(|x-y|+|z-w|)d\pi(z,w)d\pi(x,y)\\
&\lesssim \int_{G\times\mathbb{R}^3}|x-y|^p\int_{\mathbb{R}^3\times\mathbb{R}^3}\frac{\mathbb{1}_{\{x\neq z\}}}{|x-z|^2}d\pi(z,w)+\int_{\mathbb{R}^3\times\mathbb{R}^3}|z-w|^p\frac{\mathbb{1}_{\{x\neq z\}}}{|x-z|^2}d\pi(z,w)d\pi(x,y)\\
&\quad+\int_{G\times\mathbb{R}^3}|x-y|^p\int_{\mathbb{R}^3\times\mathbb{R}^3}\frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2}d\pi(z,w)+\int_{\mathbb{R}^3\times\mathbb{R}^3}|z-w|^p\frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2}d\pi(z,w)d\pi(x,y)\\
&\lesssim (S_{2,\delta}+1+\|\rho\|_{L^\infty}\,)\eta,
\end{aligned}
$$

where we applied Fubini and utilized (A.3) and (A.4).

In an analogous manner, we observe

$$
\begin{aligned}
I_2 &\lesssim \int_{D\times\mathbb{R}^3}|x-y|^p\int_{B_\delta(x)^C\times\mathbb{R}^3}|x-z|^{-2}\,d\pi(z,w)d\pi(x,y)\\
&\quad+\int_{D\times\mathbb{R}^3}|x-y|^p\int_{B_\delta(x)^C\times\mathbb{R}^3}\frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2}d\pi(z,w)d\pi(x,y)\\
&\quad+\int_{D\times\mathbb{R}^3}\int_{\mathbb{R}^3\times\mathbb{R}^3}|z-w|^p\frac{\mathbb{1}_{\{|x-z|>\delta\}}}{|x-z|^2}d\pi(z,w)d\pi(x,y)\\
&\quad+\int_{D\times\mathbb{R}^3}\int_{\mathbb{R}^3\times\mathbb{R}^3}|z-w|^p\frac{\mathbb{1}_{\{y\neq w\}}}{|y-w|^2}d\pi(z,w)d\pi(x,y)\\
&\lesssim (S_{2,\delta}+1+\|\rho\|_{L^\infty})\eta
\end{aligned}
$$

using (A.3) and (A.5). Finally, we can focus on the critical term I_3. But in preparation, we notice two helpful estimates. By assumption (A3) we know that for $x=X_i$ there is at most one $j\neq i$ with $X_j\in B_\delta(x)$. Hence,

$$
\sup_{x\in D}\int_{B_\delta(x)\times\mathbb{R}^3}|K(x-z)|d\pi(z,w)\lesssim \tfrac{1}{N}d_{min}^{-1}. \qquad (A.6)
$$

To handle the continuous part, we define the x-dependent measure σ_x on $\mathbb{R}^3$ via $\sigma_x(A)=\pi(B_\delta(x)\times A)$. Then, it holds $\|\sigma_x\|_{L^1}\leq\frac{2}{N}$ and $\|\sigma_x\|_{L^\infty}\leq\|\rho\|_{L^\infty}$ leading to

$$
\begin{aligned}
\sup_{y\in\mathbb{R}^3}\sup_{x\in D}\int_{B_\delta(x)\times\mathbb{R}^3}|K(y-w)|d\pi(z,w) &= \sup_{y\in\mathbb{R}^3}\sup_{x\in D}\int_{\mathbb{R}^3}|K(y-w)|d\sigma_x(w)\\
&\lesssim \|\sigma\|_{L^1}^{2/3}\|\sigma\|_{L^\infty}^{1/3}\lesssim\|\rho\|_{L^\infty}^{1/3}N^{-2/3} \quad (A.7)
\end{aligned}
$$

by usage of Young's convolution inequality. Combining (A.6) and (A.7), we can bound I_3 by

$$
\begin{aligned}
I_3 &\leq \int_{D\times\mathbb{R}^3} |x-y|^{p-1} \int_{B_\delta(x)\times\mathbb{R}^3} |K(x-z)| + |K(y-w)| d\pi(z,w) d\pi(x,y) \\
&\lesssim \int_{D\times\mathbb{R}^3} |x-y|^{p-1} \left(\tfrac{1}{N} d_{min}^{-1} + \|\rho\|_{L^\infty}^{1/3} N^{-2/3}\right) d\pi(x,y) \\
&\lesssim \rho_N(D)^{1/p}\, \eta^{\frac{p-1}{p}} \left(\tfrac{1}{N} d_{min}^{-1} + \|\rho\|_{L^\infty}^{1/3} N^{-2/3}\right) \\
&\lesssim \eta + \rho_N(D) \left(\tfrac{1}{N} d_{min}^{-1} + \|\rho\|_{L^\infty}^{1/3} N^{-2/3}\right)^p .
\end{aligned}
$$

Summarizing all the collected bounds, we end up with

$$
\begin{aligned}
&\int_{\mathbb{R}^3\times\mathbb{R}^3} |x-y|^{p-1}\, |K*\rho_N(x) - K*\rho(y)|\, d\pi(x,y) \\
&\qquad\lesssim (1 + S_{2,\delta} + \|\rho\|_{L^\infty})\eta + \rho_N(D) \left(\tfrac{1}{N} d_{min}^{-1} + \|\rho\|_{L^\infty}^{1/3} N^{-2/3}\right)^p .
\end{aligned}
$$

This is the claimed estimate. The case $p=\infty$ follows from taking the maximum of the estimates (A.3) - (A.7). □

A.4 Probabilistic Results

The goal of this subsection is to provide rigourous proofs of several claimed scaling behaviours of minimal distances for independently distributed particle centers.

Our main argument will rely on a Poisson-approximation generated by the Chen-Stein method. The advantage of using this method is that it goes hand in hand with bounds on the total variation distance between the real and the approximate measure.

The following theorem will form the foundation for the ongoing discussion.

Theorem A.1 ([AGG90, Thm. 1])**.** *Let I be a countable index set and $\{X_\alpha\}_{\alpha\in I}$ a sequence of Bernoulli random variables. For $\alpha \in I$ we consider $B_\alpha = \{\beta \mid X_\beta$ and X_α dependent$\}$ and set*

$$
b_1 = \sum_{\alpha\in I}\sum_{\beta\in B_\alpha} \mathbb{E}(X_\alpha)\mathbb{E}(X_\beta) \quad \text{and} \quad b_2 = \sum_{\alpha\in I}\sum_{\beta\in B_\alpha, \beta\neq\alpha} \mathbb{E}(X_\alpha X_\beta).
$$

Let $W = \sum_{\alpha\in I} X_\alpha$ and assume $\lambda = \mathbb{E}(W) < \infty$, then

$$
\|\mathcal{L}(W) - \mathcal{L}(Z)\|_{TV} \leq 2(b_1+b_2)
$$

for $Z \sim Poi(\lambda)$. Here, $\mathcal{L}(\cdot)$ refers to the distribution of a random variable. In particular, $|\mathbb{P}(W=0) - e^{-\lambda}| \leq 2(b_1+b_2)$.

We will utilize this theory by means of the following lemma.

Lemma A.6. *Assume $P = \Phi_\# Q$ with $Q = Uniform([0,1]^d)$ and $\Phi : [0,1]^d \to \mathbb{R}^d$ bi-Lipschitz with $Lip(\Phi), Lip(\Phi^{-1}) < L$ for some $L > 1$. Let $k \in \mathbb{N}$ be fixed and $N > 2k$. Consider $R_1 \leq R_2 \leq \cdots \leq R_{k-1} < \frac{1}{2L}$, then there exists a $C > 0$ just depending on k and L such that*

$$
\begin{aligned}
&P(\exists i_1, \dots i_k \in \{0,\dots,N\} \text{ pairwise different} \mid d_{i_1 i_2} < R_1, d_{i_1 i_3} < R_2, \dots, d_{i_1 i_k} < R_{k-1}) \\
&\qquad \geq 1 - e^{-C^{-1} N^k (R_1 R_2 \cdots R_{k-1})^d} - C N^k (R_1 R_2 \cdots R_{k-1})^d \left(\sum_{l=1}^{k-1} N^{k-l} (R_l R_{l+1} \cdots R_{k-1})\right)
\end{aligned}
$$

Proof. We consider the index set $I = \{\alpha \in \{0,\dots,N\}^k \mid \alpha_i \neq \alpha_j \text{ for } i \neq j\}$ and define the events $\mathcal{A}_\alpha = \{d_{\alpha_1\alpha_l} < R_{l-1}\ \forall l = 2\dots k\}$ and the Bernoulli random variables $X_\alpha = \mathbb{1}_{\mathcal{A}_\alpha}$. In this scenario $B_\alpha = \{\beta \in I \mid \beta_i = \alpha_j \text{ for some } i,j\}$. Now, we are going to bound the error terms b_1 and b_2. To do so, we notice

$$\mathbb{E}\left[X_\alpha\right] = \int_{\mathbb{R}^d}\int_{B_{R_1}(x_1)} dP(x_2)\cdots\int_{B_{R_{k-1}}(x_1)} dP(x_k)dP(x_1)\begin{cases} \leq C_L^k(R_1\cdots R_{k-1})^d \\ \geq c_L^k(R_1\cdots R_{k-1})^d \end{cases}$$

since there are $c_L > 0, C_L > 0$ with $c_L R^d \leq P(B_R(x)) = \int_{B_R(x)} dP(y) < C_L R^d$ for all $x \in \operatorname{supp}(P)$. Analogously, for $\beta \in B_\alpha \setminus \{\alpha\}$ it holds

$$\mathbb{E}\left[X_\alpha X_\beta\right] \leq C_L^k(R_1\cdots R_k)^d C^{k-l}(R_l\cdots R_k)^d \quad \text{if } \#\beta\cap\alpha = l.$$

Accordingly, we obtain

$$\begin{aligned} b_1 &= \sum_{\alpha\in I}\sum_{\beta\in B_\alpha}\mathbb{E}[X_\alpha]\mathbb{E}[X_\beta] \\ &= \frac{N!}{(N-k)!}\left(\frac{N!}{(N-k)!} - \frac{(N-k)!}{(N-2k)!}\right)\mathbb{E}[X_\alpha]\mathbb{E}[X_\beta] \leq CN^kN^{k-1}C_L^{2k}(R_1\cdots R_k)^{2d} \end{aligned}$$

and

$$\begin{aligned} b_2 = \sum_{\alpha\in I}\sum_{\beta\in B_\alpha\setminus\{\alpha\}}\mathbb{E}[X_\alpha X_\beta] &\leq \frac{N!}{(N-k)!}\sum_{l=1}^{k-1} k!\binom{k}{l}\binom{N-k}{k-l}C_L^k(R_1\cdots R_k)^d C^{k-l}(R_l\cdots R_k)^d \\ &\leq C_{L,k}N^k(R_1\cdots R_k)^d\sum_{l=1}^{k-1}N^{k-l}(R_l\cdots R_k)^d. \end{aligned}$$

Setting $W = \sum_{\alpha\in I} X_\alpha$ we remark that

$$\mathbb{E}[W] \geq \tfrac{N!}{(N-k)!}c_L^k(R_1\cdots R_k)^d \geq c_{L,k}N^k(R_1\cdots R_k)^d$$

and by Theorem A.1 we obtain

$$P[W\neq 0] \geq 1 - e^{-\mathbb{E}[W]} - 2(b_1+b_2) \geq 1 - e^{-c_{L,k}N^k(R_1\cdots R_k)^d} - C_{L,k}(N^k(R_1\cdots R_k)^d\sum_{l=1}^{k-1}N^{k-l}(R_l\cdots R_k)^d).$$

By the equivalence

$$W\neq 0 \Leftrightarrow \exists i_1,\dots i_k \in \{0,\dots,N\} \text{ pairwise different s.t. } d_{i_1i_l} < R_l \text{ for } l = 2,\dots,k-1$$

we conclude the claimed statement. □

The counter part of the previous lemma is the following brute-force estimate.

Lemma A.7. *Under the previous conditions, there exists a constant $\tilde{C}$ depending only on k, d and L such that*

$$\begin{aligned} P(\exists i_1,\dots i_k \in \{0,\dots,N\} \textit{ pairwise different} \mid d_{i_1i_2} < R_1, d_{i_1i_3} < R_2,\dots,d_{i_1i_k} < R_{k-1}) \\ \leq \tilde{C}N^k(R_1\dots R_{k-1})^d. \end{aligned}$$

Proof. We remark that the number of choices of indices $i_1,\dots,i_k$ is bounded by N^k. Accordingly, we obtain

$$\begin{aligned} P(\exists i_1,\dots i_k \in \{0,\dots,N\} \text{ pairwise different} \mid d_{i_1i_2} < R_1, d_{i_1i_3} < R_2,\dots,d_{i_1i_{k-1}} < R_{k-1}) \\ \leq N^kP(d_{12} < R_1, d_{13} < R_2,\dots,d_{1k} < R_{k-1}) \leq \tilde{C}N^k(R_1\dots R_{k-1})^d. \end{aligned}$$

□

The following corollary closes the gap in the current statement about the validity of assumption (A4) for $d = 3$. In fact, it proves that the probability of not fulfilling (A4) is uniformly lower bounded.

Corollary A.1. *In the case $d = 3$ it holds for $\delta_N \sim N^{-2/3+\varepsilon}$ and $\varepsilon \in (0, 1/6)$*

$$\liminf_{N\to\infty} P(\exists i \neq k \neq j \neq i \mid d_{ij} \leq \delta_N \wedge N^{-1}\, d_{ij}^{-1} d_{ik}^{-1} \geq 1) > 0.$$

Proof. We will utilize the previous lemma with $d = 3$, $k = 3$ and the appropiate choices $R_1 = N^{-2/3+\varepsilon/2}$ and $R_2 = N^{-1/3-\varepsilon/2}$. In this setting, we obtain $N R_1 R_2 = 1$ and

$$N^k (R_1 \cdots R_k)^d \sum_{l=1}^{k-1} N^{k-l} (R_l \cdots R_k)^d = N^2 N^{-3} + N N^{-1-\frac{3\varepsilon}{2}} \to 0.$$

Since

$$d_{ij} < R_1 \wedge d_{ik} < R_2 \quad \Rightarrow \quad d_{ij} \leq \delta_N \wedge N^{-1} d_{ij}^{-1} d_{ik}^{-1} \geq 1,$$

we deduce

$$\begin{aligned}
&\liminf_{N\to\infty} P(\exists i \neq k \neq j \neq i \mid d_{ij} \leq \delta_N \wedge N^{-1}\, d_{ij}^{-1} d_{ik}^{-1} \geq 1) \\
&\qquad\qquad \geq \liminf_{N\to\infty} P(\exists i \neq k \neq j \neq i \mid d_{ij} < R_1, d_{ik} < R_2) > 1 - e^{-C^{-1}} > 0
\end{aligned}$$

□

We end with proving several results about the validity of the extended condition (A4').

Corollary A.2. *In the case $d = 3$ and $k = 4$, it holds*

$$\liminf_{N\to\infty} P(\exists i_1, \dots i_4 \in \{0, \dots, N\} \text{ p.w. different} \mid d_{i_1 i_2} < R_1, \dots, d_{i_1 i_4} < R_3) > 0 \Rightarrow R_1 R_2 R_3 \gtrsim N^{-4/3}.$$

Proof. This follows directly from Lemma A.7. □

On the contrary, we also obtain the following.

Corollary A.3. *In the case $d = 3$ and $k = 4$, assume $R_1 \leq R_2 \leq R_3$ with $\lambda^{-1}\, N^{-4/3} \leq R_1 R_2 R_3 \leq \lambda N^{-4/3}$ for some $\lambda > 1$, then there exists some constant c with the following property: If $R_3 \leq c N^{-1/3}$, then*

$$\liminf_{N\to\infty} P(\exists i_1, \dots i_4 \in \{0, \dots, N\} \text{ pairwise different} \mid d_{i_1 i_2} < R_1, \dots, d_{i_1 i_4} < R_3) > 0.$$

Proof. We are going to apply Lemma A.6 and therefore we are going to bound the error term. It holds

$$C N^4 (R_1 R_2 R_3)^3 \left(\sum_{l=1}^{3} N^{4-l} (R_l R_{l+1} \cdots R_3) \right) \leq \lambda C (\lambda N^3 N^{-4} + N^2 (R_2 R_3)^3 + N R_3^3) \leq \lambda C (3c)$$

for N large enough. Accordingly, we obtain

$$P(\exists i_1, \dots i_4 \in \{0, \dots, N\} \text{ pairwise different} \mid d_{i_1 i_2} < R_1, \dots, d_{i_1 i_4} < R_3) \geq 1 - e^{-(\lambda C)^{-1}} - \lambda C (3c)$$

for N large enough. Choosing $c < (3\lambda C)^{-1}(1 - e^{-(\lambda C)^{-1}})$, we deduce the claim. □

Remark A.1. The previous two lemma indicate two facts introduced in section 4:

- By choosing $R_1 = R_2 = R_3 = N^{-4/9}$ we deduce that the minimal third-closest distance scales like $N^{-4/9}$ and thus we should choose $\delta_N \lesssim N^{-4/9-\varepsilon}$ in the modified three-particle case.

- Assuming $R_1 R_2 \sim N^{-1}$ we conclude that just in the case $R_1 \sim N^{-2/3}$ a similar scaling of R_2 and R_3 is possible. Hence, only in this case the distance to the third-nearest neighbour is not essential different to the next-to-nearest particle. Consequently, condition (A4') is fulfilled in all the other cases.

Finally, we prove the problematic scaling behaviour of the triangle size in the two-dimensional setting.

Corollary A.4. *In the setting of the foregoing lemma we consider the case $d = 2$. Denote by* $\text{area}(i, j, k)$ *the area of the triangle spanned by X_i, X_j and X_k. Then, it holds*

$$\liminf_{N\to\infty} P(\exists i \neq j \neq k \neq i \mid \text{area}(i, j, k) \leq N^{-3/2}) > 0.$$

Proof. Again, we are going to utilize Lemma A.6. This time we choose $k = 3$ and consider the choices $R_1 = R_2 = N^{-3/4}$. We notice that

$$N^k (R_1 \cdots R_k)^d \sum_{l=1}^{k-1} N^{k-l} (R_l \cdots R_k)^d = N^2 N^{-3} + N^1 N^{-3/2} \to 0$$

and

$$d_{ij} < R_1 \wedge d_{ik} < R_2 \quad \Rightarrow \quad \text{area}(i, j, k) < R_1 R_2 = N^{-3/4}.$$

Accordingly, we end up with

$$\begin{aligned}\liminf_{N\to\infty} & P(\exists i \neq j \neq k \neq i \mid \text{area}(i, j, k) \leq N^{-3/2}) \\ & \geq \liminf_{N\to\infty} P(\exists i \neq j \neq k \neq i \mid d_{ij} < R_1, d_{ik} < R_2) \geq 1 - e^{-C} > 0.\end{aligned}$$

□

References

[AGG90] Richard Arratia, Larry Goldstein, and Louis Gordon. Poisson approximation and the chen-stein method. *Statistical Science*, 5(4):403 – 424, 1990. doi:10.1214/ss/1177012015.

[CLLS88] Russel E. Caflisch, Chjan Lim, Jonathan H. C. Luke, and Ashok S. Sangani. Periodic solutions for three sedimenting spheres. *The Physics of Fluids*, 31(11):3175–3179, 11 1988. doi:10.1063/1.866928.

[Dob79] R. L. Dobrušin. Vlasov equations. *Funktsional. Anal. i Prilozhen.*, 13(2):48–58, 96, 1979.

[Gal11] Giovanni P. Galdi. *An Introduction to the Mathematical Theory of the Navier-Stokes Equations: Steady-State Problems*. Springer Tracts in Natural Philosophy. Springer Science & Business Media, New York, NY, 2nd edition, 2011. doi:10.1007/978-1-4614-6348-1.

[GMP11] Élisabeth Guazzelli, Jeffrey F. Morris, and Sylvie Pic. *A Physical Introduction to Suspension Dynamics*. Cambridge Texts in Applied Mathematics. Cambridge University Press, 2011.

[GVH20] David Gérard-Varet and Richard M. Höfer. Mild assumptions for the derivation of einstein's effective viscosity formula, 2020, 2002.04846. URL `https://arxiv.org/abs/2002.04846`.

[Hau09] Maxime Hauray. Wasserstein distances for vortices approximation of euler-type equations. *Mathematical Models and Methods in Applied Sciences*, 19(08):1357–1384, 2009. doi:10.1142/S0218202509003814.

[Hoc64] L. M. Hocking. The behaviour of clusters of spheres falling in a viscous fluid part 2. slow motion theory. *Journal of Fluid Mechanics*, 20(1):129–139, 1964. doi:10.1017/S0022112064001070.

[HS21] Richard M. Höfer and Richard Schubert. The influence of einstein's effective viscosity on sedimentation at very small particle volume fraction, 2021, 2008.04813. URL `https://arxiv.org/abs/2008.04813`.

[HS23] Richard M. Höfer and Richard Schubert. Sedimentation of particles with very small inertia ii: Derivation, cauchy problem and hydrodynamic limit of the vlasov-stokes equation, 2023, 2311.01891. URL `https://arxiv.org/abs/2311.01891`.

[HS25] Richard M. Höfer and Richard Schubert. Propagation of chaos for first-order mean-field systems with non-attractive moderately singular interaction, 2025, 2509.14022. URL `https://arxiv.org/abs/2509.14022`.

[HV18] Richard Höfer and Juan-Jose Velazquez. The method of reflections, homogenization and screening for poisson and stokes equations in perforated domains. *Archive for Rational Mechanics and Analysis*, 227, 03 2018. doi:10.1007/s00205-017-1182-4.

[Hö18] Richard Höfer. Sedimentation of inertialess particles in stokes flows. *Communications in Mathematical Physics*, 360, 05 2018. doi:10.1007/s00220-018-3131-y.

[Hö23] Richard M. Höfer. Convergence of the method of reflections for particle suspensions in stokes flows, 2023, 1912.04388. URL `https://arxiv.org/abs/1912.04388`.

[JO04] Pierre Emmanuel Jabin and Felix Otto. Identification of the dilute regime in particle sedimentation. *Communications In Mathematical Physics*, 250(2):415–432, September 2004. doi:10.1007/s00220-004-1126-3.

[Luk89] Jonathan H. C. Luke. Convergence of a multiple reflection method for calculating stokes flow in a suspension. *SIAM Journal on Applied Mathematics*, 49(6):1635–1651, 1989. doi:10.1137/0149099.

[Mec20] Amina Mecherbet. Sedimentation of particles in stokes flow, 2020, 1806.07795. URL https://arxiv.org/abs/1806.07795.

[MR94] Paolo Maremonti and Remigio Russo. On the maximum modulus theorem for the Stokes system. *Annali della Scuola Normale Superiore di Pisa - Classe di Scienze*, Ser. 4, 21(4):629–643, 1994. URL https://www.numdam.org/item/ASNSP_1994_4_21_4_629_0/.

[NS20] Barbara Niethammer and Richard Schubert. A local version of einstein's formula for the effective viscosity of suspensions. *SIAM Journal on Mathematical Analysis*, 52(3):2561–2591, 2020. doi:10.1137/19M1251229.

[Smo11] M. Smoluchowski. Über die Wechselwirkung von Kugeln, die sich in einer zähen Flüssigkeit bewegen. Krak. Anz. (A) 1911, 28-39 (1911)., 1911.